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Supplementary Materials for

Soft hydrogel semiconductors with augmented biointeractive functions

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Materials and Methods

Materials

All the chemicals in this work were used from purchase without further purification. 2-hydroxy-2-methyl-propiophenon, 2,5-bis(trimethylstannyl)thiophene, bromobenzene, deuterium oxide, chloroform, ethyl acetate, hexane, acetone, dimethyl sulfoxid, poly(vinylidenefluoride-co-hexafluoropropylene), tris(dibenzylideneacetone)dipalladium, tri(o-tolyl)phosphine, sodium chloride, poly(sodium 4-styrenesulfonate) solution (M_w=200 kDa, 30 wt.% in H_2O), trimethyl(phenyl)tin, glucose oxidase (from *Aspergillus niger*, Type X-S), D-(+)-Glucose, choline chloride, acrylamide, ammonium hexafluorophosphate, ferrocene, dextran ($M_r \sim 70000$), 1-ethyl-3-methylimidazolium bis(trifluoromethylsulfonyl)imide were purchased from Sigma-Aldrich. Acrylic acid was purchased from Alfa Aesar. Tetraethylene glycol dimethylacrylate was purchased from Polysciences Inc. 5,5'-dibromo-3,3'-bis(2-(2-(2-methoxyethoxy)ethoxy)ethoxy)-2,2'-bithiophene was purchased from SunaTech Inc. Styrene-Ethylene-Butylene-Styrene was obtained from Asahi Kasei. Polydimethylsiloxane (Sylgard 184) was purchased from Ellsworth Adhesives. Ecoflex 00-30 was purchased from Smooth-on, Inc. Polyethylene glycol-fluorescein isothiocyanate (M_w = 20 kDa) was purchased from Wuhan Borenpharm Co.

Synthesis of p(g2T-T)

The polymerization of p(g2T-T) was carried out according to the reported method (*41*), with the Biotage Initiator$^+$ microwave reactor. Briefly speaking, in a clean 5.0 mL microwave vial, 63.8 mg of 2,5-bis(trimethylstannyl)thiophene (155.7 µmol) and 100.0 mg of 5,5'-dibromo-3,3'-bisalkoxy-2,2'-bithiophene (155.7 µmol) were dissolved in 2.0 mL of anhydrous degassed chlorobenzene. $Pd_2(dba)_3$ (2.48 mg, 2.71 µmol) and $P(o\text{-}tol)_3$ (3.76 mg, 12.3 µmol) were added and the vial was sealed under nitrogen. The vial was subjected to heating: 5 min at 100 °C, 5 min at 140 °C, 5 min at 160 °C, 5 min at 180 °C, and 25 min at 200 °C. After polymerization, the vial was cooled, 40 µL of trimethyl(phenyl)tin was added and the contents were subjected to microwave heating: 2 min at 100 °C, 2 min at 140 °C, 2 min at 160 °C, 2 min at 180 °C, and 5 min at 200 °C. Finally, 100 µL of 2-bromobenzene was added and the reaction was subjected to microwave heating: 2 min at 100 °C, 2 min at 140 °C, 2 min at 160 °C, 2 min at 180 °C, and 5 min at 200 °C. Then the reaction mixture was cooled to room temperature and precipitated in methanol. Formed blue solids were filtered by a filtering paper, and Soxhlet extraction was carried out with hexane, ethyl acetate, acetone, and chloroform for 6 h at each step. The polymer was then dissolved in hot chloroform. Finally, the polymer chloroform solution was concentrated and dried under a high vacuum.

Preparation of p(g2T-T) and hydro-SC thin films

The preparation of hydro-SC thin films was carried out inside a glovebox except for the solvent exchange step. To prepare the precursor solution of hydro-SC, p(g2T-T) polymer was dissolved in DMSO with a default concentration of 20 mg/ml unless specified. The solution was heated under 90 °C for 4 hours to yield a viscous solution. Then the hydrogel precursors were added to p(g2T-T) solution with the default acrylic acid (AAc) concentration of 200 mg/ml, together with tetraethylene glycol dimethylacrylate and 2-hydroxy-2-methyl-propiophenon, both in a concentration of 0.5 wt% of AAc, unless specified otherwise. The precursor solution was mildly heated at 60 °C with stirring for 2 min to well mix all compounds. Before spin-coating, the precursor solution was cooled to room temperature to minimize the DMSO evaporation during the coating process. The spin coating was at the rate of 1000 rpm for 5 s to get a uniform liquid layer with minimal evaporation. Then the film was crosslinked under UV light (UV Nail Lamp, 34B0-

380 nm wavelength, 36 W) for 10 min. Afterward, the DMSO-gel films were transferred out of the glove box and immersed in DI water for 2 min to accomplish the solvent exchange process.

Cryogenic electron microscopy (cryo-EM) characterizations

The Cryo-EM sample was prepared using Vitrobot. The chamber was preheated to 60 ℃ and then 1 μL of heated p(g2T-T) solution 20 mg/mL in DMSO was added to the carbon-coated TEM grid, which was immediately immersed in liquid ethane to freeze the solution structure. The samples were transferred under the liquid nitrogen and imaged by the FEI Talos (S)TEM, operated at 200 kV.

Scanning electron microscopy (SEM) characterizations

The DMSO gel and hydro-SC were prepared using the aforementioned protocol. The hydro-SC was kept in water to ensure sufficient swelling. Both samples were immediately immersed in liquid nitrogen for freezing, and further dried under vacuum at the frozen state with the solvent trap kept at -90 ℃ for 7 days. The samples were imaged by the Carl Zeiss Merlin SEM, operated at 3 kV.

Scanning transmission electron microscopy (STEM) for energy dispersive X-ray spectroscopy (EDS) mapping

The DMSO gel and hydro-SC samples were prepared on the carbon-coated copper grid. Briefly speaking, 1 μL p(g2T-T)/AAc precursor solution was dropped to the grid, and the excessive solution was removed by filter paper to create an ultrathin liquid layer. The precursor solution was then polymerized to a DMSO gel under UV light for 10 min. To create the hydro-SC, the ultrathin DMSO gel was immersed in DI water for 10 min. Both samples were naturally dried for STEM imaging and EDS elemental mapping, which were carried out by the ThermoFisher Spectra 200 probe-corrected STEM equipped with 4 EDS detectors. Although S is the different element between p(g2T-T) and PAAc, the residues of DMSO could have S signals even after the solvent exchange. Hence, Pd and Sn were used for elemental mapping, where Pd came from the catalyst residue during p(g2T-T) synthesis (*42*), and Sn was introduced by not capping the end group of p(g2T-T) in purpose. Typically, the HAADF images were acquired with an electron acceleration energy of 80 kV, a gun lens size of 4, a spot size of 7, and a camera length of 98 mm. When acquiring the EDS data, the gun lens size was reduced to 1 and the spot size was adjusted to 5 to achieve a beam current of over 200 pA, with all four EDX detectors inserted to collect the X-ray signal. During the EDS mapping, the scanning resolution was set to match the corresponding HAADF image resolution (post-process reduced the pixel density), with a dwell time of 2 μs per pixel. To obtain a proper signal-to-noise ratio, a typical mapping averaged over 100 frames of consecutive scans with live drift correction.

Conductive atomic force microscopy (C-AFM) characterizations

The C-AFM measurement was performed by using the C-AFM mode of Cypher ES Environmental AFM with ASTELEC-01 probe and the dual gain C-AFM probe holder (gain = 1μA/1nA/V). Hydro-SC and DMSO gel samples were prepared on Au-covered SiO_2 substrates, and then dried in vacuum for 24 hours. Magnets were placed separately on sample surfaces and exposed areas of Au, and further sealed with Ag paste for reliable electrical contact. Before measurement, the sample bias was zeroed by applying an offset voltage from the AFM. During the measurement,

1V bias was applied. The produced current was converted to a voltage and then amplified and recorded by the AFM hardware.

OECT measurement

OECT devices were fabricated by firstly depositing source/drain electrodes (5 nm Ti and 100 nm Au) on glass substrates via shadow masks. The channel length and width are 1 mm and 2 mm, respectively. Then, hydro-SC films or neat p(g2T-T) films were deposited on top using the aforementioned protocol. The excessive semiconducting films outside the channel area were carefully removed using a blade under an optical microscope to isolate the device for measurement. For OECT measurement, poly(dimethylsiloxane) (PDMS) strips were placed on both sides of the channel to confine the electrolyte. Then, aqueous NaCl solution of 100 mM (unless specified otherwise) was carefully dropped on the channel. During the test, an Ag/AgCl pellet (Warner instruments, E210) was immersed in the electrolyte as the gate electrode. The device measurement was carried out using a Keithley 4200 semiconductor system. Both transfer and output curves were acquired with the scan speed of 40 mV/s unless specified.

Grazing incidence X-ray diffraction (GIXD) measurement

GIXD measurements were conducted on beamline 7.3.3 at the Advanced Light Source in Berkeley Lawrence National Laboratory. Data were collected under a Helium environment with an incident beam energy of 10 keV and an incidence angle of 0.15°. Thin-film samples were prepared and dried on Si substrates. The scattering signals were collected by a Pilatus 2M detector and processed using Igor 8 software combined with the Nika package and WAXSTools.

Electrochemical cyclic voltammetry (CV) and electrochemical impedance spectroscopy (EIS) measurements

CV and EIS measurements were carried out using a PalmSens electrochemical workstation with polymer semiconductors thin films deposited on Au as the working electrode. Specifically, neat p(g2T-T) films were coated on Au electrodes, and hydro-SC films were transferred to Au electrodes using poly(sodium 4-styrenesulfonate) (PSS) as the sacrificial layer. Ag/AgCl was used as the reference electrode and a Pt plate was used as the counter electrode. The measurements were conducted in two types of electrolytes: ammonium hexafluorophosphate in acetonitrile, and NaCl in water, both with a concentration of 100 mM. All CV tests were performed with a scan rate of 0.1 V/s. EIS measurements were performed in 100 mM aqueous NaCl solution over a frequency range of 1 kHz to 0.1 Hz with an AC amplitude of 10 mV, and a DC offset of 0.2 V. The analysis of EIS data was carried out using Multitrace 4.2 software. The capacitance was extracted by fitting the EIS results with a simplified Randle model, and then normalized by film dimensions.

Operando UV-vis measurements

The Operando UV-vis measurements were carried out using a Shimadzu UV-3600 Plus UV-VIS-NIR spectrophotometer together with a manually built electrochemical cell controlled by the aforementioned PalmSens electrochemical workstation. An Ag/AgCl wire, Pt wire, and film-coated indium tin oxide (ITO) glass were used as the reference, counter, and working electrode, respectively. To acquire the UV-vis spectra from p(g2T-T) films and hydro-SC films at the non-doped state, a -0.2 V bias was applied to the working electrode before the acquisition. During the operando UV-vis measurements, the voltage bias was increased from -0.2 V to +0.5 V with a 0.1-V step. The UV-vis spectrum at each bias condition was acquired after the equilibrium.

OECT measurements with an ion exchange gel

The OECT measurements based on the ion exchange gel were carried out by modifying a reported method (*27*). The precursor solution for the ion gel was prepared by dissolving poly(vinylidene fluoride-co-hexafluoropropylene) (PVDF-HFP) and the ionic liquid, in a weight ratio of 1:4, in acetone with the following composition: 17.6 wt% of 1-ethyl-3-methylimidazolium bis(trifluoromethylsulfonyl)imide, 4.4 wt% of PVDF-HFP, and 78 wt% of acetone. To test the OECT performance with the ion gel, about 100 μL of ion-gel precursor solution was dropped onto the channel area, followed by 10 min of evaporation of acetone in the air. This results a thin ion gel layer formed on top the channel. Then aqueous NaCl solution of 100 mM was dropped on the ion gel, and an Ag/AgCl pellet was immersed into the aqueous solution as the gate electrode. The device measurement was carried out using a Keithley 4200 semiconductor system.

Mobility measurement based on OECT configuration

The μ of redox-active polymer semiconductors under OECT operation was obtained by measuring the hole transit time (τ_h) as reported (*43*). Briefly speaking, different constant gate currents (I_g) have been applied to the device under constant drain bias (V_{ds} = -0.1 V). Then the transient slope is plotted versus I_g to calculate the τ_h. The μ is calculated by:

$$\mu = \frac{L^2}{\tau_h \times V_{\mathrm{ds}}}$$

where the L is the channel length.

Floating-on-water (FOW) tensile tests

The floating-on-water tensile tests were carried out based on the reported method (*28*). Briefly speaking, PSS sacrificial layers were prepared by spin-coating 6 wt% PSS aqueous solution onto plasma-treated silicon substrates at 2000 r.p.m. for 1 min. Mixed p(g2T-T)/AAc solutions of different compositions in DMSO were drop-casted on top of the sacrificial layer, crosslinked under UV exposure, dried under vacuum, and cut into testing strips. Prior to the tensile tests, the sample strips were released in either aqueous NaCl solution (100 mM) or PBS buffer to form hydro-SC, and left for 5 min to reach the swelling equilibrium. Then the linear stage and load cell PDMS clamps in the FOW testing setup were attached to each end of the hydro-SC film. All the tensile tests were carried out at a stretching rate of 0.04 mm/s, and the force was measured at a sampling rate of 10 Hz. The Young's modulus was calculated by fitting the strain stress curve within the 0-1% strain range. The swelled film thickness was measured by the liquid AFM as discussed in fig. S22.

Bending test of hydro-SC films in OECT devices

Au source/drain electrodes were deposited on the PET substrates via a shadow mask (channel length = 0.5 mm, channel width = 4 mm). The hydro-SC thin film was prepared on the PET substrates. For the OECT tests, the device was bent on curved surfaces with different diameters to achieve bending radii from 4.5 mm to 23.5 mm. PDMS stamps were attached to both sides of the channel to confine the solution. 100 mM aqueous NaCl solution was dropped onto the hydro-SC, and the device performance was measured with Ag/AgCl pellet as the gate electrode.

Stretchability test of hydro-SC films in OECT devices

Au source/drain electrodes were deposited on the SEBS (1052) substrates via a shadow mask (channel length = 1 mm, channel width = 2 mm), through thermal evaporation at a rate of 0.2 Å/s. The SEBS substrates were heated at 60 ℃. This condition allows the formation of microcrack morphology on the Au electrodes under stretching. The hydro-SC thin film was transferred to the SEBS with the assistance of PSS films as the sacrificial layer. The hydro-SC films were then dried and gently baked under 60 ℃ to improve the adhesion on the SEBS substrates. For the OECT tests, 100 mM aqueous NaCl solution was dropped onto the hydro-SC, and the device was stretched to different strain levels prior to electrical measurement of the OECT performance.

Conformability test

The tendon replica was fabricated by molding a beef tendon twice with EcoFlex. Hydro-SC and p(g2T-T) films were prepared on the PTFE substrate and picked up using a PET rigid frame. Both films were laminated onto the tendon replica to let them neutrally conform to the replica surface, without applying additional pressure. The surface morphologies after the film attachment were imaged under an Olympus OLS5000 LEXT 3D microscope.

Preparation of PAAc and PAAm hydrogel

To prepare PAAc or PAAm hydrogel as the implant or substrate, 250 mg/ml monomers (AAc or AAm), 5 mg/ml tetraethylene glycol dimethylacrylate, 2.5 mg/ml 2-hydroxy-2-methyl-propiophenon were mixed in deionized water and purged with N_2 for 15 min. The mixture was then injected into a rubber cell sandwiched by two glass substrates and crosslinked under UV light (UV Nail Lamp, 340- 380 nm wavelength, 36 W) for 10 min to form the hydrogel.

Animal implantation surgeries

Male C57BL/6 mice of 6-8 weeks old were purchased from Jackson Laboratories. Preoperatively, all mice received subcutaneous injections of 5 mg/kg of Meloxicam for pre-surgery analgesia. All mice were anesthetized using 3% isoflurane in oxygen, and the backs of all mice were shaved. All mice received eye lubricants to prevent dehydration-induced blindness. Then, the shaved backs of all mice were sterilized by scrubbing with betadine and alcohol pads in an alternating fashion, repeated two times. Two transverse incisions, 1 cm in length, were made in the upper and lower backs. The blade tip of a pair of scissors was inserted into the subcutaneous space through the incisions, and two subcutaneous pockets, flanking each incision, were made via blunt tissue dissection using the blunt sides of a pair of scissors. Two implants (5 mm in diameter and 1 mm in thickness) were inserted in the subcutaneous pockets on both sides of each incision, towards the limbs of the mice, for a total of four implants per mouse. Following surgery, all mice were given subcutaneous injections of 5 mg/kg of Meloxicam once every 24 hours for two days. Wound clips were removed after 10 days.

Histological processing for H&E and Masson's trichrome staining

After 4 weeks post-implantation, all mice were euthanized by CO_2 asphyxiation, followed by cervical dislocation. Afterward, the backs of all the mice were shaved. The implants and the surrounding tissues were explanted and fixed in 10% neutral buffered formalin for 24 h. After fixation, the implants and tissues were transferred to 70% ethanol for 24 hrs. The materials were then paraffin-embedded, sectioned (5 μm), and stained for H&E using standard procedures by the Human Tissue Resource Center at the University of Chicago. Masson's trichrome staining was

carried out in-house using standard procedures. Whole-section scans of H&E and MTS-stained tissue sections were conducted using the Olympus VS200 Slideview Slide Scanner.

Immunofluorescence analysis

Materials were fixed, embedded, and sectioned as described above. Tissue sections were de-paraffinized in xylene and rehydrated using an ethanol-to-water gradient. Antigen retrieval was conducted in citrate buffer (pH 5) for 20 minutes at >95°C. Samples were then rinsed in water, permeabilized by incubating in 10% DMSO in PBS for 5 minutes, washed in PBS-tween 20 (PBS-T), and rinsed in water again. Thereafter, the samples were blocked for 1 hour using a 1% bovine serum albumin (BSA) solution. The blocking buffer was removed, and the sections were incubated in an antibody cocktail consisting of rat anti-mouse F4/80 (1:100 dilution, Abcam, Cat. #AB6640) and goat anti-mouse αSMA (1:100 dilution, Novus Biologicals, Cat. #NB300-978) overnight at 4°C. The next day, the sections were washed twice in PBS-T and once in PBS. The washed sections were incubated for 2 hours at room temperature in a secondary antibody cocktail, consisting of donkey anti-rat IgG AlexaFluor 647 (1:200 dilution, Southern Biotech, Cat. #OB643031) and donkey anti-goat IgG AlexaFluor 594 (1:500 dilution, Fisher Scientific, Cat. #A21432). Afterward, the sections were washed twice with PBS-T and once with PBS. The washed sections were counterstained with 4',6-diamidino-2-phenylindole (DAPI, 1:1000 dilution, Thermo Scientific) for 2 minutes and washed with PBS. Prolong Antifade Gold mounting medium (Invitrogen) was added, and the sections were sealed with coverslips and nail polish. Images were acquired using a Zeiss AxioObserver 7 inverted epifluorescence microscope with a Zeiss Axiocam 705 or 305 camera run by Zeiss Zen software.

Measurements of photoelectrochemical effects from hydrogel and polymer semiconductors

A three-electrode electrochemical cell system was employed for photoelectrochemical measurements using Corrtest Instruments multichannel potentiostat (CS310X). The working electrode was the as-prepared p(g2T-T) or hydro-SC coated on an Au film, the counter electrode was a Pt coil, and the reference electrode was an Ag/AgCl (in 3M NaCl). All tests were conducted in PBS buffer with a 0 V bias applied vs. the reference electrode. Illumination was provided by a 625 nm LED (Thorlabs, M625L4-C1) at intensities of 114.3 mW cm^{-2}. The steady-state faradaic current was measured after 60 seconds of continuous illumination.

Measurements of photothermal effects from hydrogel and polymer semiconductors

Hydro-SC and p(g2T-T) films were prepared on PET substrates and cut to 5 mm diameter plates. Before the photothermal test, the hydro-SC films were hydrated in 100 mM NaCl solution and then covered by a layer of thin PET foil to prevent water evaporation during the illumination. Both films were heated under a 635 nm laser (Laserglow) with a 5 mm diameter spot, with different irradiation powers. For the photothermal tests on the skin, a piece of the hydro-SC film was laminated to a piece of wet pig skin with the other side encapsulated by PDMS. The heating and recording were performed in the similar way with a radiation power of 318 mW cm^{-2}. The temperature measurement was performed by a Optris PI 640i thermal camera.

Tests of macromolecule diffusion ability through hydrogel and polymer semiconductor films

Hydro-SC and p(g2T-T) films were prepared on the polytetrafluoroethylene (PTFE) surfaces and peeled off using PET tapes as the rigid frames. Then, the peeled films were carefully sandwiched between two glass tubes and sealed with epoxy resin. Polyethylene glycol-fluorescein

isothiocyanate (PEG-FITC) solution (50 μM in PBS buffer) and PBS buffer were injected on two sides of the tube. The tubes were left horizontally for 24 hrs. The fluorescent images were taken under UV light. After the diffusion test, the solution in the PBS buffer side was collected and measured by a HORIBA Fluorolog-3 Spectrofluorometer.

GOx immobilization and glucose sensing

The preparation of hydro-SC started from the solution of p(g2T-T) of 20 mg/mL, AAm of 300 mg/mL, tetraethylene glycol dimethylacrylate of 3 mg/mL, and 2-hydroxy-2-methyl-propiophenon concentration of 3 mg/mL, in DMSO. Then, the solution was spin-coated on glass substrates with pre-deposited Au source/drain electrodes (channel length = 0.2 mm, channel width = 4 mm), followed by UV crosslinking and solvent exchange in water. P(g2T-T)-based OECT devices were fabricated by spin-coating p(g2T-T) solution of 10 mg/mL in chloroform onto Au-deposited substrates. The device channels were then incubated in PBS buffer containing 10 mg/ml GOx and 5 mg/ml glutaraldehyde for 24 hrs at room temperature. Then, the devices were rinsed with PBS buffer 3 times, dried by nitrogen blow, and carefully isolated into individual channels using a blade under an optical microscope. For glucose sensing, the glucose solutions were prepared with PBS buffer and carefully added to the channel. Following the addition of glucose, the device was incubated for 10 min and then the transfer curve was acquired from +0.4 V to -0.4 V with a scan rate of 40 mV/s. For glucose sensing tests in biological environments, including fetal bovine serum (FBS) and human sweat, the hydro-SC-based OECTs were incubated in the biofluids for 1 h before tests for the stabilization of transfer curves, and then the glucose solution was added and the transfer curves were acquired in sequence.

X-ray photoelectron spectroscopy (XPS)

All testing samples were prepared on Si substrate. Both N1s and C1s spectra of GOx-modified films were acquired using XPS. The depth-profiling XPS was done Kratos AXIS Nova with a monochromatic Al Kα X-ray source and a delay line detector system with Ar1000+ with 10 keV to etch until the Si element was detected.

Hydro-SCs preparation and measurement using other polymer semiconductors

Poly(3,3'-dialkoxybithiophene) (p(gT2)), poly(2-(3,3′-bis(2-(2-(2-methoxyethoxy)-ethoxy)ethoxy)-[2,2′-bithiophen]-5-yl)thieno[3,2-b] thiophene) (p(g2T-TT)), and poly(N,N′-bis(7-glycol)-naphthalene-1,4,5,8-bis(dicarboximide)-co-2,2′-bithiophene (P-100) were synthesized according to the reported methods(*44-46*). The protocols for preparing p(gT2)/PAAc and p(g2T-TT)/PAAc hydro-SCs were identical to the p(g2T-T)/PAAc, except some adjustments in composition. For p(gT2)/PAAc, p(gT2) of 40 mg/ml and AAc monomer of 150 mg/ml were used. For p(g2T-TT)/PAAc, p(g2T-TT) of 10 mg/ml and AAc monomer of 200 mg/ml were used. To prepare P-100/PAAc hydro-SC, the P-100 was first dissolved in the 1:1 mixture solvent of DMSO and $CHCl_3$ with a concentration of 15 mg/ml. The AAc monomer concentration was set as 250 mg/ml. After the spin-coating of precursor thin films, the substrate was mildly heated at 60 °C for 10 s to remove $CHCl_3$. The subsequent curing and solvent exchange processes were the same as the p(g2T-T)/PAAc hydro-SC. For all hydro-SCs, the crosslinker (tetraethylene glycol dimethylacrylate) and photoinitiator (2-hydroxy-2-methyl-propiophenon) were kept in a concentration of 0.5 wt% of AAc. For OECT tests, 2 mm channel width and 1 mm channel length were used for p(gT2)/PAAc and p(g2T-TT)/PAAc hydro-SCs, and 4 mm channel width and 0.2 mm channel length were used for P-100/PAAc hydro-SC.

Other characterizations

Fourier-transform infrared spectroscopy (FTIR) spectra were recorded using the Shimadzu IRTracer100 FTIR spectrometer. Proton Nuclear Magnetic Resonance (^{1}H NMR) spectra were acquired on 400 MHz Bruker Avance III HD nanobay spectrometer equipped with an iProbe SmartProbe, using Topspin 2.1. The conjugated polymers or hydrogel monomers were dissolved in $CDCl_3$ or D_2O, respectively. Size exclusion chromatography (SEC) was performed on a Tosoh EcoSEC size exclusion chromatography system equipped with a Wyatt 3 angle MiniDAWN light scattering detector and a single channel UV detector. DMF with 0.01M LiBr was used as the eluent and the column sets were Tosoh SuperAW3000 and Tosoh SuperAW4000. Molecular weight and dispersity were determined by Polystyrene calibration standards. The mass of the film and the functionalization extent were characterized using Gamry Instruments quartz crystal microbalance (QCM) equipped with a 5 MHz Au-coated quartz crystal. The shifts of equilibrium resonant frequency were analyzed using the Sauerbry equation to calculate the corresponding mass changes.

OECT test with the DMSO electrolyte

100 mM choline chloride DMSO solution was prepared as the electrolyte. The Ag/AgCl pellet was immersed in the DMSO electrolyte overnight to equilibrate. To test the OECT performance of the DMSO solution stage, the precursor solution was spin-coated onto the glass substrate pre-deposited with D/S electrodes to yield a highly viscous liquid layer. Then the DMSO electrolyte was immediately added on top of the viscous layer with caution to not disturb its integrity. The slow dissolution rate of p(g2T-T) in DMSO at room temperature allows the viscous liquid film to remain intact for more than 2 min (fig. S15). Following the addition of the DMSO electrolyte, the Ag/AgCl was immersed and the electrical test was carried out immediately to minimize the dissolution of p(g2T-T). The test of the DMSO gel stage was carried out in the same way.

Measurement of the swelling ratio of hydro-SC thin films

The swelling ratio of hydro-SC film is calculated by the ratio between the thickness under swelling and the thickness of the dry film. The dry film thickness is measured by the KLA Tencor P-7 Profilometer. The thickness under swelling is measured by the liquid-phase AFM to mitigate the dehydration of hydro-SC thin films in the air. The measurement is conducted using Bruker Multimode 8 AFM under the scanAsyst-liquid mode. Since the height of the swelled gel exceeds the limit of AFM height scanning, the thickness is measured by the method shown in fig. S22A. Briefly speaking, the AFM is used as the high-precision motor to detect the height differences at three different positions (P1, P2 and P3), for which one is on the hydro-SC, and two are on the Si substrate. The AFM probe is elevated by H_1 at P1 and then moved to P2 and P3 separately to approach the surface, the H_2 and H_3 are recorded by the Bruker Nanoscope 9.0 software. The swelled film thickness is calculated accordingly.

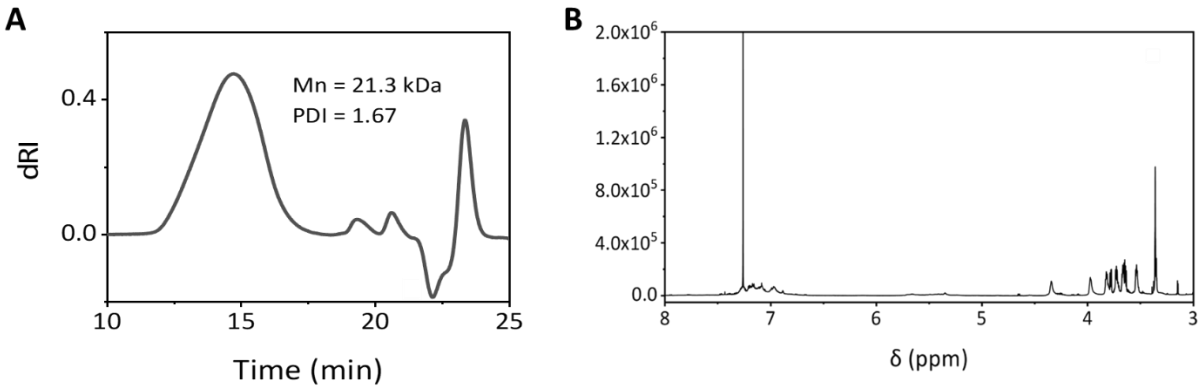


Fig. S1. SEC (**A**) and ^{1}H-NMR (**B**) of p(g2T-T).

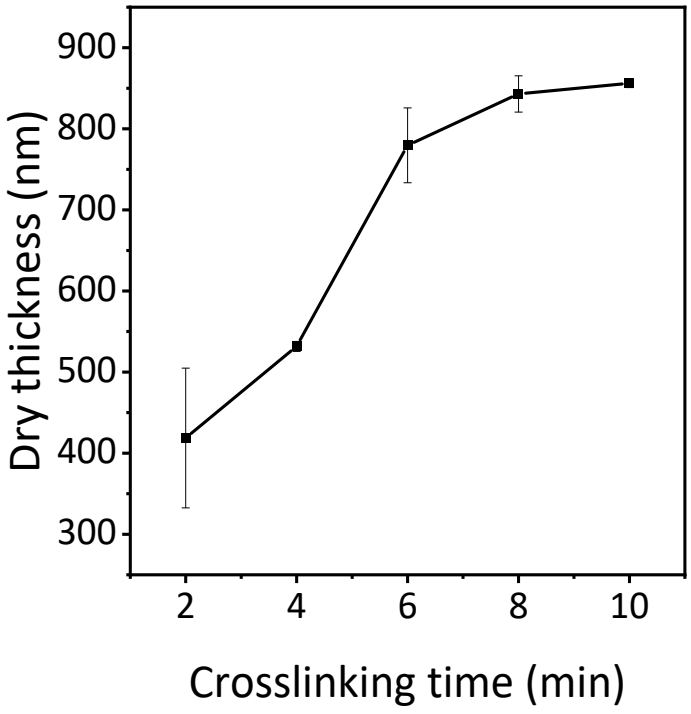


Fig. S2. Thickness measurement on hydro-SC crosslinked under different UV exposure times. The crosslinking of the hydrogel network is validated by the gradual thickness increase, which reaches a fully crosslinked stage after 8 min (n = 3 independent measurements).

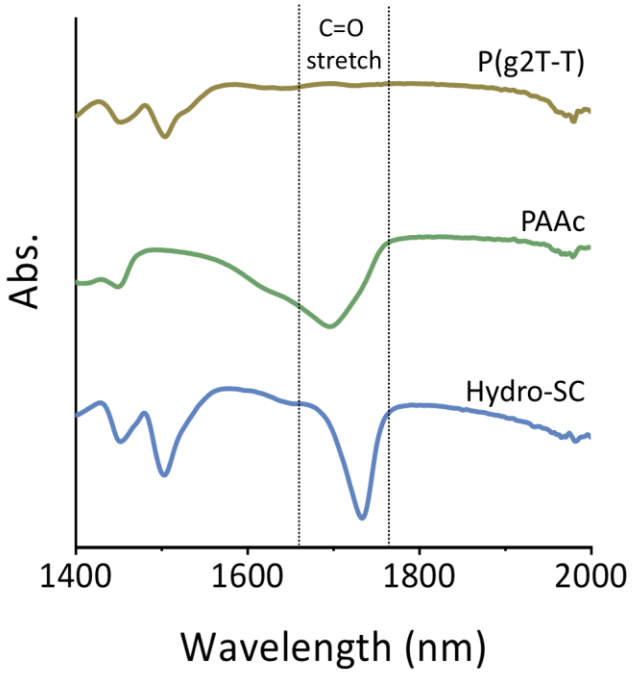


Fig. S3. Fourier-transform infrared spectroscopy (FTIR) spectra of p(g2T-T) and hydro-SC films. The successful incorporation of PAAc network is confirmed by the appearance of C=O stretch peak after UV crosslinking.

Measure the polymerization efficiency of AAc during hydro-SC preparation

The polymerization efficiency of AAc during hydro-SC preparation was measured by using D_2O to wash out unreacted AAc from a prepared hydro-SC, then quantifying the amount of AAc dissolved in D_2O by H-NMR (fig. S4). Specifically, 100 μL DMSO solution of p(g2T-T) and AAc precursors (with the p(g2T-T) to PAAc ratio of 10/100 and crosslinking density of 0.2%) was cast on the substrate and cured under UV light for 10 min. As-prepared DMSO gel was soaked in the 5 ml D_2O. The released AAc concentration did not further increase after 1 min, as evidenced by fig. S4. 500 uL of soaking solution was then extracted, with the addition of 0.1 mg sodium benzenesulfonate (SBS) used as the internal standard for ^{1}H-NMR test. ^{1}H peaks corresponding to the phenyl ring on the benzenesulfonate were used to quantify the concentration of acrylic acid (fig. S4), which gives the concentration of unreacted AAc in the soaking solution of 6.4×10^{-4} mol/L. This yields a polymerization efficiency of 98.8%. Here is the detailed calculation process:

$$\boldsymbol{m(\mathbf{AAc_{total}}) = 200\ \mathbf{mg/mL} \times 100\ \mathbf{\mu L} = 20\ \mathbf{mg}}$$

$$\boldsymbol{n(\mathbf{AAc_{unreacted}}) : n(\mathbf{SBS}) = 0.5 : 0.29 = 1.72}$$

$$\boldsymbol{n(\mathbf{SBS}) = \frac{0.0001\ \mathbf{g}}{180\ \mathbf{g\ mol^{-1}}} = 5.6 \times 10^{-7}\ \mathbf{mol}}$$

$$\boldsymbol{n(\mathbf{AAc_{unreacted}}) = 3.2 \times 10^{-7}\ \mathbf{mol}}$$

$$\boldsymbol{c(\mathbf{AAc_{unreacted}}) = 3.2 \times \frac{10^{-7}\ \mathbf{mol}}{0.0005\ \mathbf{L}} = 6.4 \times 10^{-4}\ mol\ L^{-1}}$$

$$\boldsymbol{m(AAc_{unreacted}) = 6.4 \times 10^{-4}\mathbf{mol/L} \times 5\ mL \times 72\mathbf{g/mol} = 0.23\ \mathbf{mg}}$$

$$\textbf{Polymerization effciency} = \frac{m(\mathbf{AAc_{total}}) - m(\mathbf{AAc_{unreacted}})}{m(\mathbf{AAc_{total}})} = 98.8\ \%$$

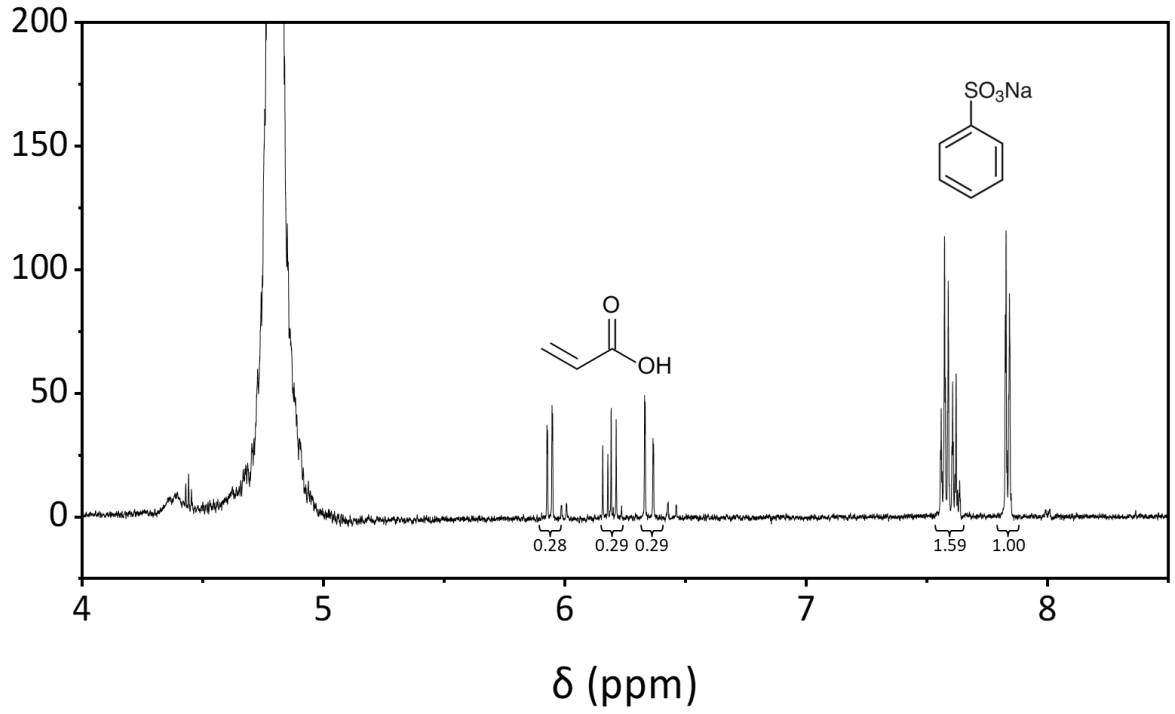


Fig. S4. ^{1}H-NMR for quantifying the unreacted AAc monomer during DMSO gel preparation. Sodium benzenesulfonate was used as the internal calibration.

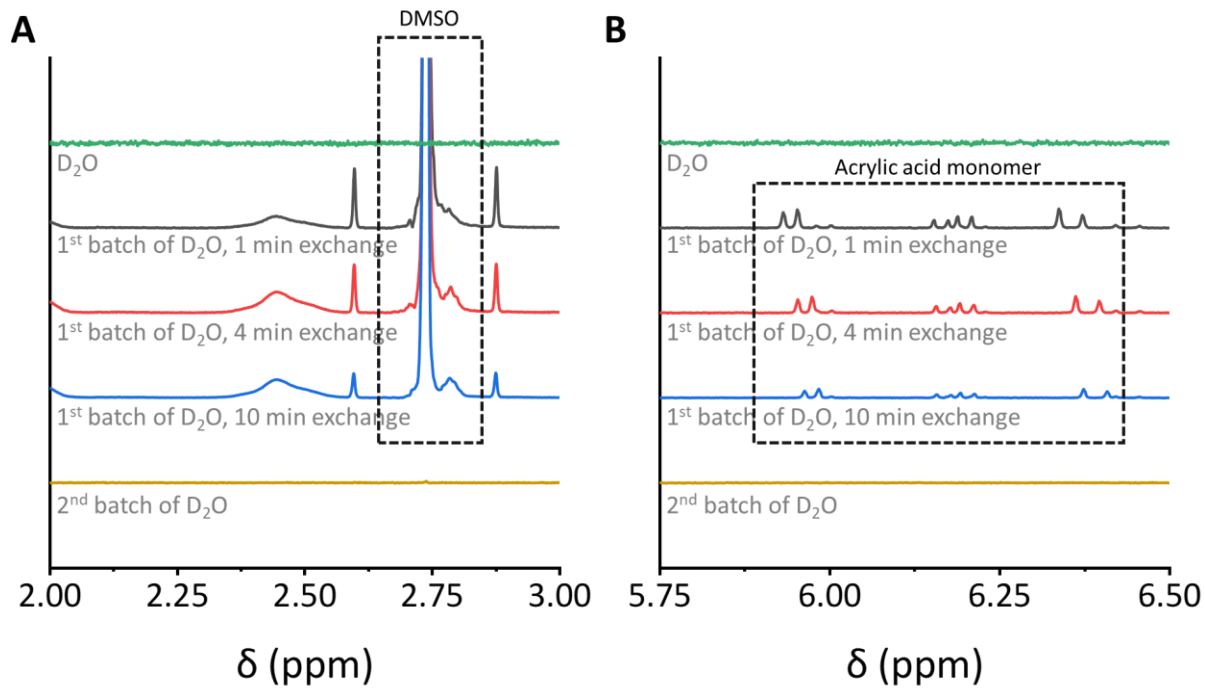


Fig. S5. Solvent exchange efficiency monitored by ^{1}H NMR. The DMSO gel was prepared with DMSO and then exchanged in D_2O. The D_2O was sampled in different time intervals and measured by ^{1}H NMR. The signal from the (**A**) DMSO residue and (**B**) unreacted AAc reaches equilibrium within 1 min in the 1st batch of D_2O, and diminishes after the 1st batch solvent exchange, validating the quick and thorough solvent exchange process. The minor shifts of AAc peaks are due to the change in solvent pH. The slight decrease in peak intensity with increasing exchange time is likely due to the non-uniform distribution of AAc in D_2O.

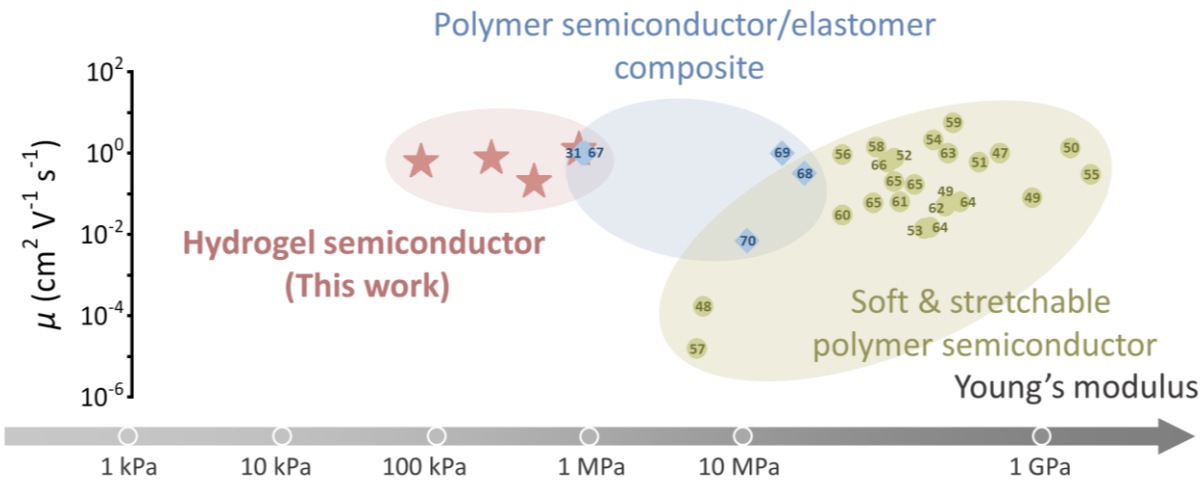


Fig. S6. Comparisons of Young's modulus and charge-carrier mobility from the hydro-SCs (dark red) with other reported soft and stretchable semiconductors.

Table S1. Summary of the charge carrier mobility and modulus of the representative soft and stretchable semiconductors.

Semiconductor type	Material	Mobility ($cm^2 V^{-1} s^{-1}$)	Young's modulus	Mechanical test method	Reference
Soft & stretchable polymer semiconductor	P(DPPTVT)	~ 1	~ 550 MPa	Buckling	(*47*)
	P3HT-b-PMA-b-P3HT	1.7×10^{-4}	6 MPa	Tensile test	(*48*)
	P3HT	0.06	250 MPa	Buckling	(*49*)
	PBTTT	0.08	900 MPa	Buckling	(*49*)
	PDPPTVT-PIID2T	1.33	1.61 GPa	Nanoindentation	(*50*)
	P(DPPTVT)-PDCA	~ 0.6	~ 400 MPa	Buckling	(*51*)
	P(DPPTT) with CPP	0.75	110 MPa	Buckling	(*52*)
	PDPPBT with brakers	0.014	174 MPa	Buckling	(*53*)
	PDPPTVT with brakers	2.1	200 MPa	Buckling	(*54*)
	PBDT-3T	0.3	2.2 GPa	Buckling	(*55*)
	P(g2T-T)	0.92	50 MPa	Buckling	(*56*)
	20% T-10-T/20% dtdDPP	1.57×10^{-5}	5.45 MPa	FOW	(*57*)
	P(DPP-BT - piro[cycloalkane-1,9′-fluorene])	1.41	83.7 MPa	Buckling	(*58*)
	PII2T-C8	5.56	270 MPa	Buckling	(*59*)
	$PDPP_{urethane}$	0.03	50 MPa	Tensile test	(*60*)
	PII2T-PBA20	0.063	120 MPa	Buckling	(*61*)
	P3HT-PE	~ 0.05	240 MPa	Tensile	(*62*)
	PII2T-C10	~ 1	250 MPa	Buckling	(*63*)
	P3HT/PBA12K	0.015	190 MPa	Buckling	(*64*)
	P3HT/PBA6K	0.07	~ 300 MPa	Buckling	(*64*)
	PIDTTPD	0.2	110 MPa	Buckling	(*65*)
	PIDTBTD	0.17	150 MPa	Buckling	(*65*)
	PIDYTBPD	0.06	80 MPa	Buckling	(*65*)
	Crosslinked PDPPTT	~ 0.6	~ 100 MPa	Buckling	(*66*)
Polymer semiconductor /elastomer composite	PDPPTVT/Butyl rubber	1	1 MPa	FOW	(*67*)
	P2TDPP2TFT4/SEBS	0.32	28 MPa	FOW	(*68*)
	P(DPPTT)/SEBS	~ 1	~ 20 MPa	Buckling	(*69*)
	P(g2T-T)/Adhesive gel	1.01	0.95 MPa	FOW	(*31*)
	P3HT/SEBS	0.007	11.7 MPa	Tensile test	(*70*)
Hydrogel semiconductor	10/100 0.5% crosslinking	0.61	81.8 kPa	FOW	This work
	10/100 2.0% crosslinking	0.752	238.6 kPa	FOW	
	7.5/100 0.2% crosslinking	0.196	456.8 kPa	FOW	
	10/100 0.2% crosslinking	1.22	905.7 kPa	FOW	

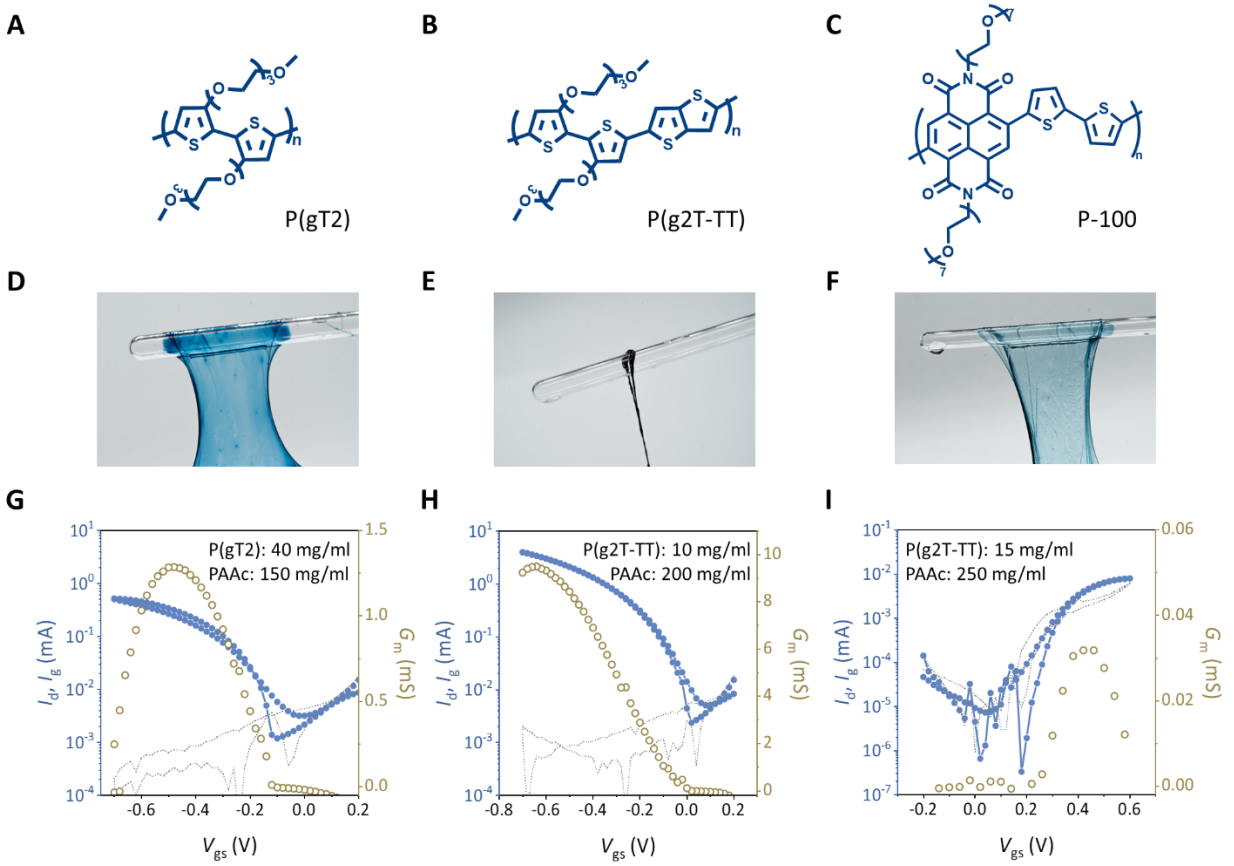


Fig. S7. Hydro-SCs prepared using other polymer semiconductors via solvent-exchange approach. (**A**-**C**) Molecular structure of p(gT2), p(g2T-TT), and P-100. (**D**-**F**) Photos of prepared hydro-SCs using p(gT2), p(g2T-TT), and P-100. (**C**) OECT tests of hydro-SCs made from p(gT2), p(g2T-TT), and P-100. The blue and grey curves refer to I_d and I_g, respectively. V_{ds} = -0.8 V. All hydro-SCs give ideal transistor performance.

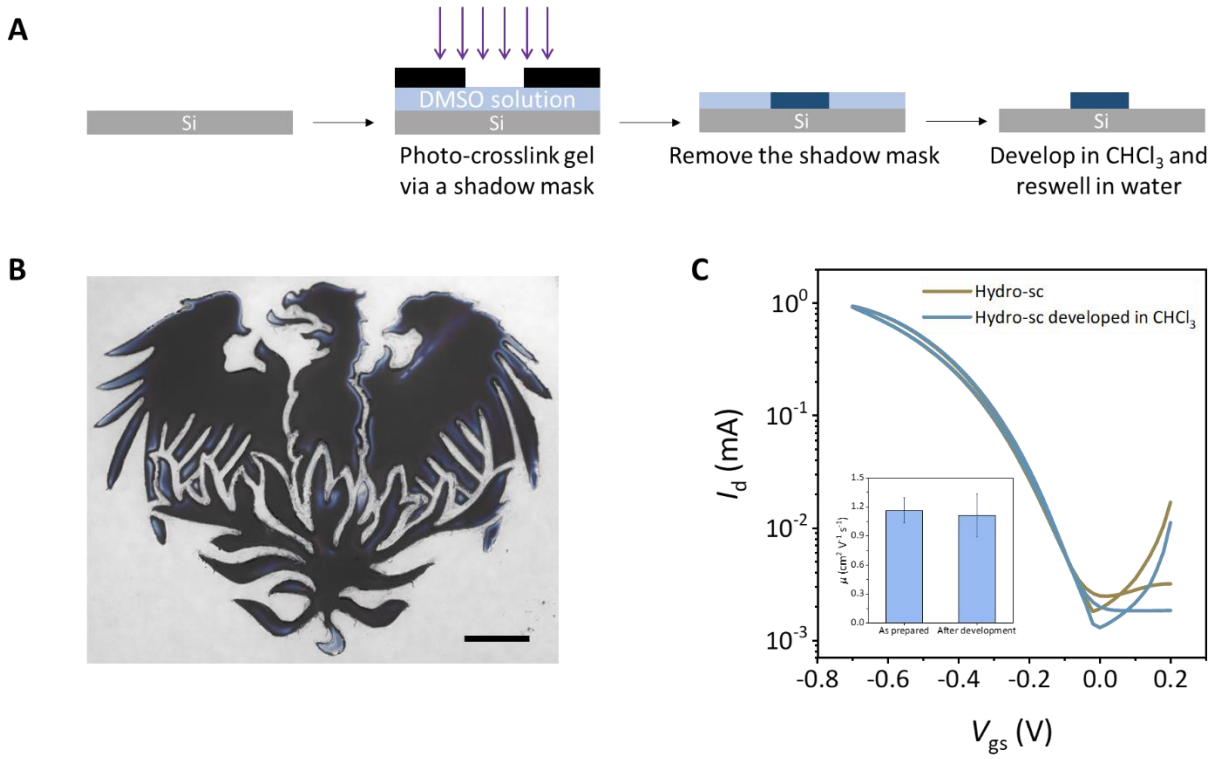


Fig. S8. Photopatterning of hydro-SC. (**A**) Workflow of photopatterning process. (**B**) Picture of a patterned hydro-SC film. Scale bar, 1 cm. (**C**) OECT tests and μ of hydro-SC before and after $CHCl_3$ development, validating hydro-SC's capability for maintaining the electrical performance after development (n = 3 independent measurements). V_{ds} = -0.8 V. Hydro-SC was made with the 10/100 p(g2T-T)-to-PAAc ratio and 2% crosslinking density.

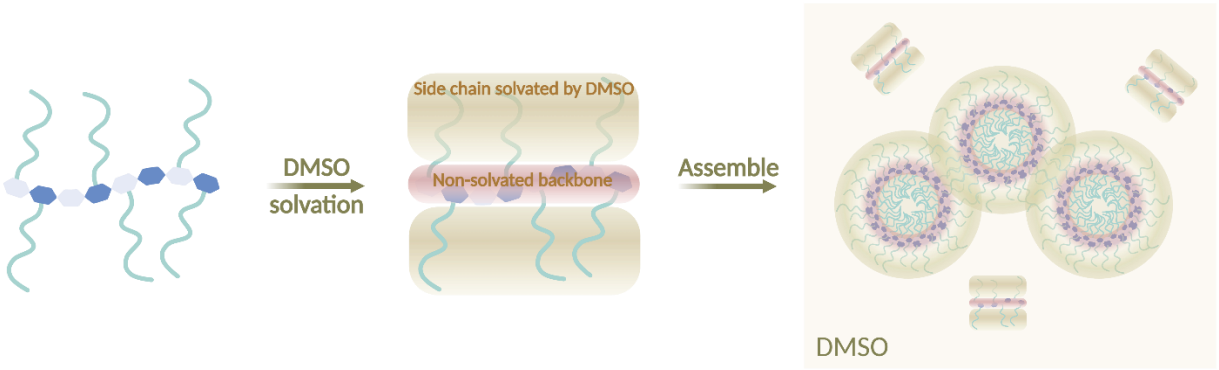


Fig. S9. P(g2T-T) nanoaggregates induced by the selective solvation of DMSO.

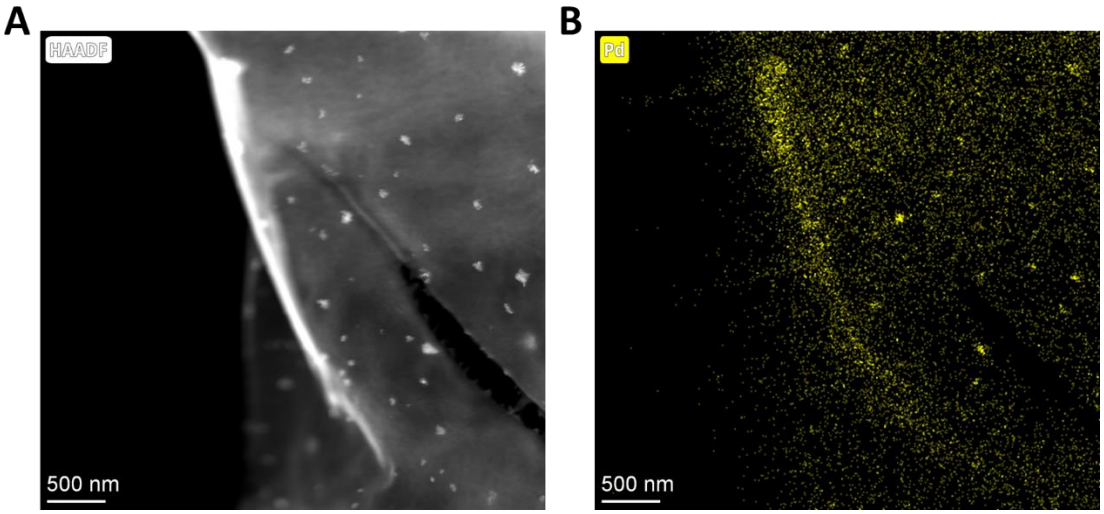


Fig. S10. STEM image (A) and EDS mapping (B) of a DMSO gel. The presence of p(g2T-T) particles was validated by the Pd element. The size of the nanoparticles agrees with the SEM image result.

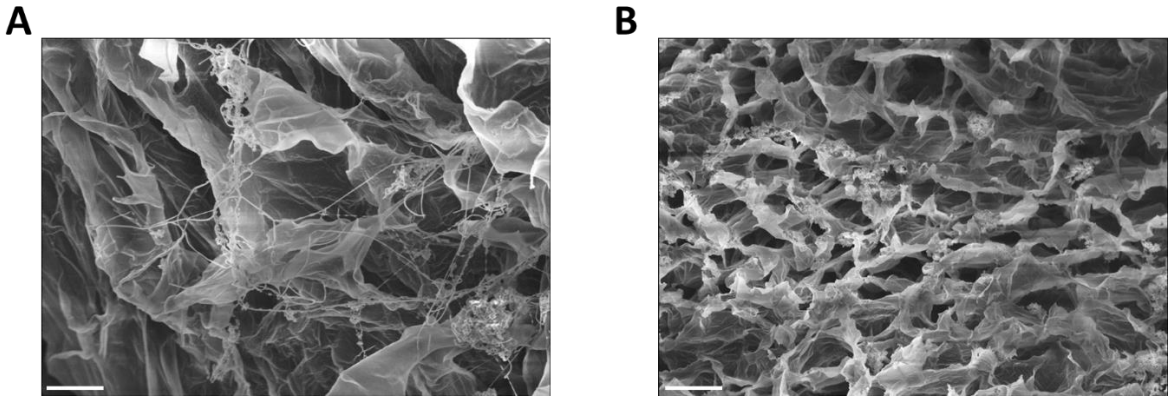


Fig. S11. SEM images of freeze-dried hydro-SC. (**A**) scale bar, 6 μm. (**B**) scale bar, 40 μm.

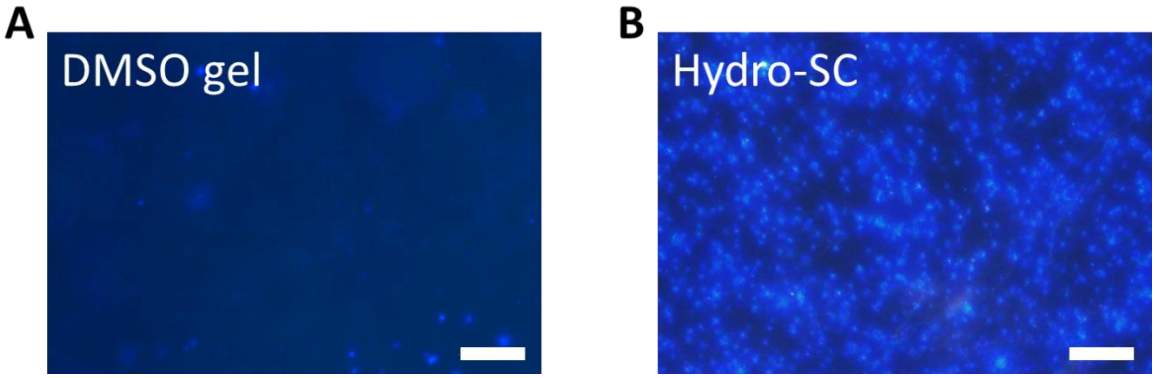


Fig. S12. Dark-field microscope images of DMSO gel and hydro-SC films. Both films were as prepared and imaged at the swelling state. The solvent exchange process significantly increases the size and density of p(g2T-T) clusters within the gel matrix. Scale bars, 20 μm.

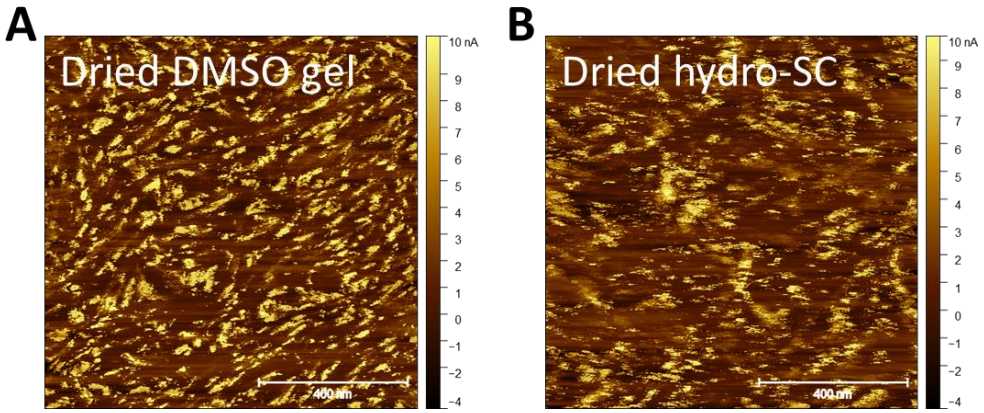


Fig. S13. C-AFM images of dried DMSO and hydro-SC films. The solvent exchange process induces a more non-uniform distribution of p(g2T-T) within the gel matrix.

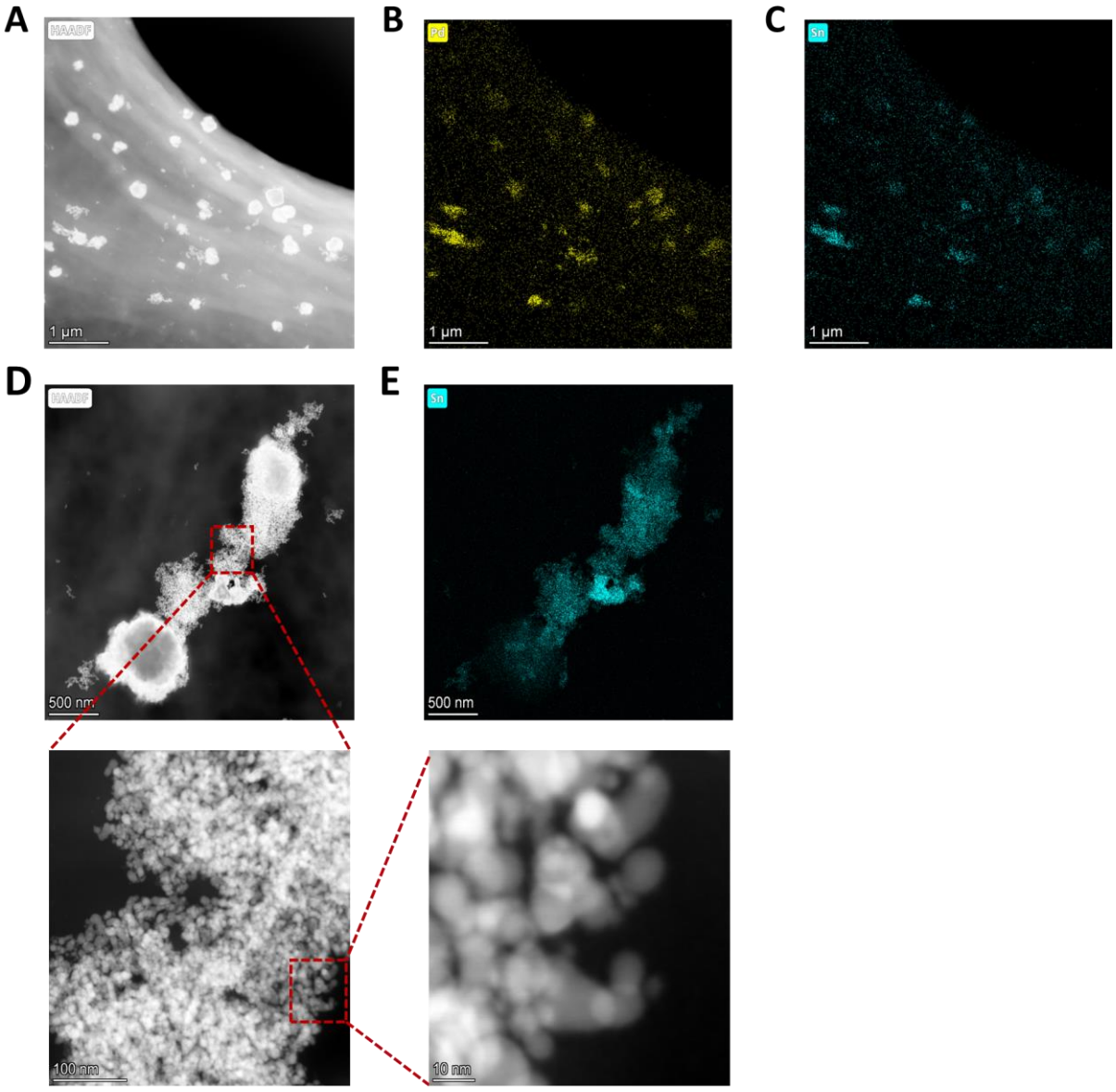


Fig. S14. STEM image and EDS mapping of a very thin hydro-SC prepared on a TEM copper grid. (A-C) Images of p(g2T-T) aggregates, which show much larger sizes compared to the ones in DMSO gel, validating the effect of solvent-exchange-induced assembly. (**D-E**) Images showing the formation of long-range p(g2T-T) percolation, which consists of both the large particles with diameters of hundreds of nanometers, and small nano aggregates that are well dispersed in the DMSO.

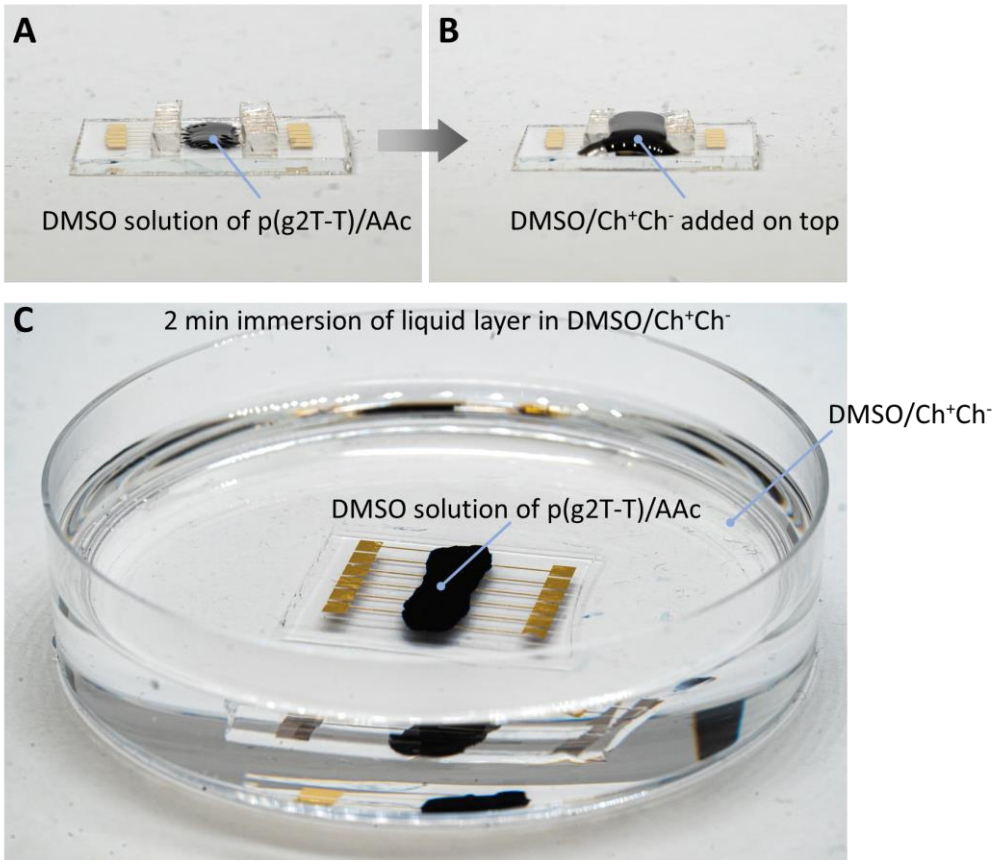


Fig. S15. OECT test of p(g2T-T)/AAc solution in DMSO electrolyte. (**A**) Photo of the OECT after dropping p(g2T-T)/PAAc-DMSO solution to the channel area. (**B**) Photo after further dropping DMSO electrolyte on top. (**C**) Photo of OECT immersing in DMSO electrolyte after 2 min. The p(g2T-T)/AAc layer visually remains intact, which is sufficient for OECT testing.

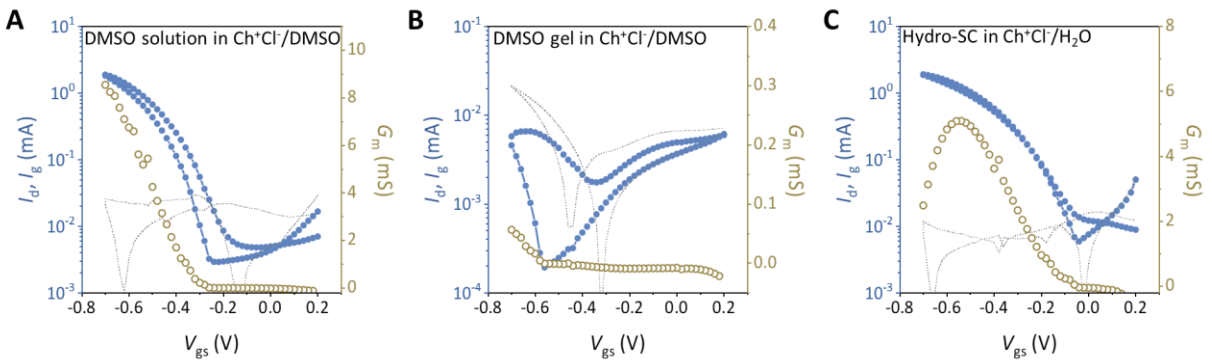


Fig. S16. OECT performance of hydro-SC films at different preparation stages. (**A**) Stage 1: DMSO solution in DMSO electrolyte. (**B**) Stage 2: DMSO gel in DMSO electrolyte. (**C**) Stage 3: hydro-SC in aqueous electrolyte. The blue and grey curves refer to I_d and I_g, respectively. V_{ds} = -0.8 V.

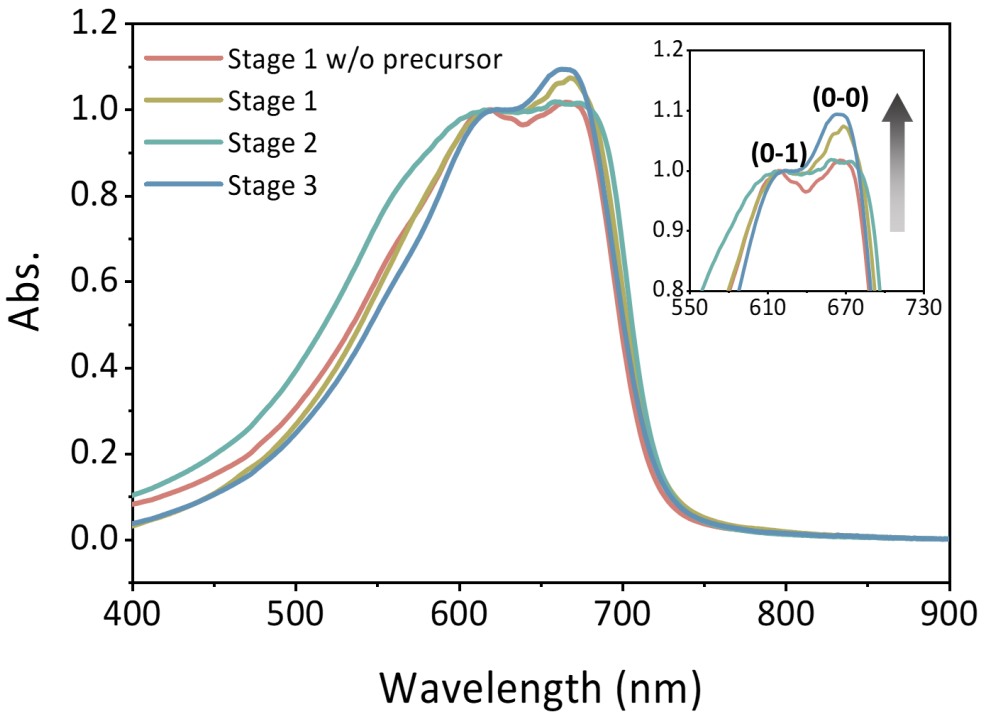


Fig. S17. UV-vis spectra of hydro-SCs at different preparation stages, including the original p(g2T-T) solution without adding hydrogel precursors. The addition of AAc monomer induced increased short-range aggregation of p(g2T-T) in the solution, while the photo-crosslinking of the AAc monomers into the 3D network in DMSO led to some decrease in the short-range ordering, approaching a similar level of the initial neat p(g2T-T) DMSO solution. The final solvent exchange step gave the highest aggregation level, which came from the rapid assembly process of all the p(g2T-T) polymer chains into a percolated network. Hydro-SC film (stage 3) was electrochemically de-doped to -0.2 V vs. Ag/AgCl during the measurement.

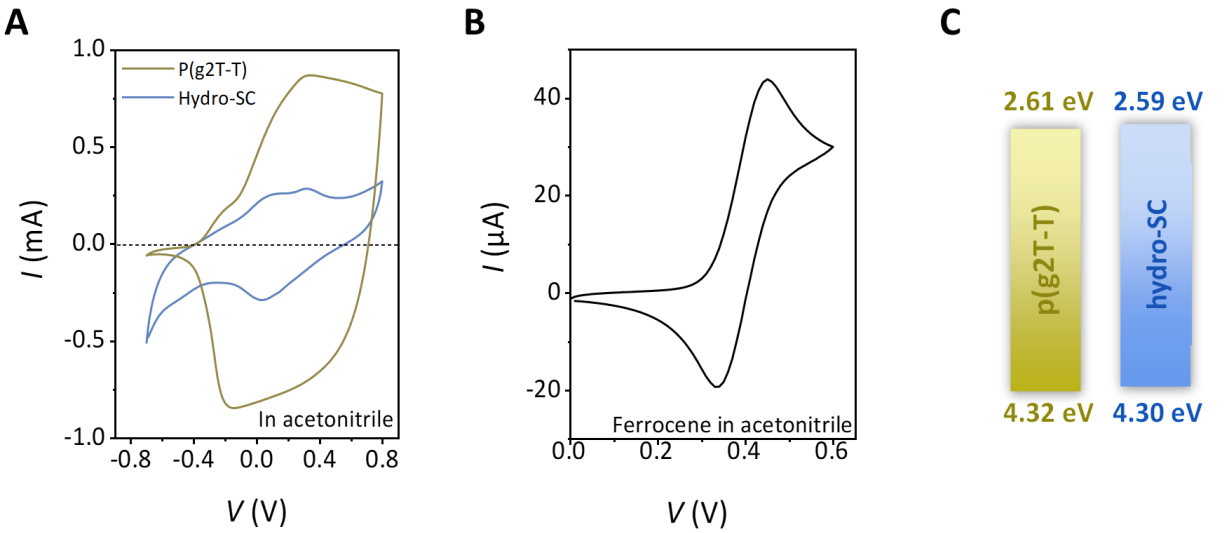


Fig. S18. Electronic structures of the hydro-SC. (**A**) Cyclic voltammetry of p(g2T-T) and hydro-SC films measured in an acetonitrile electrolyte. (**B**) Cyclic voltammetry of ferrocene in the same acetonitrile electrolyte for calibration. (**C**) HOMO and LUMO level of p(g2T-T) and hydro-SC. The HOMO level was calculated from the oxidation onset from cyclic voltammetry. The bandgap was calculated from the absorption onset on the long-wavelength side. The LUMO level was calculated according to the bandgap measured by UV-vis spectroscopy.

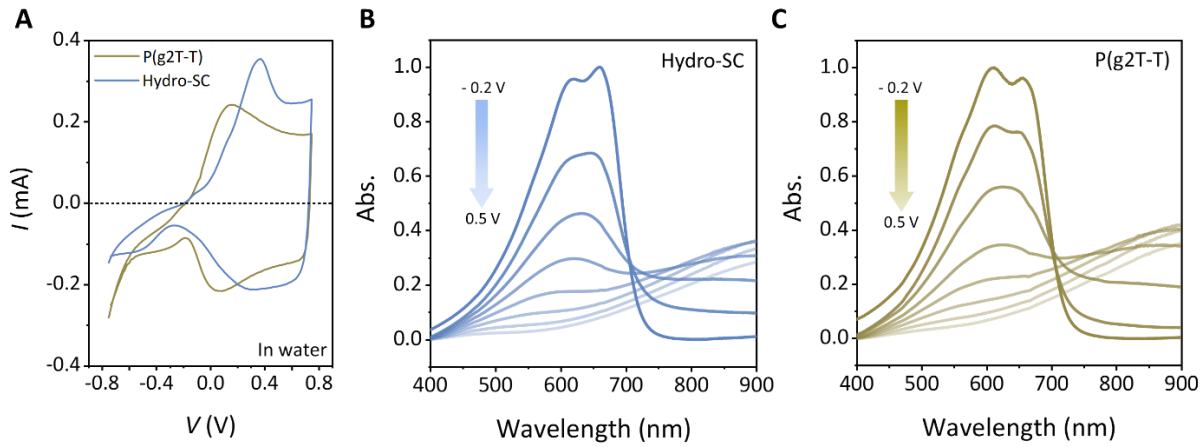


Fig. S19. Redox-activity of hydro-SC in aqueous electrolyte. (**A**) Cyclic voltammetry of p(g2T-T) and hydro-SC films in an aqueous electrolyte. (**B**) Operando UV-vis measurements on a hydro-SC film. (**C**) Operando UV-vis measurement on a p(g2T-T) film.

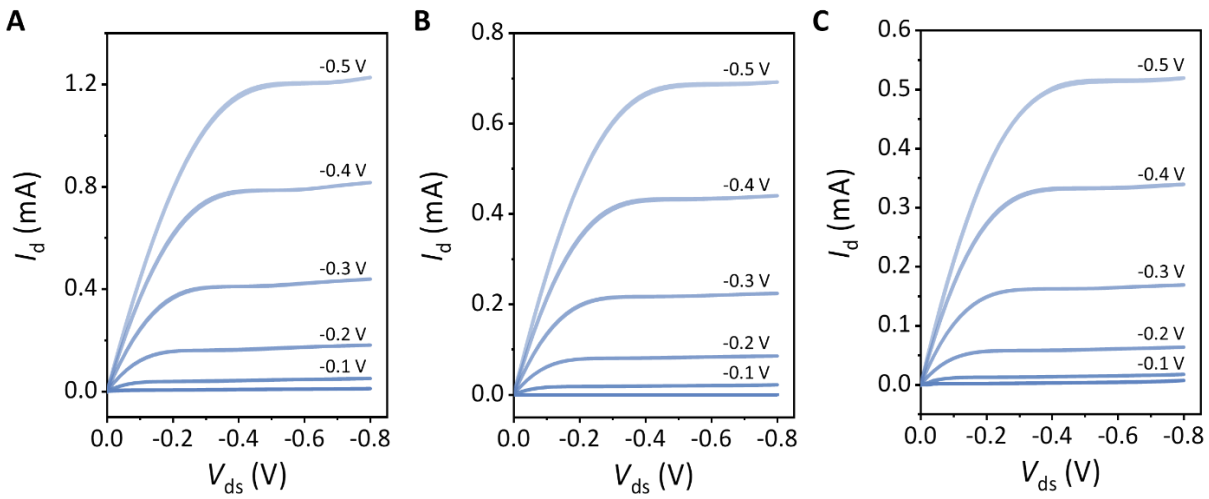


Fig. S20. Representative output curves of OECT with different hydro-SC films. (**A**) 0.2% crosslinked hydro-SC. (**B**) 0.5% crosslinked hydro-SC. (**C**) 2.0% crosslinked hydro-SC.

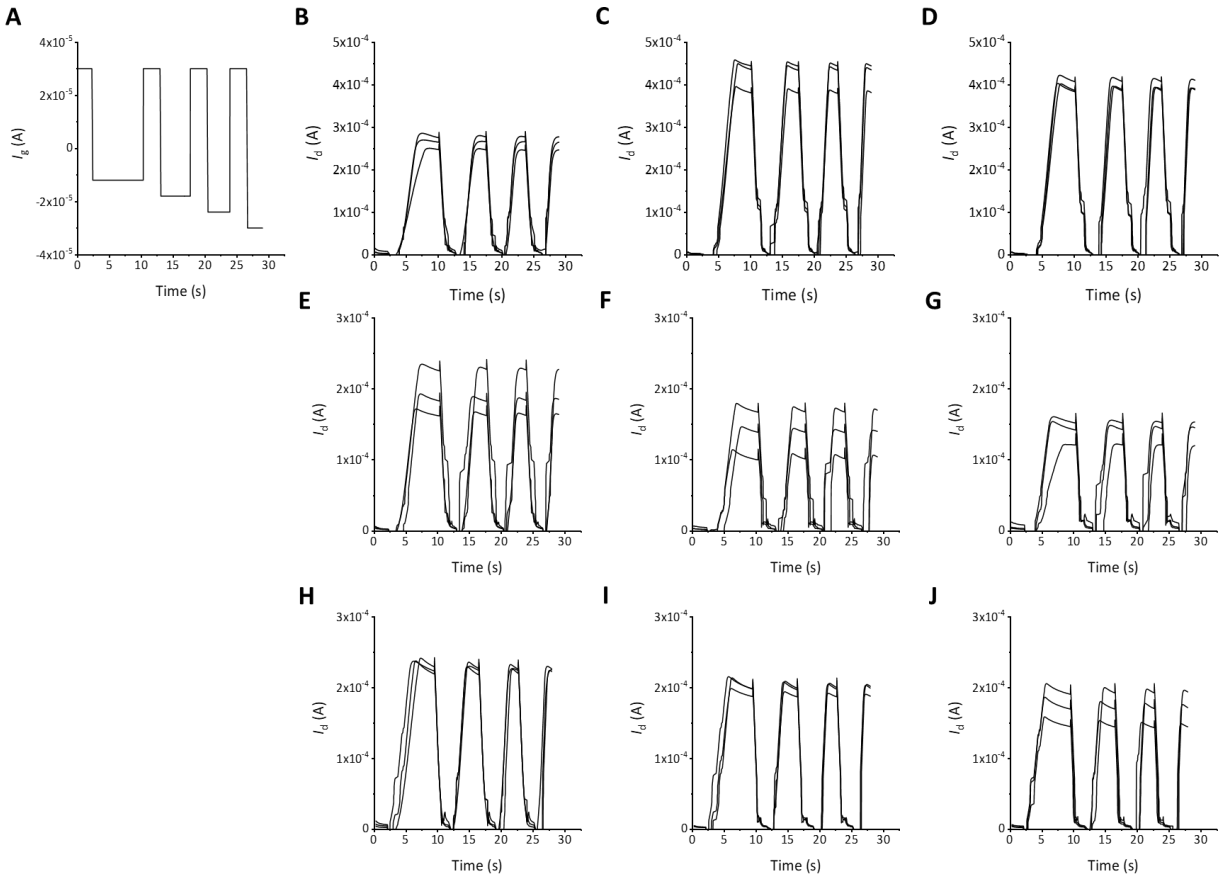


Fig. S21. Mobility measurements of hydro-SC films with different crosslinking ratios. (**A**) Programmed I_g as the function of time. (**B** to **D**), Measurements of 0.2% crosslinked hydro-SC film. (**E** to **G**) Measurements of 0.5% crosslinked hydro-SC film. (**H** to **J**) Measurements of 2.0% crosslinked hydro-SC film. The mobility of each material was calculated by averaging 9 channels from 3 different chips. V_{ds} = -0.1 V, the compliance of the gate terminal was set at 0.5 V.

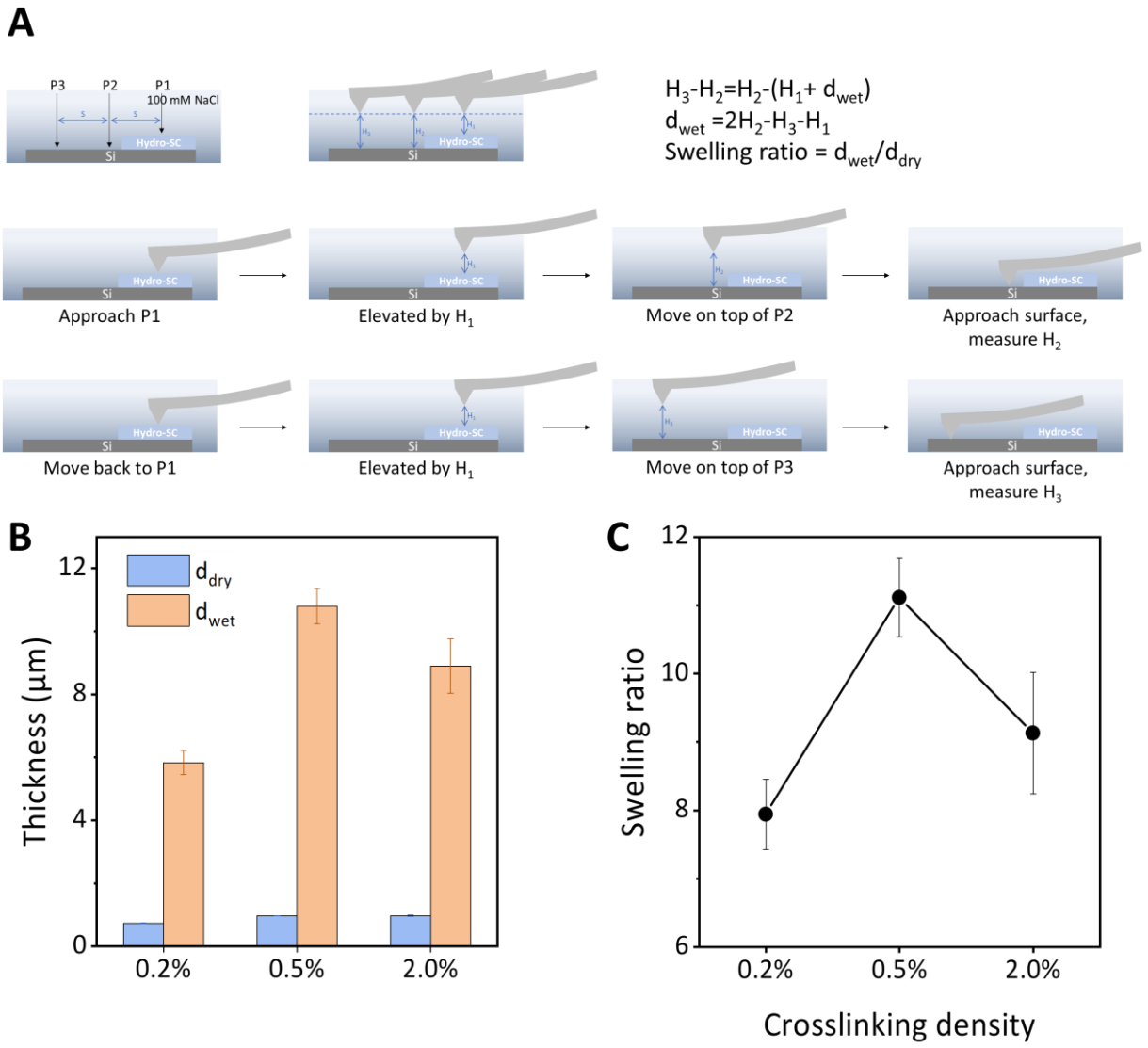


Fig. S22. Measurements of the swelling ratio of hydro-SC. (**A**) Schematics showing the flow for measuring the thickness of swelled hydro-SC films. (**B**) Dry and wet thickness of hydro-SC films with different crosslinking densities (p(g2T-T)/PAAc = 10/100, n = 3 independent measurements). (**C**) Swelling ratio of hydro-SC films in 100 mM NaCl solution with different crosslinking densities (n = 3 independent measurements).

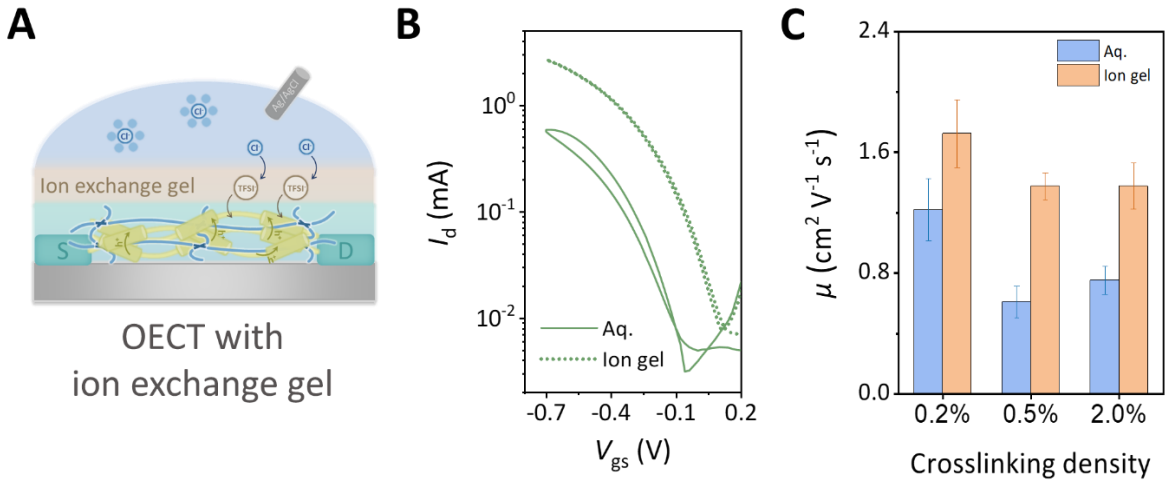


Fig. S23. Ion exchange gel-based OECT measurements of hydro-SC films. (**A**) A schematic showing the OECT operation with the ion exchange gel. (**B**) Transfer curves of 0.5% crosslinked hydro-SC-based OECT with aqueous and ion gel as the dielectric, V_{ds} = -0.8 V. (**C**) Mobility of hydro-SC films with different crosslinking densities at swelled and non-swelled states (n = 9 independent measurements).

We implemented a parallel OECT test adopting an ion exchange gel (*27*) to evaluate the electrical performance of hydro-SC in the dry state (fig. S23A), which yielded higher electrical performance (fig. S23B) due to the mitigated swelling effect. Different from the trend of wet-state μ obtained with the aqueous electrolyte, the dry-state μ shows a consistent decrease with denser crosslinking (fig. S23C), indicating the lowest wet-state μ observed at 0.5% mainly comes from the swelling effect.

Table S2. Concentrations of p(g2T-T) and AAc used to prepare hydro-SCs with different p(g2T-T)-to-PAAc weight ratios.

Ratio	P(g2T-T) concentration	AAc concentration
5/100	10 mg/ml	200 mg/ml
7.5/100	15 mg/ml	200 mg/ml
10/100	20 mg/ml	200 mg/ml
13.3/100	20 mg/ml	150 mg/ml

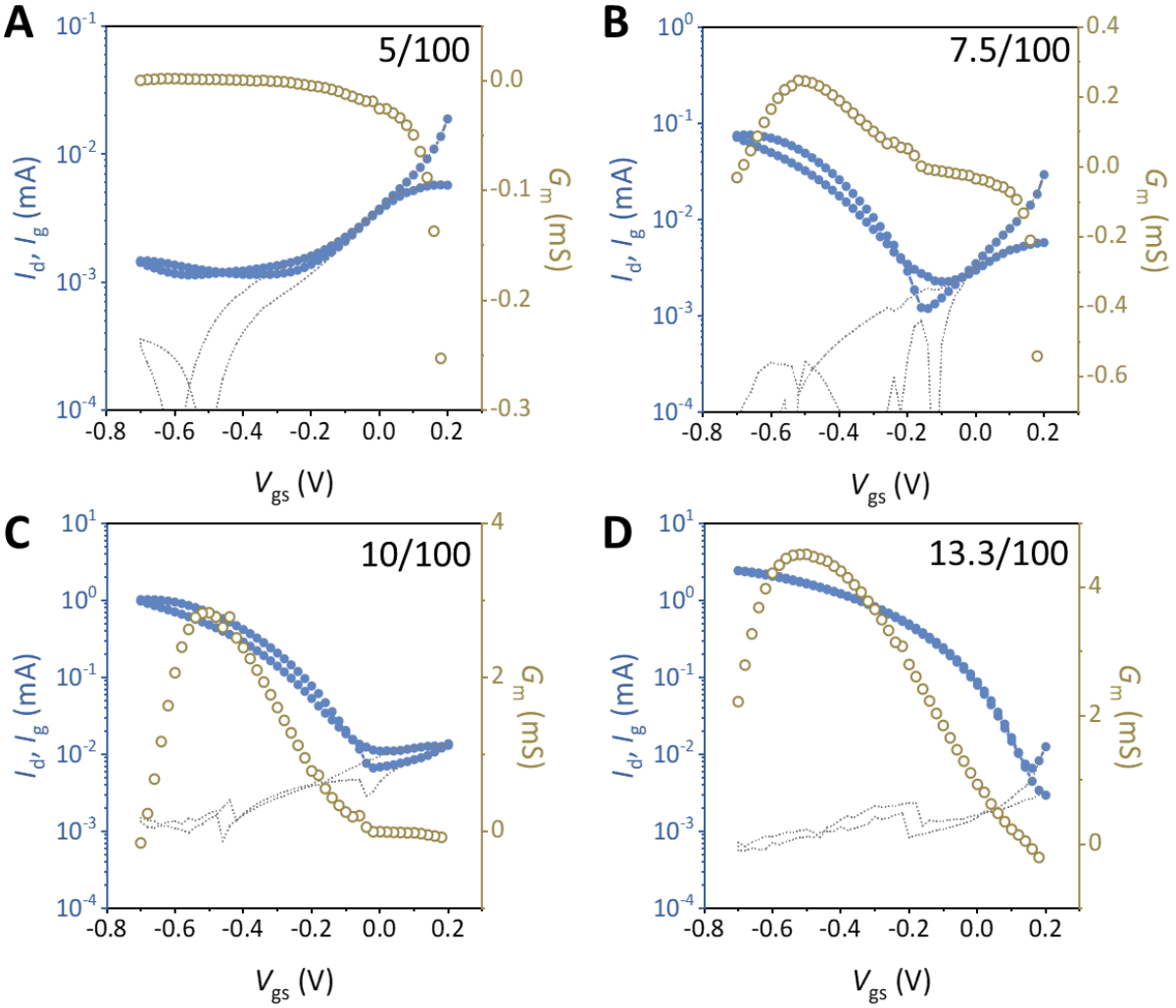


Fig. S24. Representative transfer curves of OECTs based on hydro-SC films with different p(g2T-T)-to-PAAc ratios. (**A**) 5/100 hydro-SC. (**B**) 7.5/100 hydro-SC. (**C**) 10/100 hydro-SC. (**D**) 13.3/100 hydro-SC. V_{ds} = -0.8 V. The blue and grey curves refer to I_d and I_g, respectively. V_{ds} = -0.8 V.

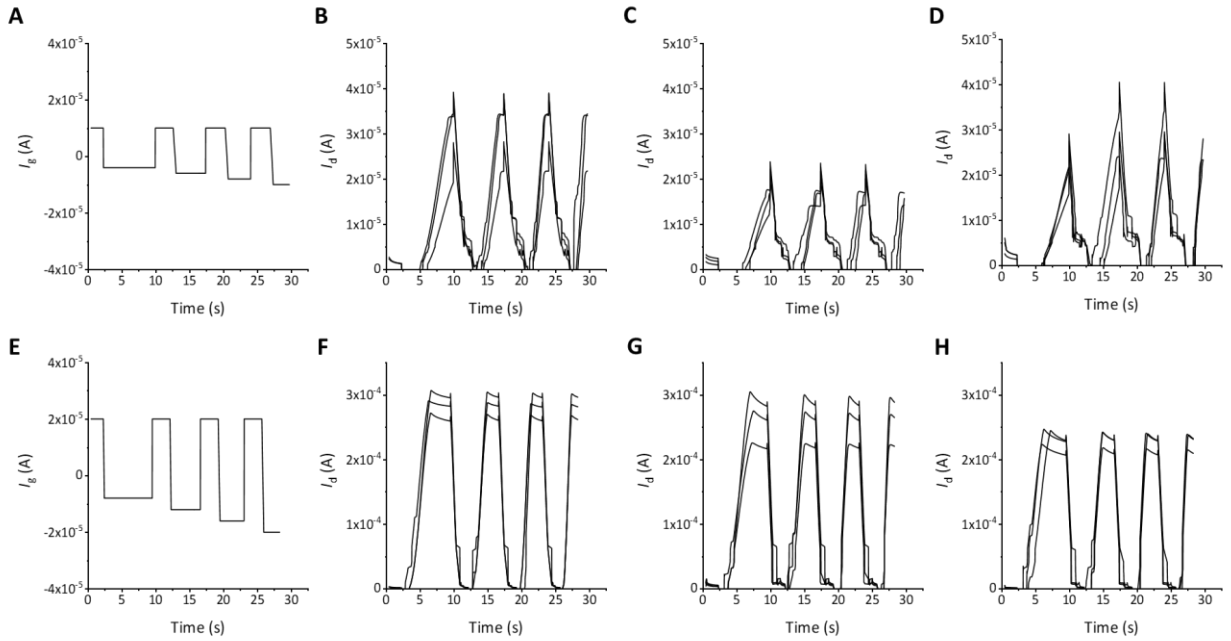


Fig. S25. Mobility measurements of hydro-SC with different p(g2T-T)-to-PAAc ratios. (**A**) Programmed I_g as the function of time for measuring 7.5/100 hydro-SC film. (**B** to **D**) Measurements of 7.5/100 hydro-SC film. (**E**) Programmed I_g as the function of time for measuring 13.3/100 hydro-SC film. (**F** to **H**) Measurements of 13.3/100 hydro-SC film. V_{ds} = -0.1 V, the compliance of gate terminal was set at 0.5 V.

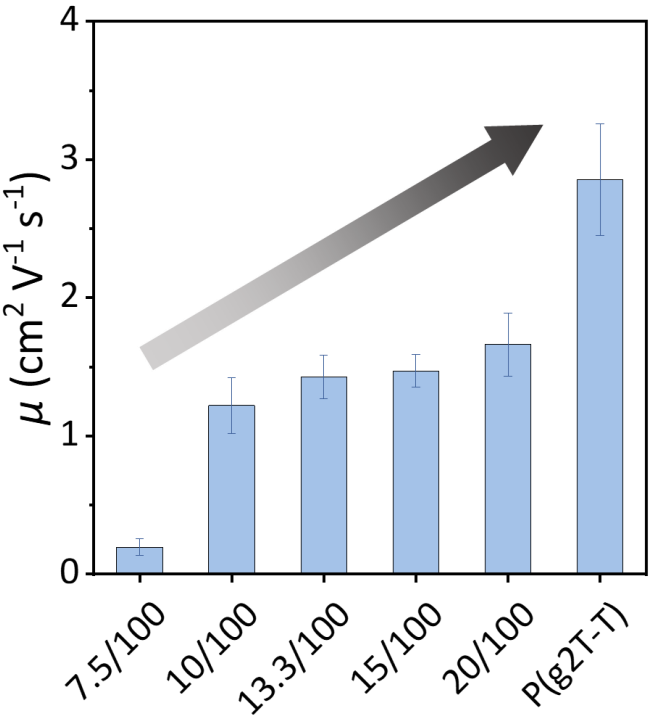


Fig. S26. Mobility of hydro-SCs with increasing p(g2T-T) loading ratio with a fixed crosslinking ratio of 0.2%. n = 6 independent measurements.

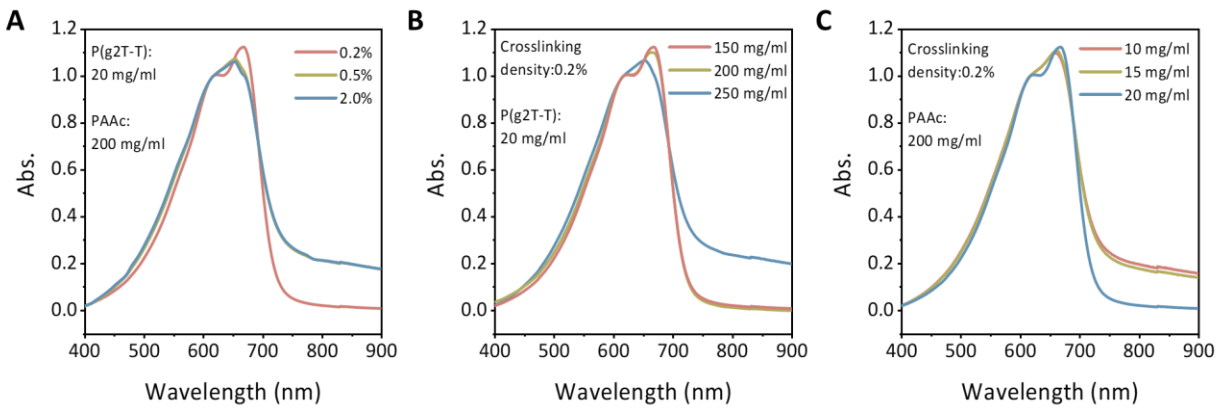


Fig. S27. UV-vis spectra of hydro-SCs prepared with different (A) crosslinking ratio, (B) PAAc concentration, and (C) p(g2T-T) concentration. Lower crosslinking density, higher p(g2T-T) concentration, and lower PAAc concentration all lead to higher short-range orderings, as evidenced by the higher (0-0)/(0-1) ratio. All hydro-SC films were electrochemically de-doped to -0.2 V vs. Ag/AgCl during the measurement.

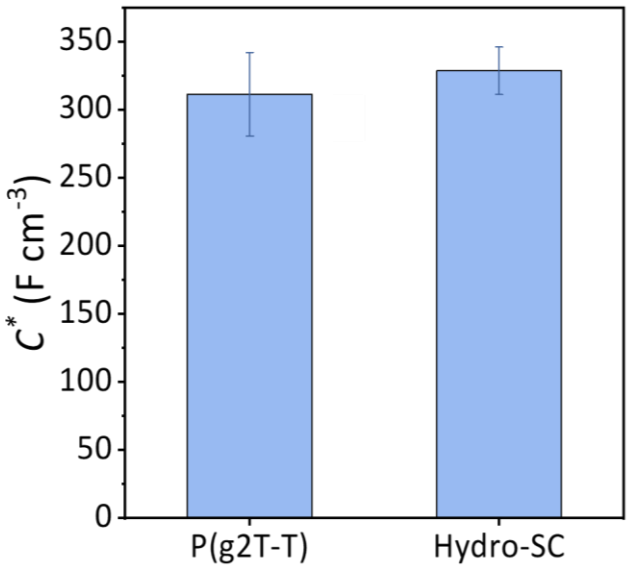


Fig. S28. C^* of p(g2T-T) and hydro-SC films. The C^* values were obtained by the normalization to the volume of the p(g2T-T) phase. n= 3 independent measurements. Hydro-SCs were made with the 10/100 p(g2T-T)-to-PAAc ratio and 0.5 % crosslinking density.

Table S3. Summary of mobility and Young's modulus of hydro-SCs with different compositions.

Ratio	P(g2T-T) concentration	AAc concentration	Crosslinking density	Mobility ($cm^2 V^{-1} s^{-1}$)	Young's modulus (kPa)
5/100	10 mg/ml	200 mg/ml	0.2%	NA	120.3
7.5/100	15 mg/ml	200 mg/ml	0.2%	0.196	456.8
10/100	20 mg/ml	200 mg/ml	0.2%	1.22	1120
13.3/100	20 mg/ml	150 mg/ml	0.2%	1.428	2710.2
10/100	20 mg/ml	200 mg/ml	0.5%	0.62	81.8
10/100	20 mg/ml	200 mg/ml	2.0%	0.75	238.6

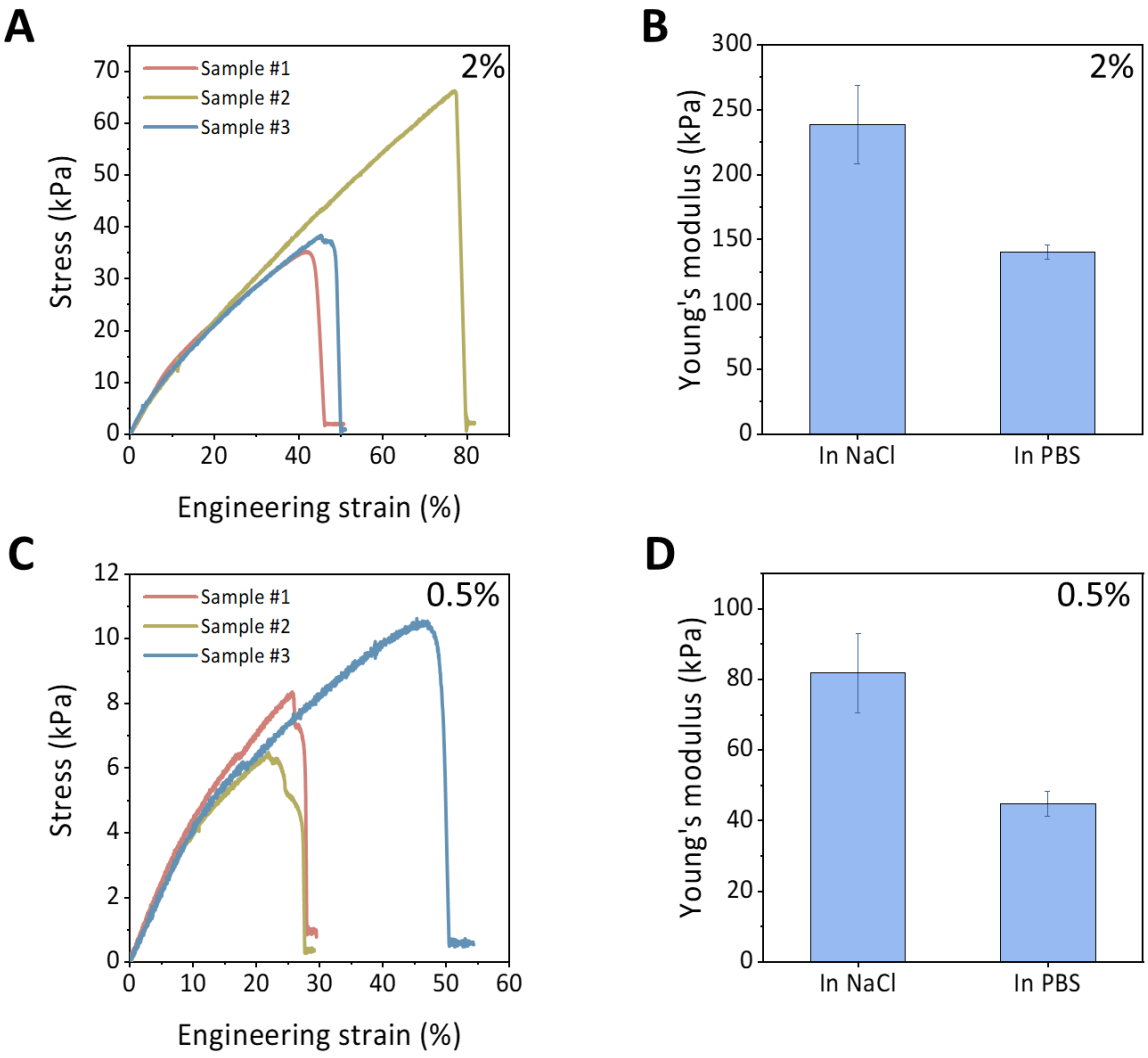


Fig. S29. FOW tests of hydro-SC thin films with different crosslinking ratios in PBS (A and C) and the comparisons of Young's modulus measured in different electrolytes (B and D). n = 6 independent measurements. The lower modulus in the PBS buffer was attributed to its capability to maintain a neutral pH environment that helps the protonation of PAAc, which increases the swelling ratio of hydro-SCs.

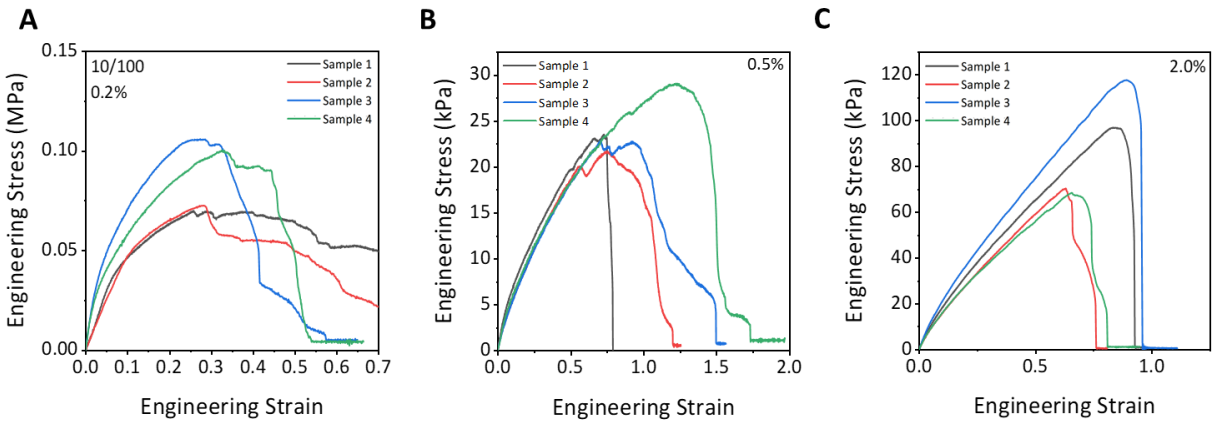


Fig. S30. FOW tests of hydro-SC thin films with different crosslinking ratios in 100 mM NaCl solution. (**A**) 0.2% crosslinked hydro-SC film. (**B**) 0.5 % crosslinked hydro-SC film. (**C**) 2.0% crosslinked hydro-SC film.

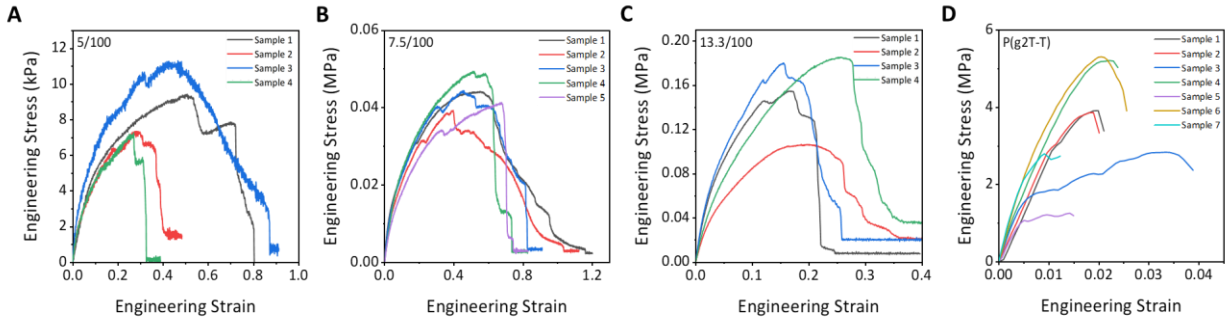


Fig. S31. FOW tests of p(g2T-T) and hydro-SC thin films with different p(g2T-T)-to-PAAc ratios in 100 mM NaCl solution. (**A**) 5/100 hydro-SC film. (**B**) 7.5/100 hydro-SC film. (**C**) 13.3/100 crosslinked hydro-SC film. (**D**) neat p(g2T-T) film.

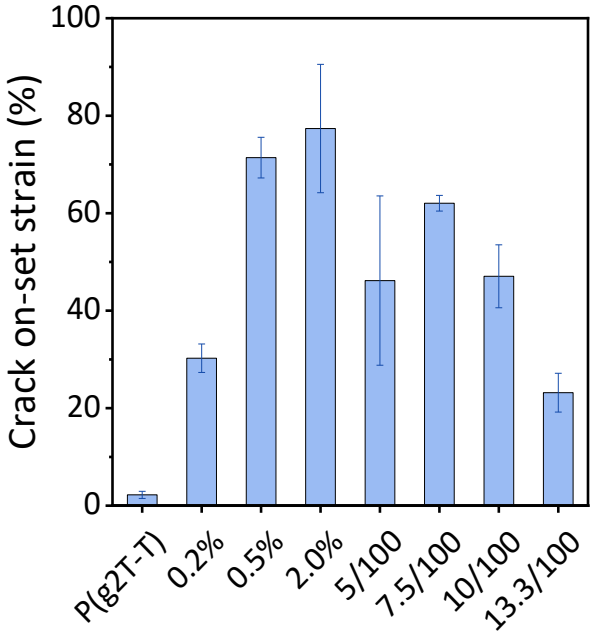


Fig. S32. Crack onset strains of p(g2T-T) and hydro-SC thin films measured using the FOW method in 100 mM NaCl solution (n = 4 independent measurements)**.** It should be noted that the crack onset strains from freestanding films measured using the FOW method are generally lower than the crack onset strains from thin films supported on elastomeric substrates.

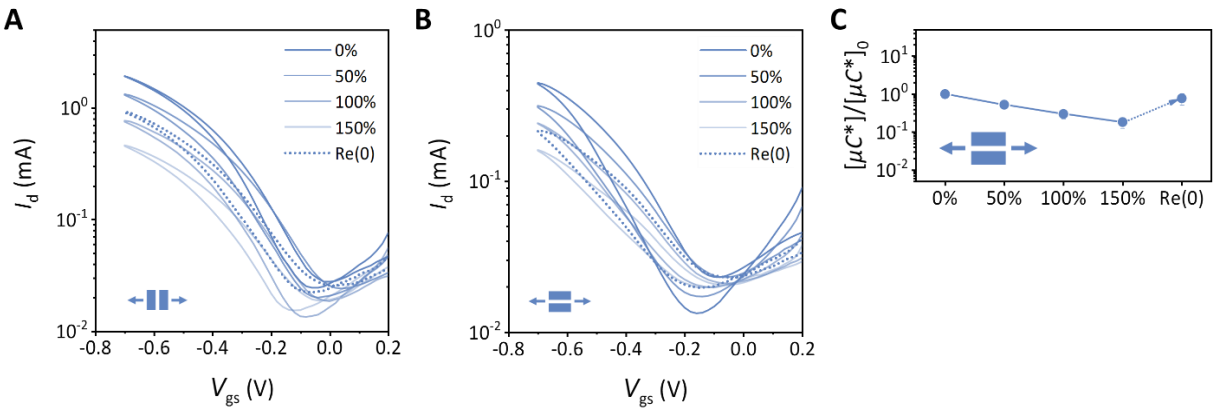


Fig. S33. OECT performance of the stretched hydro-SC thin films. (**A**) Transfer curves of OECT based on the stretched hydro-SC thin film. The film was stretched in the direction parallel to the charge transport. (**B**) Transfer curves of OECT based on the stretched hydro-SC thin film. The film was stretched in the direction perpendicular to the charge transport. V_{ds} = -0.8 V for both measurements. (**C**) Extracted and normalized μC^* as the function of strain, the films were stretched in the direction perpendicular to the charge transport (n = 9 independent measurements).

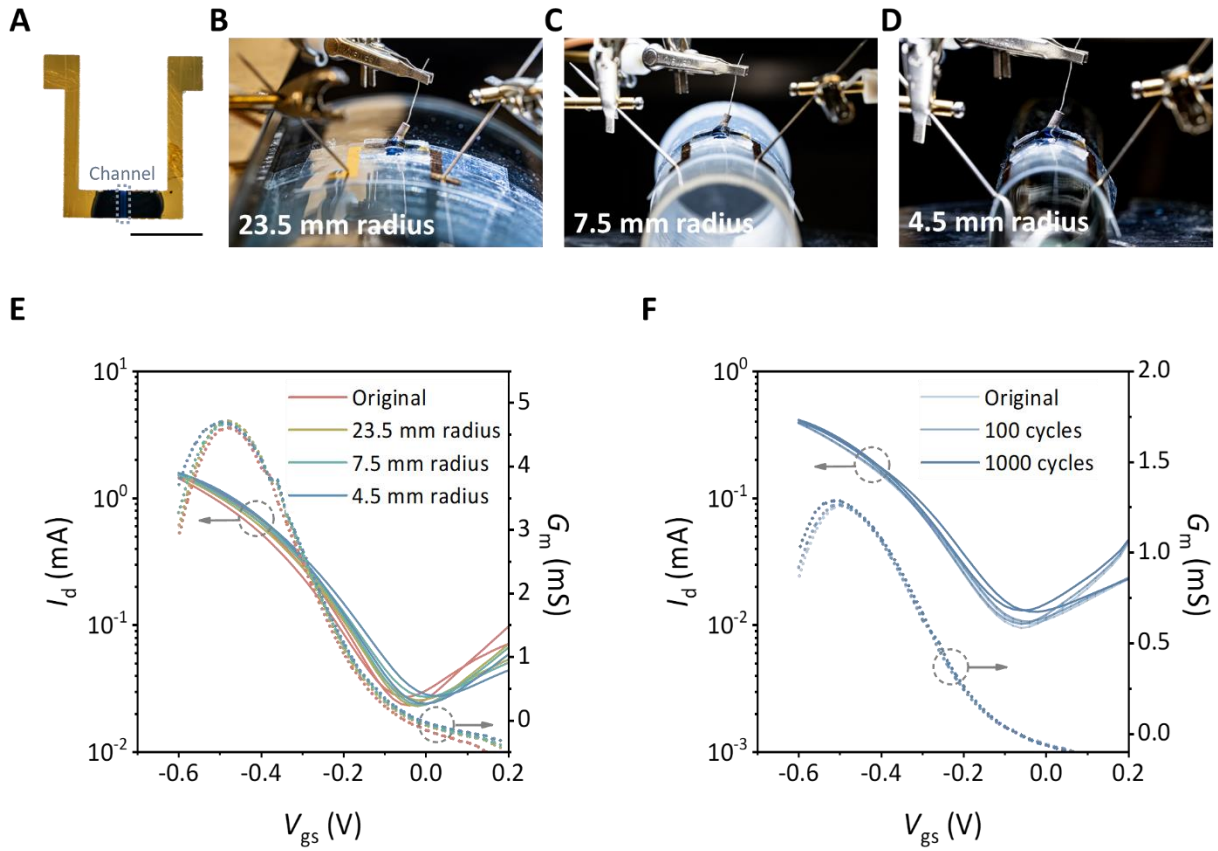


Fig. S34. Bending test of hydro-SC. (**A**) Photo of the OECT device for bending test. Scale bar, 10 mm. (**B-D**) Photos of bending tests under different bending radii. (**E-F**) OECT performance of hydro-SC under (**E**) different bending radii and (**F**) bending cycles at a 4.5 mm bending radius. V_{ds} = -0.8 V.

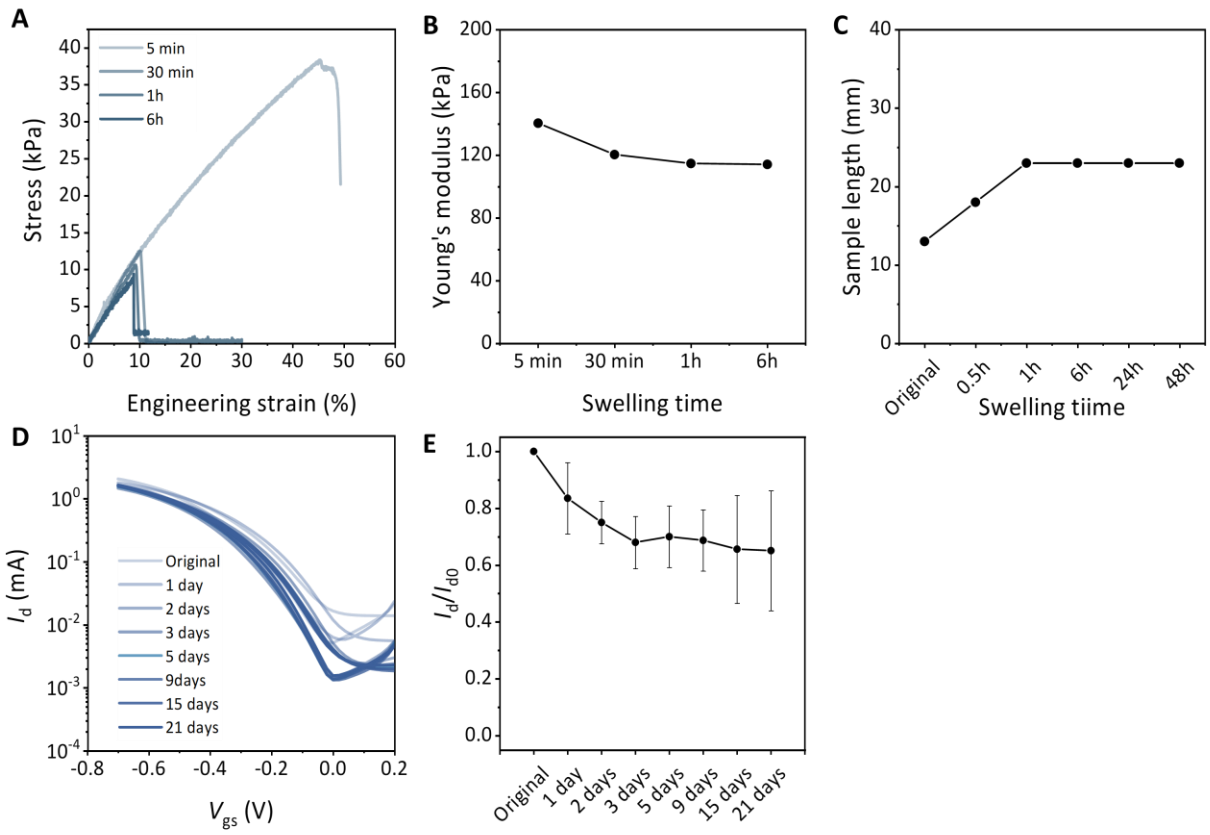


Fig. S35. Mechanical and electrical tests of hydro-SC thin films during long-term swelling in PBS. (**A**) Stress-strain curves of hydro-SC films with different swelling times. (**B**) Young's modulus of hydro-SC films, which reached a constant value after 1h swelling. (**C**) Sample length of hydro-SC films under different swelling times, which reached an equilibrium after 1h. (**D**) Transfer curves of hydro-SC-based OECTs after different days of immersion. V_{ds} = -0.8 V. (**E**) On-current retention of hydro-SC-based OECTs. n =3 independent measurements. Hydro-SCs were made with the 10/100 p(g2T-T)-to-PAAc ratio and 2% crosslinking density.

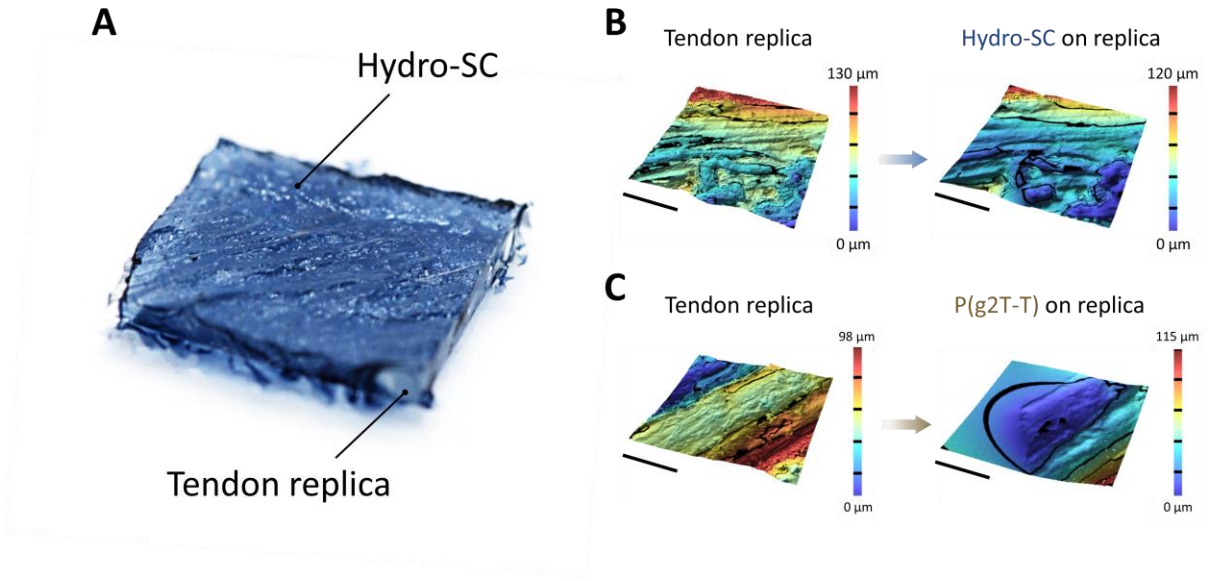


Fig. S36. 3D height profile images of tendon replica before and after the lamination of a neat p(g2T-T) film. (**A**) Picture showing a hydro-SC film laminated on a piece of tendon replica. (**B**) 3D height profile images of tendon replica before and after the lamination of the hydro-SC film. (**C**) 3D height profile images of tendon replica before and after the lamination of a neat p(g2T-T) film, as a comparison. Scale bars, 200 µm.

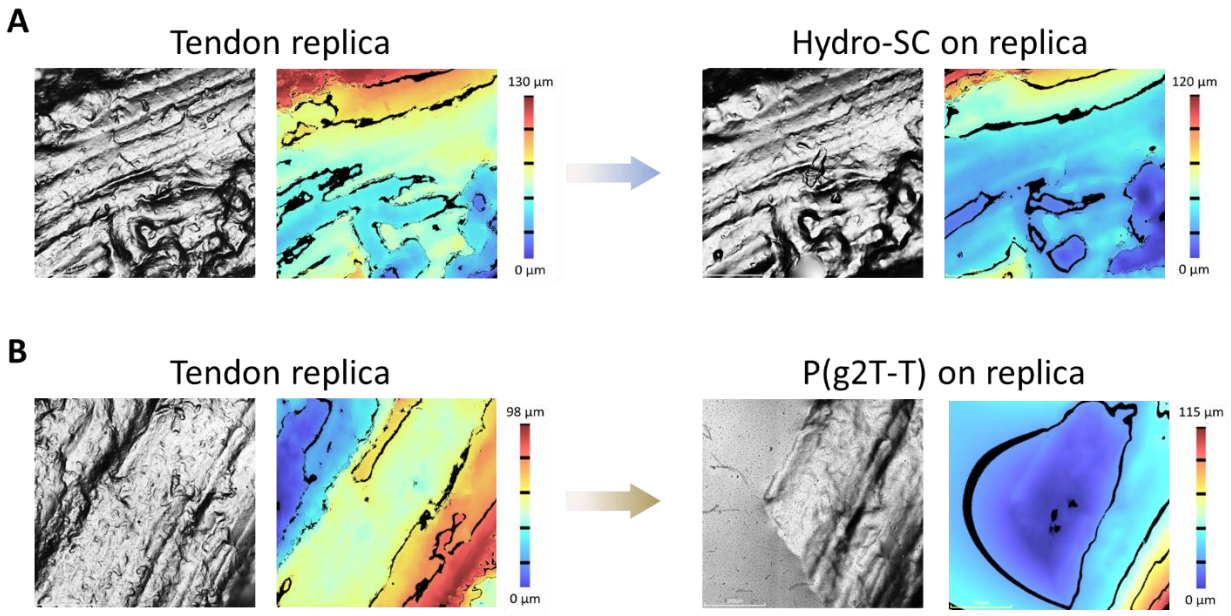


Fig. S37. Conformability test of semiconductor thin films. (**A**) 2D color and the corresponding height images of tendon replica before and after the lamination of a hydro-SC film. (**B**) 2D color and the corresponding height images of tendon replica before and after the lamination of a neat p(g2T-T) film.

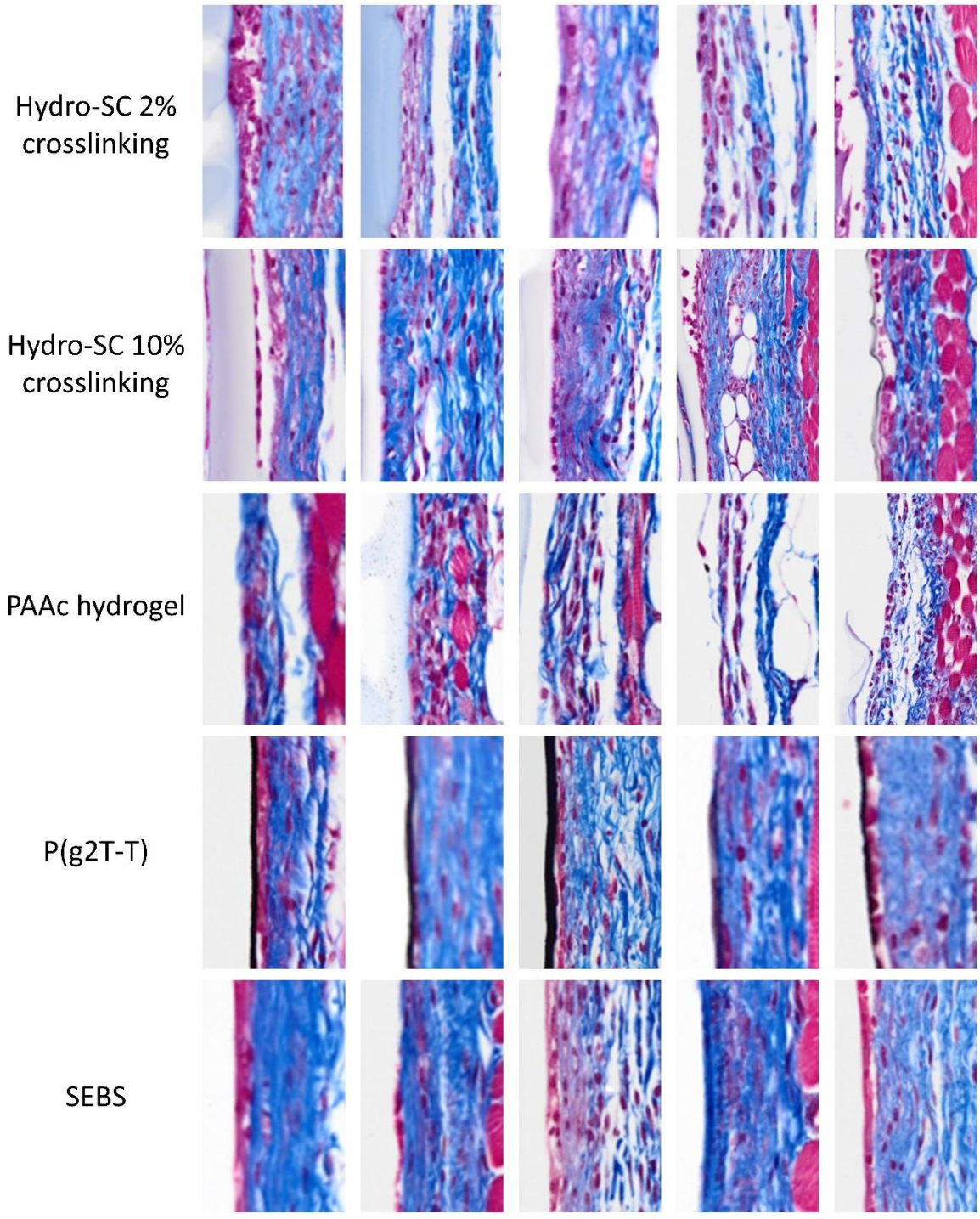


Fig. S38. Masson's trichrome staining results of tissues around different implants.

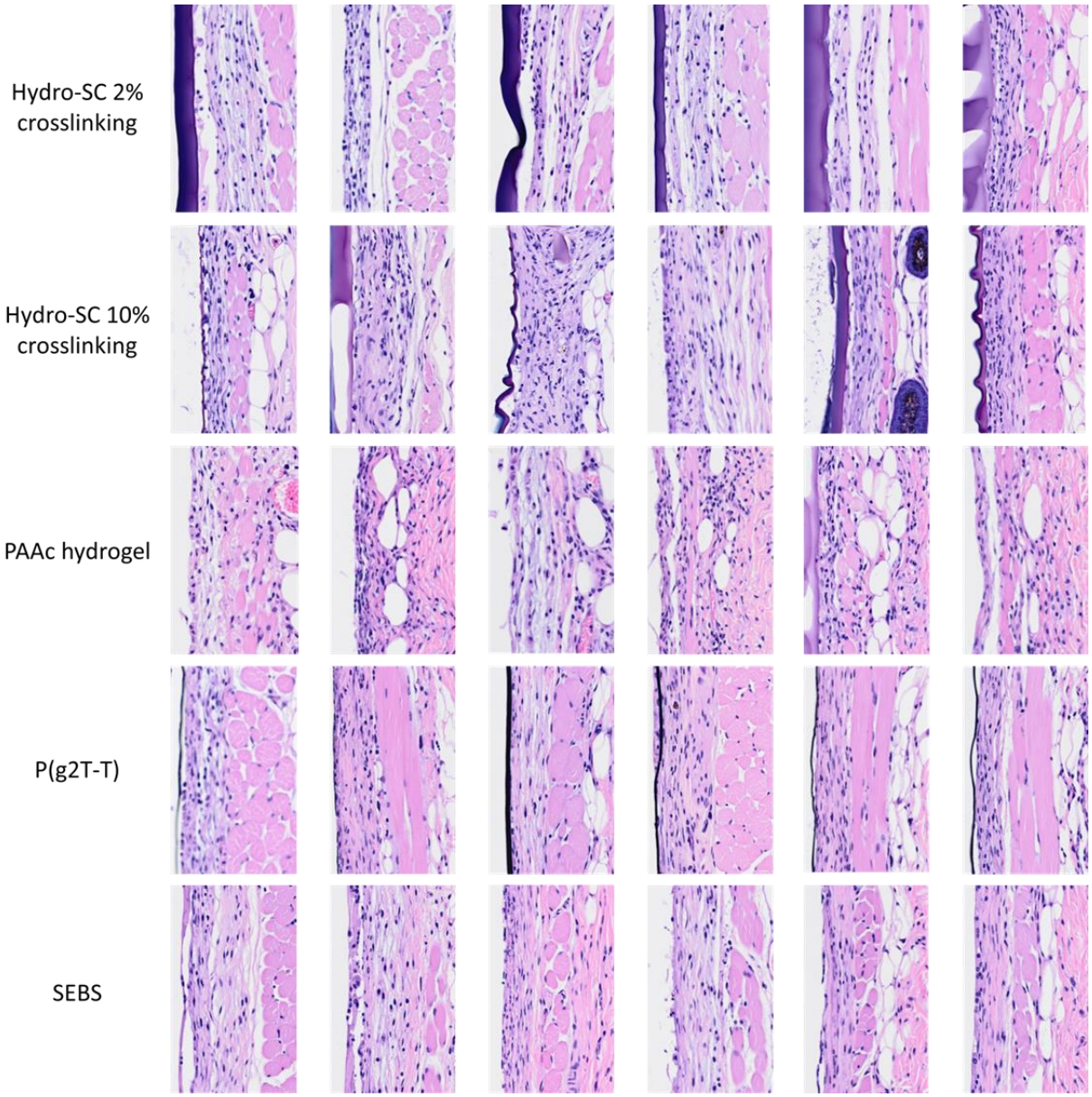


Fig. S39. H&E staining results of tissues around different implants. The modulus change of implants did not significantly impact cell infiltration.

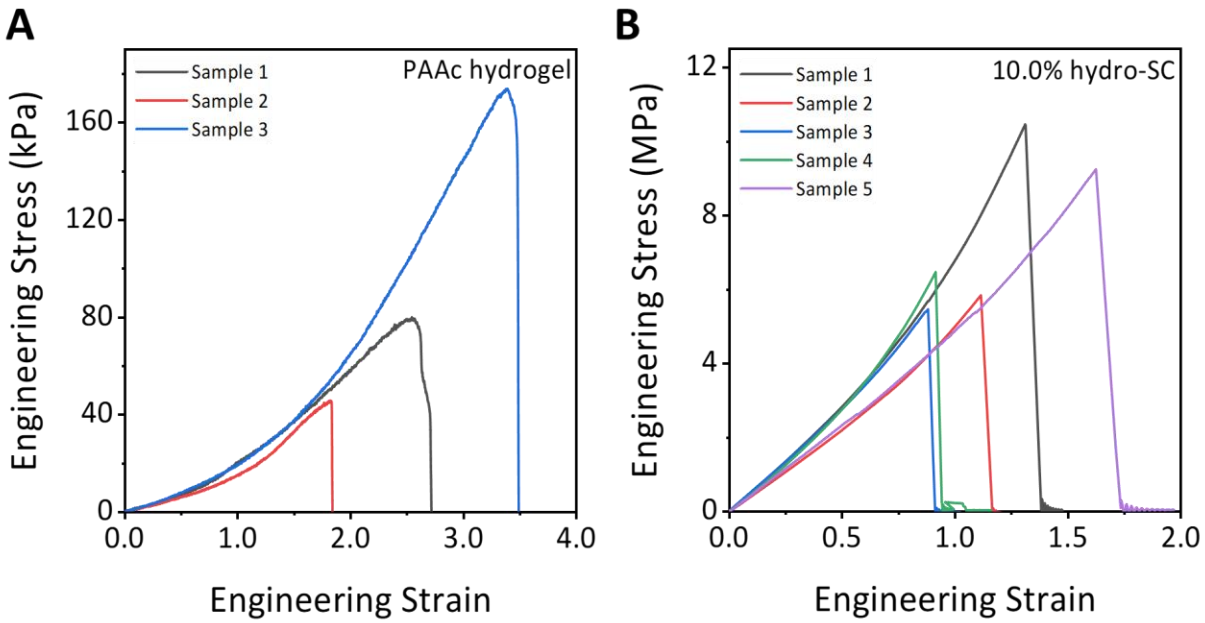


Fig. S40. Stress-strain curves of PAAc hydrogel (by tensile test) and hydro-SC films with 10% crosslinking density (by FOW tests, in 100 mM NaCl solution).

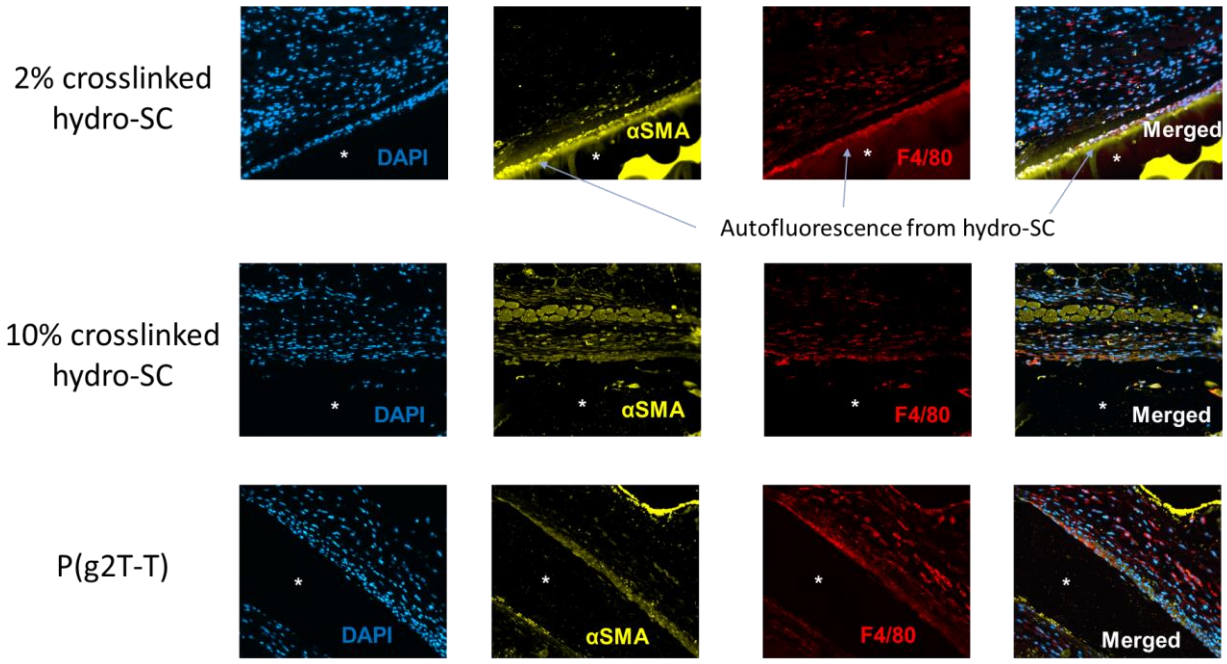


Fig. S41. Immunofluorescence staining of α-SMA for fibroblasts (yellow) and F4/80 for macrophages (red). The 2% crosslinked hydro-SC shows lower amounts of myofibroblasts deposited on the surface than more rigid hydro-SC and pristine p(g2T-T) (the fluorescence at the edge of 2% crosslinked hydro-SC is mostly attributed to the fluorescence of hydro-SC.

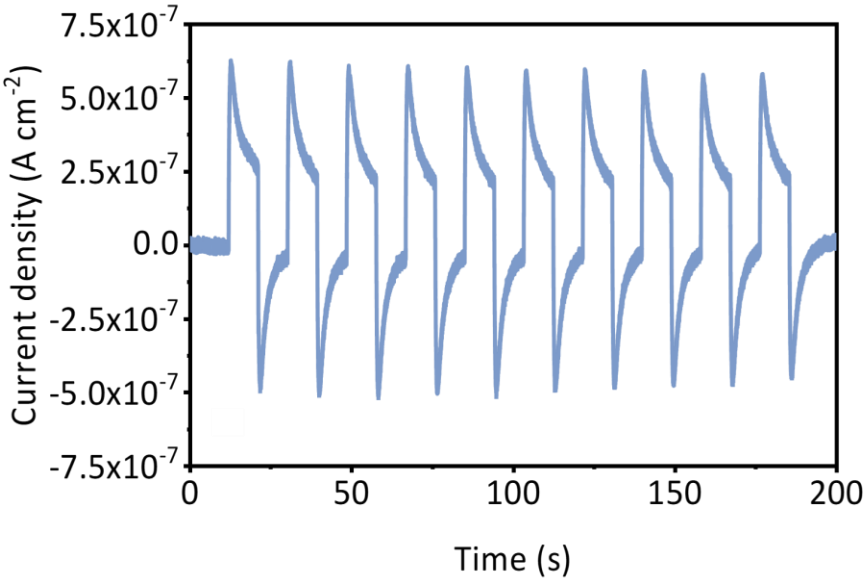


Fig. S42. Cyclic photoelectrochemical response of a hydro-SC film with 10 s pulse lengths. Wavelength, 625 nm. Radiation power, radiation power of 114.3 mW cm^{-2}. The test was conducted with a 0 V bias applied vs. the Ag/AgCl electrode. The positive photocurrent values stand for the current flowing out of the tested films, and the baseline current was subtracted.

Estimation of the incident photon conversion efficiency of the photofaradaic process of hydro-SC thin films

The incident photon conversion efficiency (IPCE) was calculated using the equation (*71*):

$$IPCE = \frac{1240(nm \cdot V) \cdot J(\frac{mA}{cm^2})}{\lambda(nm) \cdot P_{in}(\frac{mW}{cm^2})} \cdot 100\%$$

where J is the current density, λ is the wavelength, and P_{in} is the illumination power, this gives the IPCE of hydro-SC of 0.00035%.

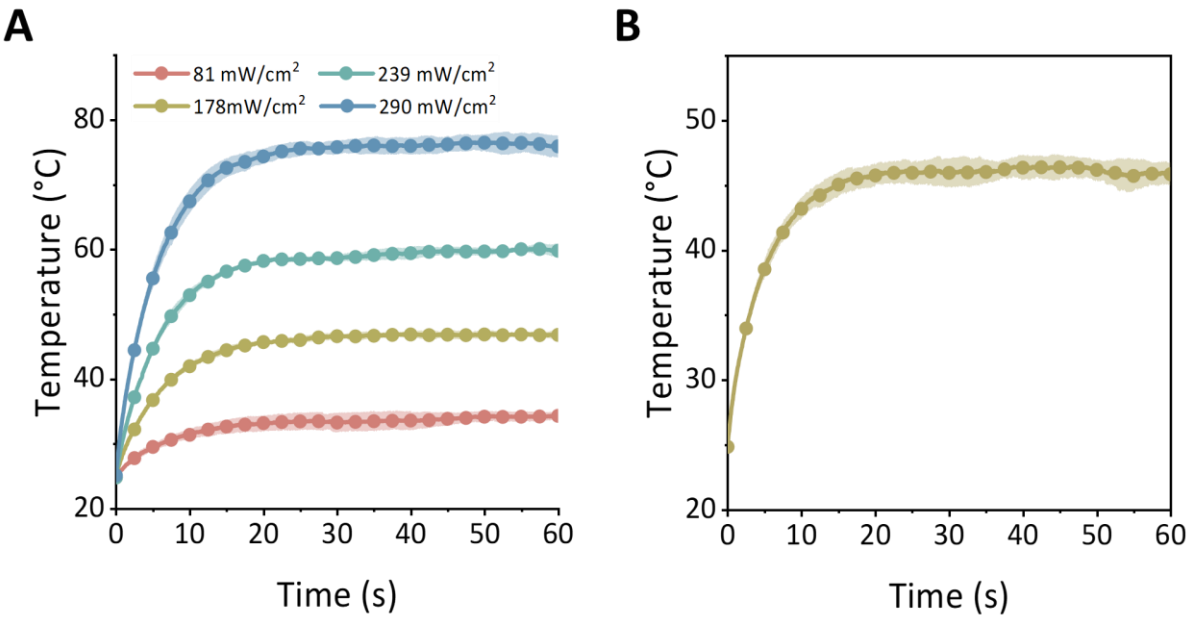


Fig. S43. Photothermal effect measurements. (**A**) Temperature increase measured on a fully-hydrated hydro-SC film under different powers of radiation. (**B**) Temperature increase measured on a dry p(g2T-T) film under a radiation power of 178 mW cm^{-2}. Wavelength, 635 nm.

Calculation of heat accumulation and photothermal conversion efficiency (PCE) of semiconductor films upon irradiation

From the temperature measurements shown above, we further calculated the heat (in Joule) accumulation process on the thin films (i.e., hydro-SC or neat p(g2T-T)) and surrounding substrates, which are utilized to compare to the photothermal effect in Fig. 4G. The calculation is based on the following equation:

$$Q = \Delta T \cdot \sum m_i C_i$$

where the m_i and C_i are the mass and heat capacity of each component, and ΔT stands for the temperature difference between the sample and the environment at each time point, which is calculated from fig. S43.

For hydro-SC:

$$\sum m_i C_i = m_{water} C_{water} + m_{PET} C_{PET} + m_{P(g2T-T)} C_{P(g2T-T)} + m_{PAAc} C_{PAAc}$$

Since the p(g2T-T) and PAAc have much lower mass than the other components, the heat capacity of the system is dominated by the absorbed water within the gel matrix (0.86 mg) and PET substrate and cover (5.96 mg). Counting the heat capacity of PET (1.146 J g^{-1} °C^{-1}) and water (4.2 J g^{-1} °C^{-1}), the heat capacity of hydro-SC sample is determined as 1.04×10^{-2} J °C^{-1}.

For p(g2T-T):

$$\sum m_i C_i = m_{PET} C_{PET} + m_{P(g2T-T)} C_{P(g2T-T)}$$

Since the p(g2T-T) has a much lower mass (0.01 mg) than the PET substrate (5.15 mg), the heat capacity of the neat p(g2T-T) sample is dominated by PET and thereby determined as 5.9×10^{-3} J °C^{-1}.

The PCE and heat of p(g2T-T) and hydro-SC films were measured using the reported method (*72*). Both samples were prepared on PET substrates with a diameter of 5 mm. The samples were irradiated by a 635 nm laser (178 mW cm^{-2}) for 1 min. Then the laser was turned off, and the sample temperature was further recorded for around 30 s until it returned back to room temperature. The PCE can be calculated using the following equation:

$$\eta = \frac{hA\Delta T_{max} - Q_s}{I\,(1 - 10^{-A_\lambda})}$$

where η is the PCE, h is the heat transfer coefficient, A is the surface area of the sample, and ΔT_{max} is the maximal temperature difference between the sample and the environment during irradiation. Q_s represents the heat generated by the light *absorbance* of the solvent and PET substrate. I is the power intensity of the laser. A_λ refers to the absorbance of testing samples at 635 nm in the UV-vis spectra, which was determined as 2 for hydro-SC, and 4 for p(g2T-T) samples. The hA value was determined by:

$$hA = \frac{\sum m_i C_i}{\tau_s}$$

where the τ_s is the time constant determined by fitting the time versus θ in the cooling phase, which was obtained by:

$$\theta = \frac{\Delta T}{\Delta T_{max}}$$

The Q_s was measured by irradiating the laser to the PET substrate and water, which did not give an observable temperature increase (fig. S44E), indicating negligible absorbance. The results of the PCE test are summarized in Fig. 4H, fig. S44, and Table S4.

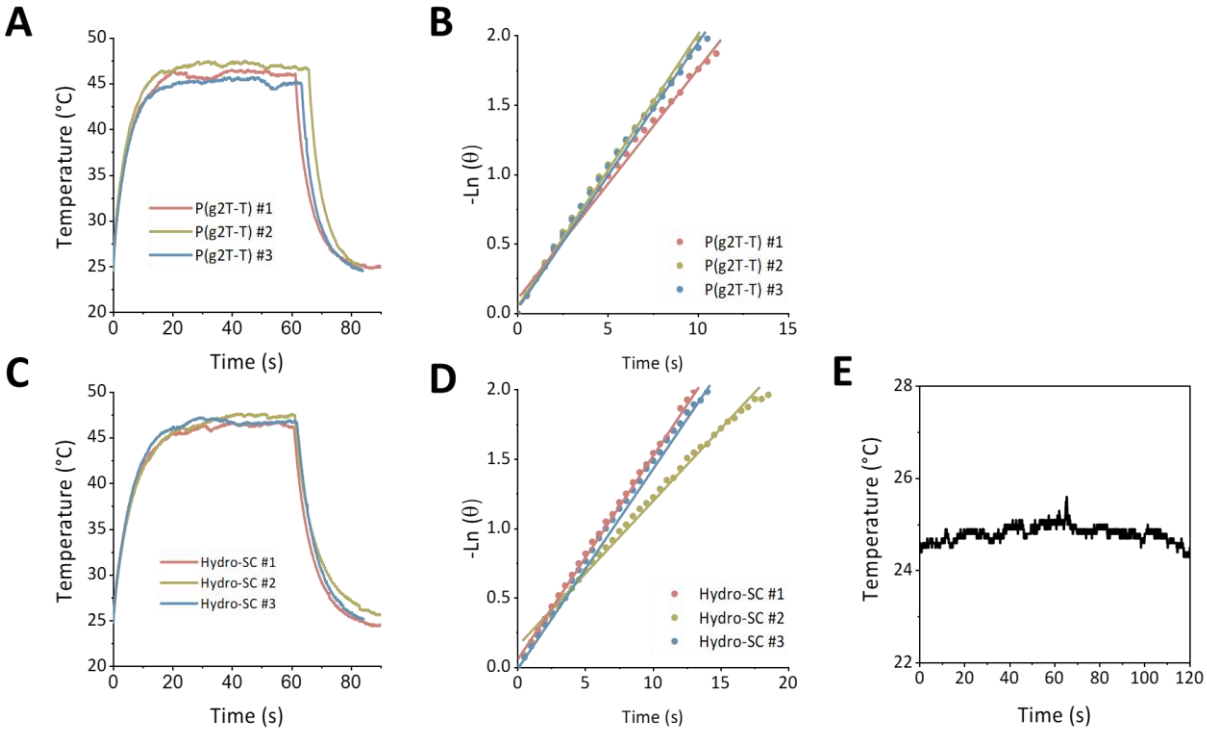


Fig. S44. Calculation of time constant based on heat dissipation. (**A** and **C**) A heating-cooling cycle of p(g2T-T) (**A**) and hydro-SC (**C**) under the radiation power of 178 mW cm^{-2}. (**B** and **D**) Calculation of time constant by fitting the -Ln (θ) versus cooling time. The results are summarized in Table S4. (**E**) Temperature response of water and PET substrate upon irradiation.

Table S4. Summary of parameters for PCE calculation for the photothermal measurements.

Composition	Sample #	ΔT_{max} (°C)	τ_s (s)	hA (J °C^{-1} s^{-1})	PCE (%)
Hydro-SC	1	19.1	6.72	$1.55*10^{-3}$	18.0 ±1.8
	2	15.9	8.64	$1.20*10^{-3}$	
	3	18.9	6.97	$1.49*10^{-3}$	
P(g2T-T)	1	22.1	5.99	$9.85*10^{-4}$	13.2 ±1.0
	2	22.5	5.22	$1.13*10^{-3}$	
	3	21	5.32	$1.11*10^{-3}$	

Further analysis of the enhanced photothermal effect on hydro-SC.
Hypothetically, such stronger photothermal effect from hydro-SC might come from the combined effects of stronger aggregation levels (Fig. 2G), increased polaron amount (fig. S45) from more efficient oxygen doping in water, and a higher amount of surface traps, which all increases the non-radiative decay under photoexcitation (*73*).

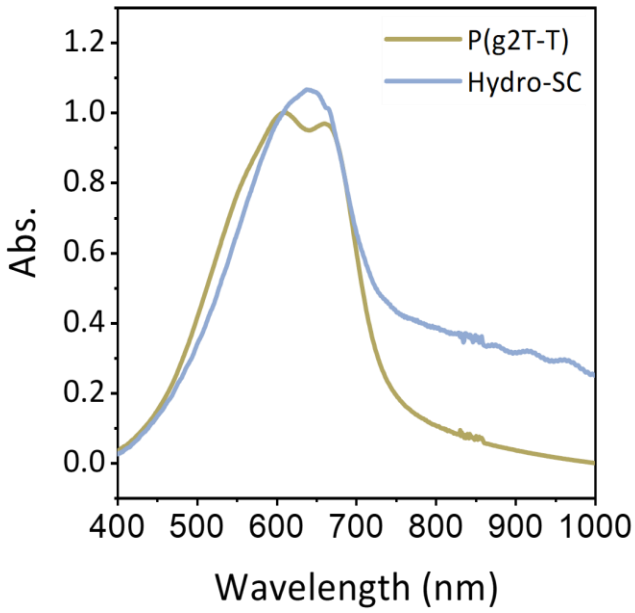


Fig. S45. UV-vis spectra of wet p(g2T-T) and a hydrated hydro-SC film. Hydro-SC shows a much higher absorption in the IR region than p(g2T-T) film, indicating the more efficient generation of polarons, as a result of oxygen doping in water.

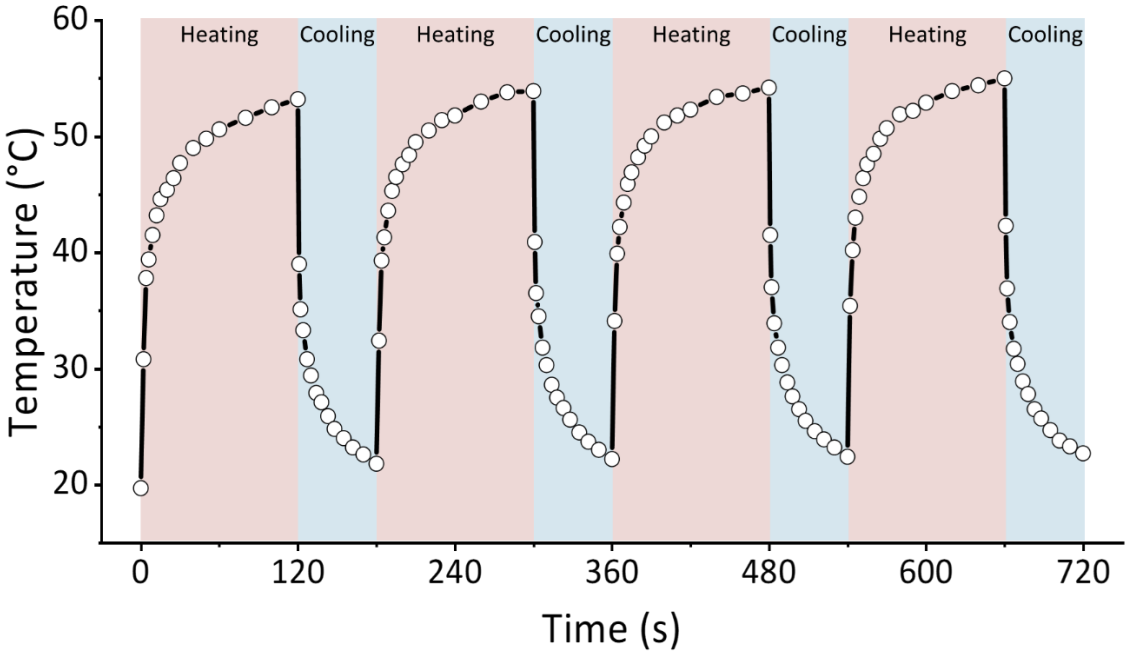


Fig. S46. Heating-cooling cycles of a hydro-SC film on wet skin through the photothermal effect. Wavelength, 635 nm. Radiation power, 318 mW cm^{-2}.

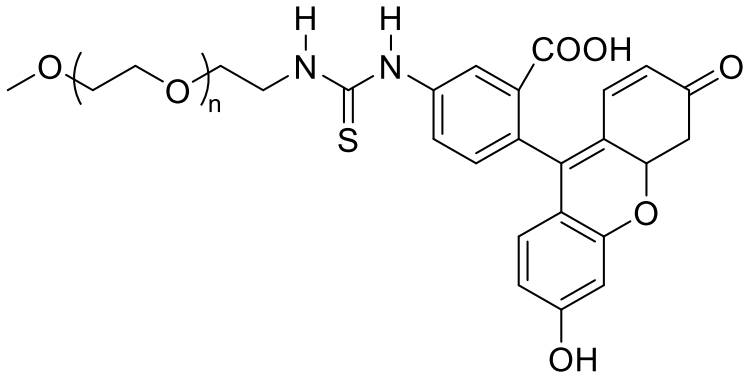


Fig. S47. Molecular structure of PEG-FITC.

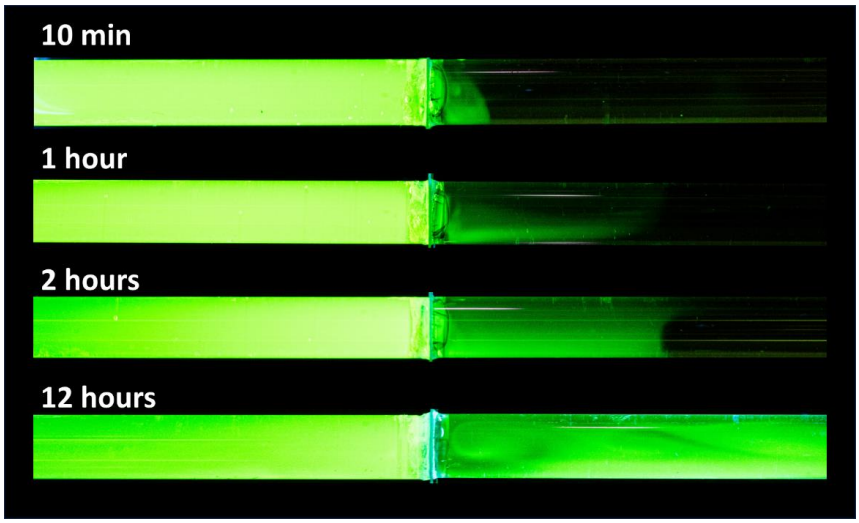


Fig. S48. Diffusion of fluorescent probes through a hydro-SC film at different time points.

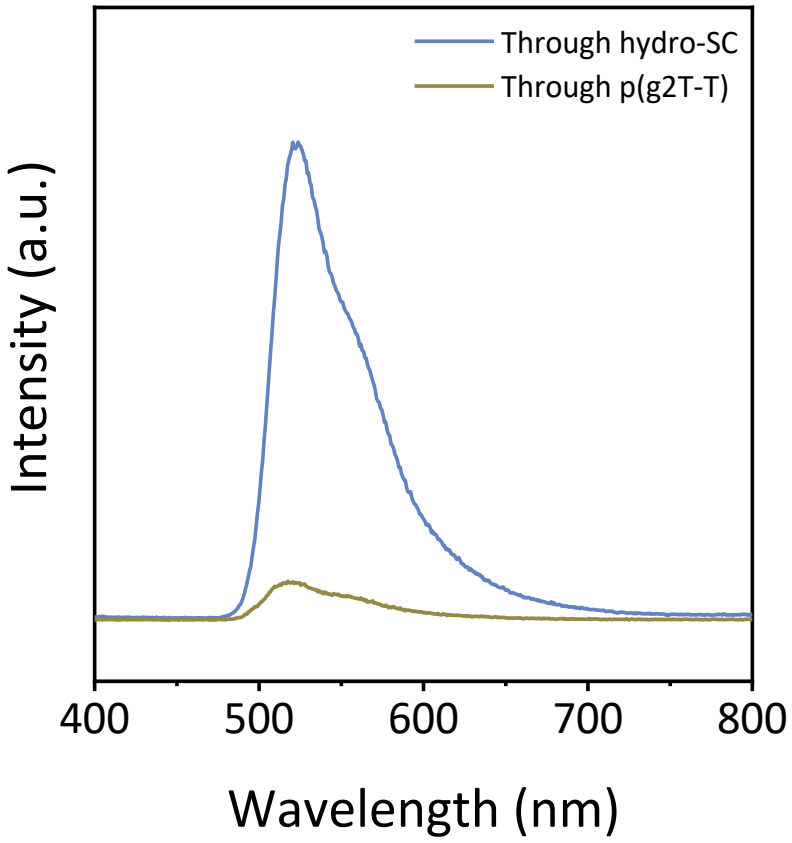


Fig. S49. Fluorescent spectra of FITC-PEG diffused through the hydro-SC film and the neat p(g2T-T) film. The fluorescent probe has a much lower diffusivity through a p(g2T-T) than a hydro-SC film.

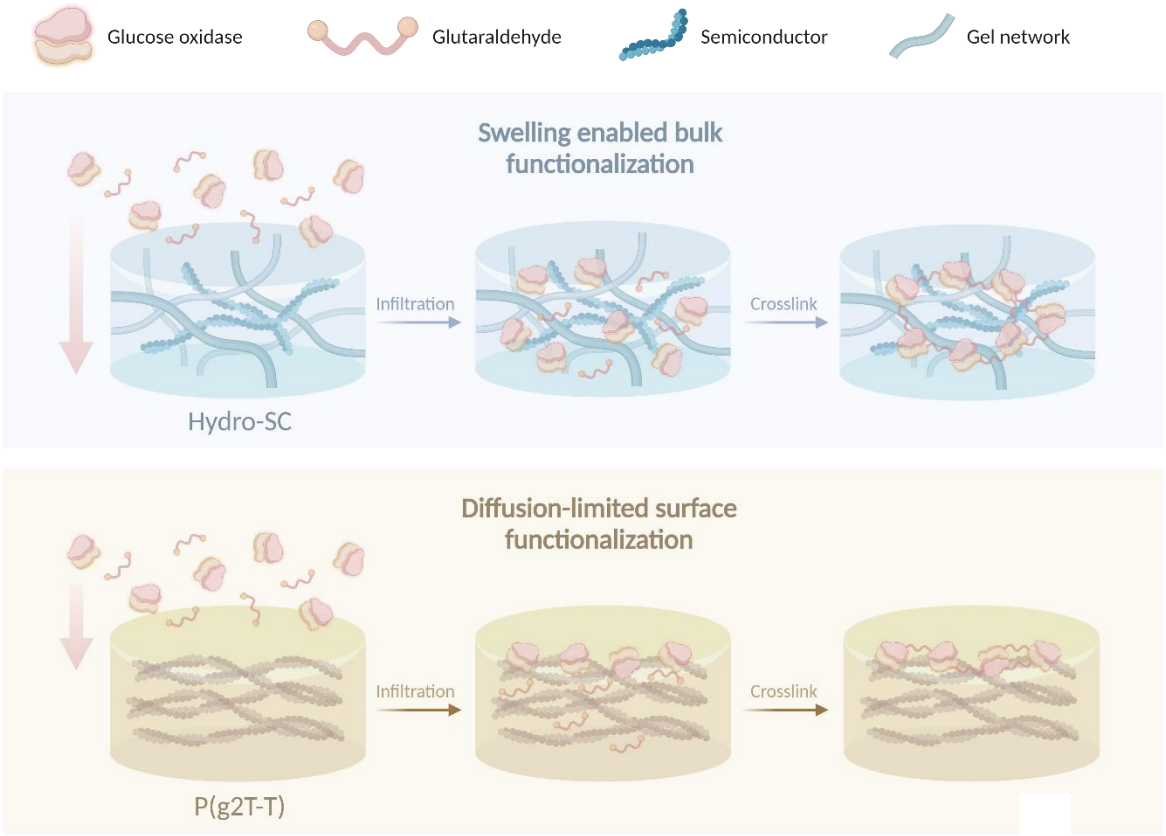


Fig. S50. GOx-immobilization based on diffusion and enzyme crosslinking into the bulk of hydro-SCs, in comparison with the case of surface-limited functionalization on neat p(g2T-T) films.

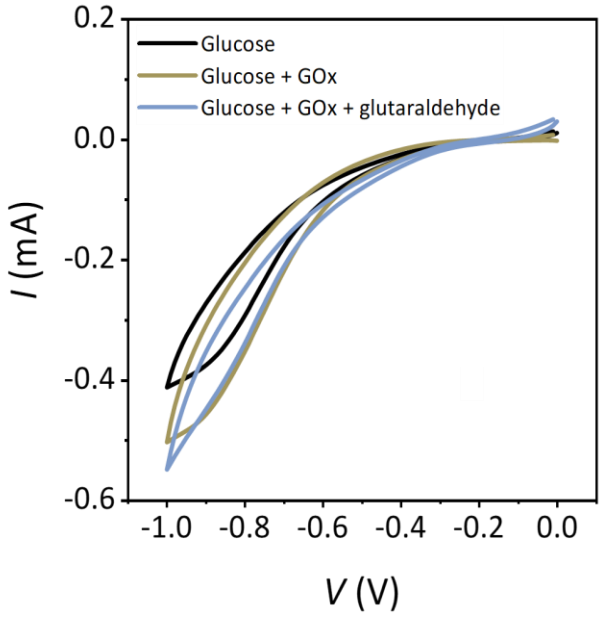


Fig. S51. CV measurements for evaluating the activity of GOx upon crosslinking. A larger reduction current was observed due to the reduction of H_2O_2, which was produced during enzymatic reaction. The GOx before/after crosslinking shows a consistent response, validating the minor influence of glutaraldehyde crosslinking on GOx activity.

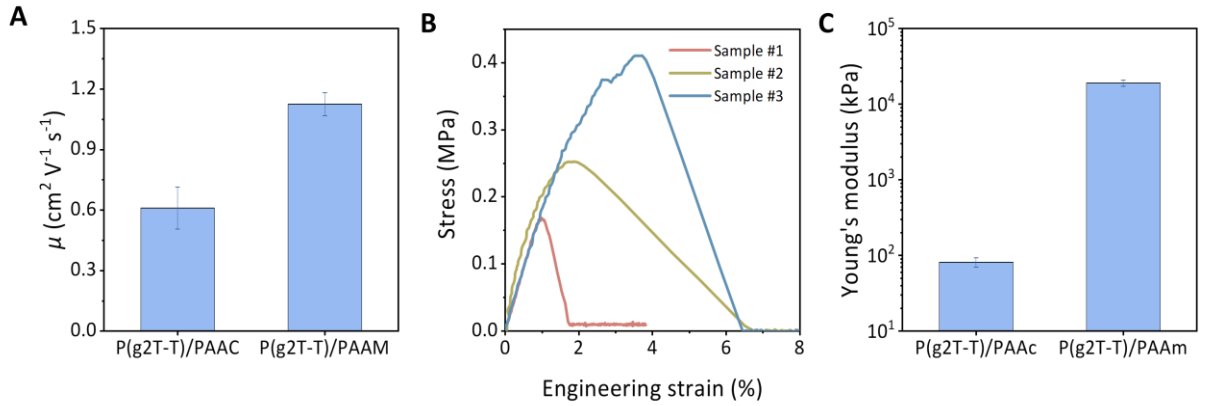


Fig. S52. Electrical and mechanical properties of p(g2T-T)/PAAM hydrogel in comparison to p(g2T-T)/PAAc hydrogel. (**A**) μ of both hydro-SCs. n = 6 independent measurements. (**B**) Strain-stress curves of p(g2T-T)/PAAm hydrogels. (**C**) Young's modulus of both hydro-SCs. n = 3 independent measurements. Both hydro-SCs were made with the 10/100 p(g2T-T)-to-hydrogel monomer ratio and 0.5% crosslinking density.

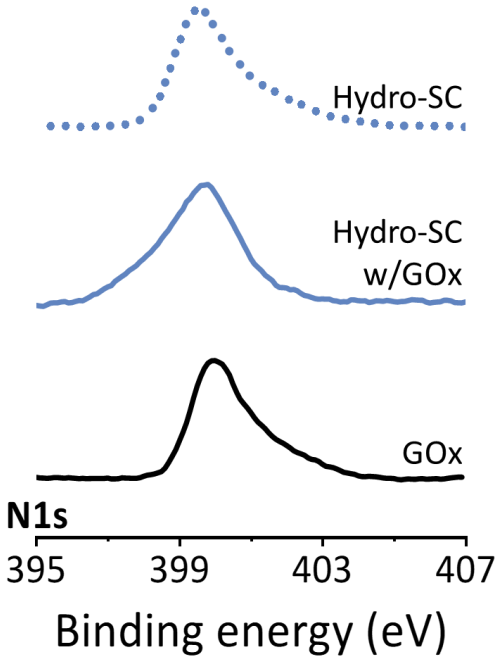


Fig. S53. N1s spectra of hydro-SC films before and after GOx immobilization. The peaks from the hydro-SC and GOx are overlapped, thereby N1s was not used for depth profiling.

Table S5. Summary of QCM test results for quantifying the density of GOx in hydro-SC.

Chip #	Original frequency (MHz)	Frequency after film deposition (MHz)	Frequency after functionalization (MHz)	Mass of Hydro-SC film (ng)	Mass of GOx (ng)	Mass increase ratio (%)	GOx density (nmol cm^{-3})
Chip 1	4.9980075	4.9857155	4.983785	52100	8185.32	15.71078695	782
Chip 2	4.9975285	4.9865184	4.9854935	46700	4345.576	9.305301927	

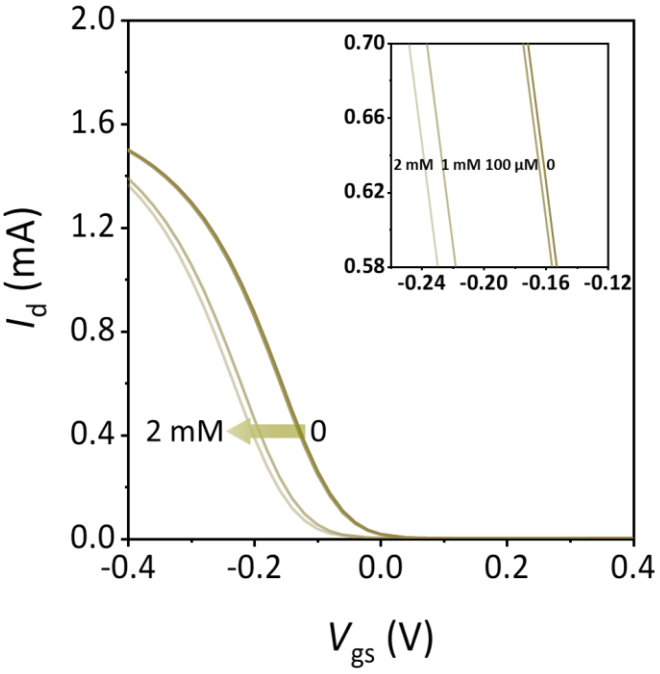


Fig. S54. Shifts of transfer curves in response to the addition of glucose. V_{ds} = -0.1 V. The OECT was made by a neat p(g2T-T) film without GOx functionalization. The negative shifts were similar to the results from GOx-functionalized p(g2T-T) films shown in Fig. 5K, indicating the response was not induced by the enzymatic reactions.

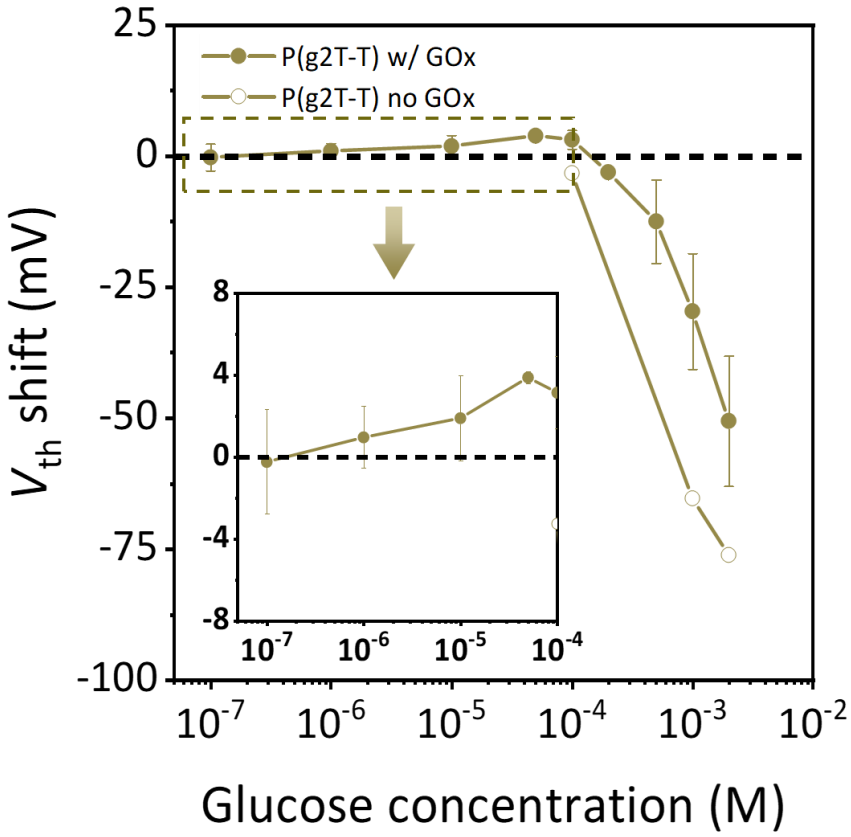


Fig. S55. Voltage shift of p(g2T-T)-based OECT as a function of glucose concentration. Both films show similar response behaviors with regardless of GOx functionalization (n = 3 independent measurements).

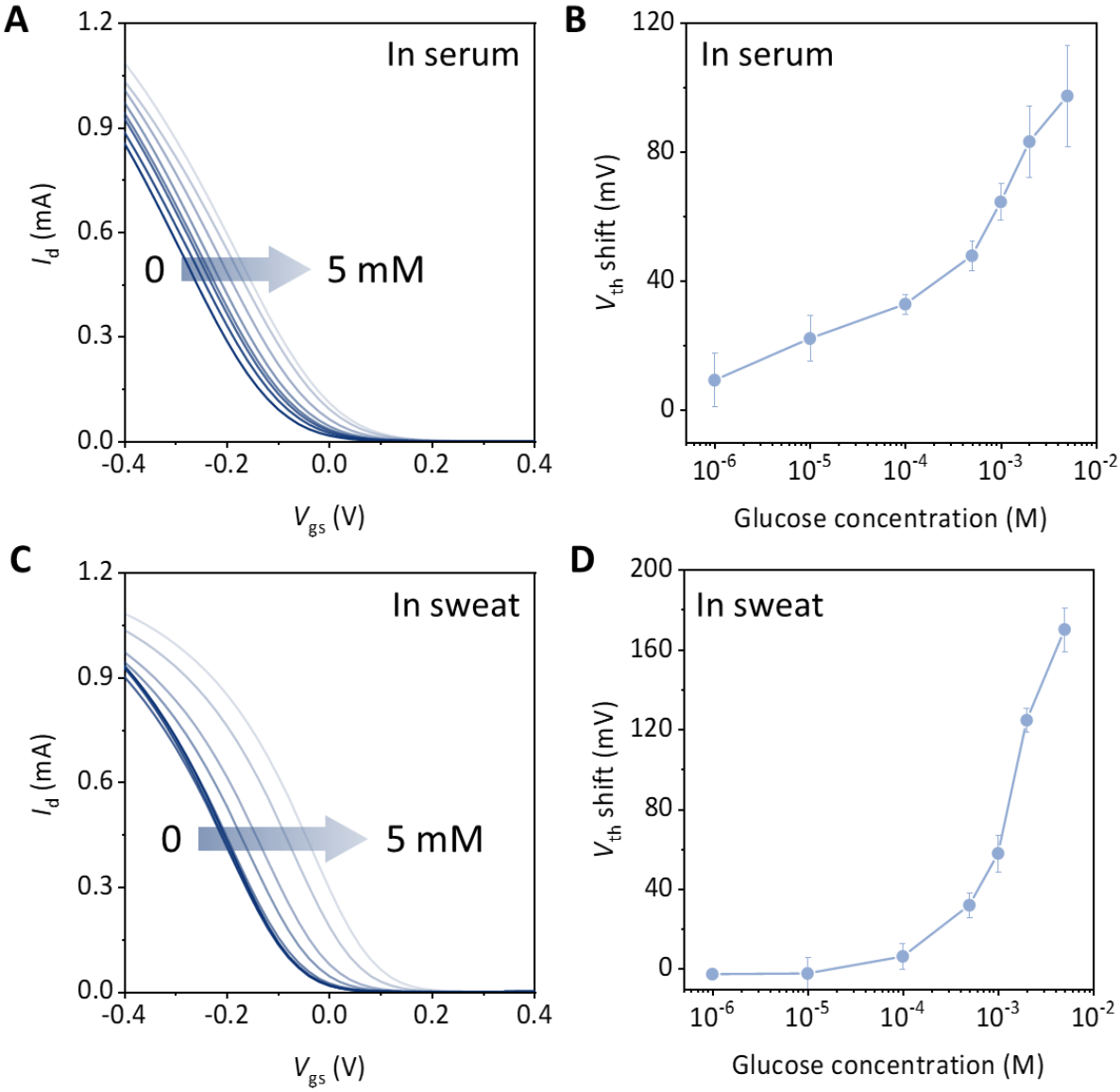


Fig. S56. Volumetric glucose sensing tests in the fetal bovine serum (FBS) (A and B) and human sweat (C and D) using hydro-SC-based OECTs. (**A** and **C**) Shifts of transfer curves in response to the addition of glucose in FBS (**A**) and human sweat (**C**). (**B** and **D**) Threshold voltage shifts of OECT as a function of glucose concentration in FBS (**A**) and human sweat (**C**) (n = 3 independent measurements). V_{ds} = -0.1 V.

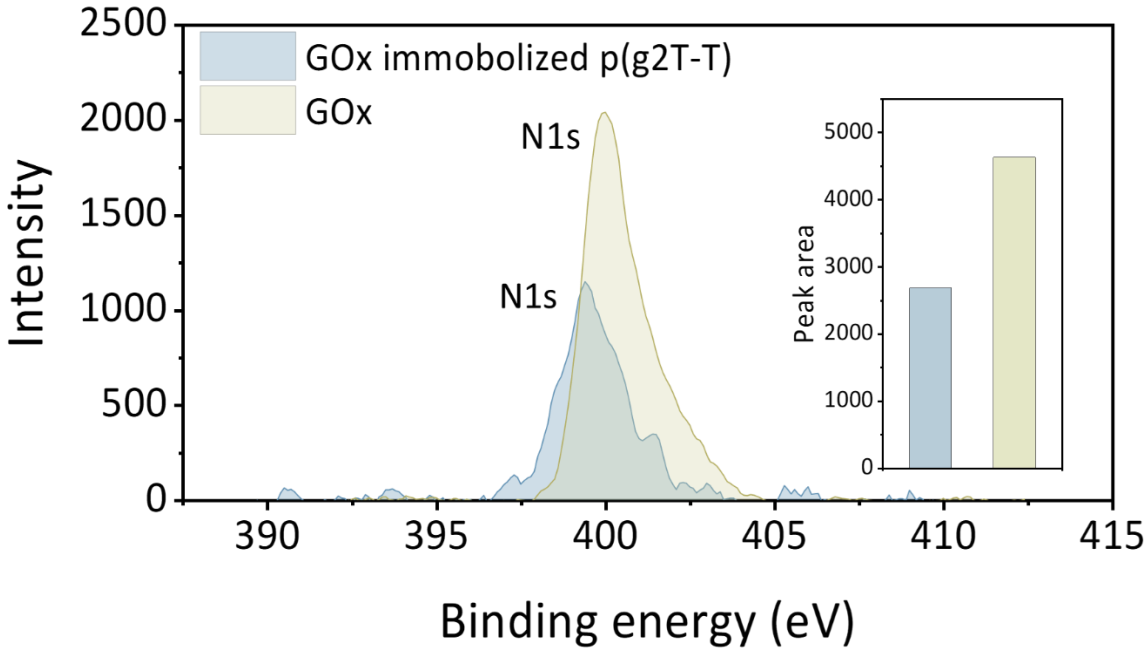


Fig. S57. Calculation of surface coverage of GOx on p(g2T-T) film. The XPS spectra of a layer of as-casted GOx on a Si substrate were used as the reference for 100% coverage. The coverage of GOx on p(g2T-T) was calculated by integrating and normalizing the N1s peak, which gives a ratio of 58%.

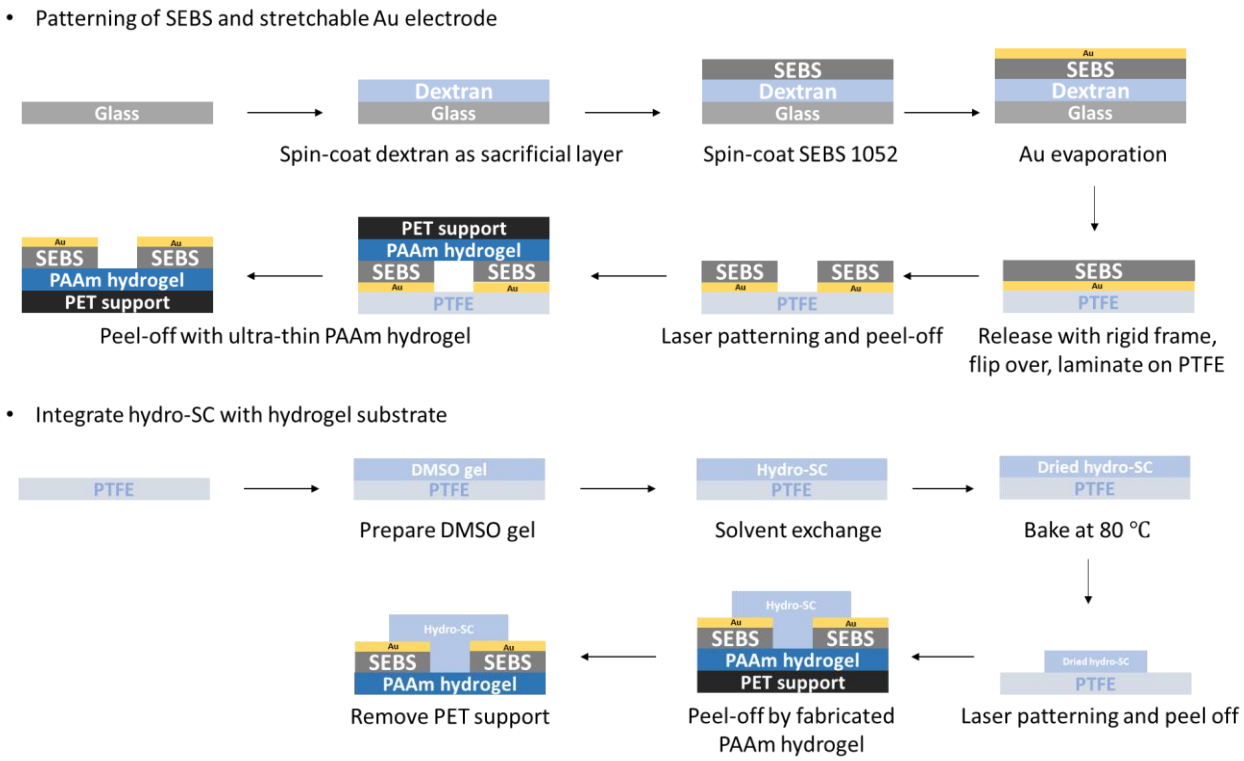


Fig. S58. Fabrication flow of hydrogel-based OECT.

The hydrogel-based OECT was fabricated by sequentially transferring the electrodes and hydro-SC to the hydrogel substrate. Briefly speaking, the microcracked Au film was fabricated on SEBS 1052 and released in water. The SEBS layer was picked up using a PET rigid frame and flipped over on a non-adhesive polytetrafluoroethylene (PTFE) substrate. The SEBS and Au electrodes were then laser-patterned and peeled off by the PAAm hydrogel substrate supported by a PET film. The hydro-SC film was prepared on a PTFE substrate and laser-patterned, and then peeled off by the hydrogel substrate. The final removal of the PET support yielded a hydrogel-based OECT.

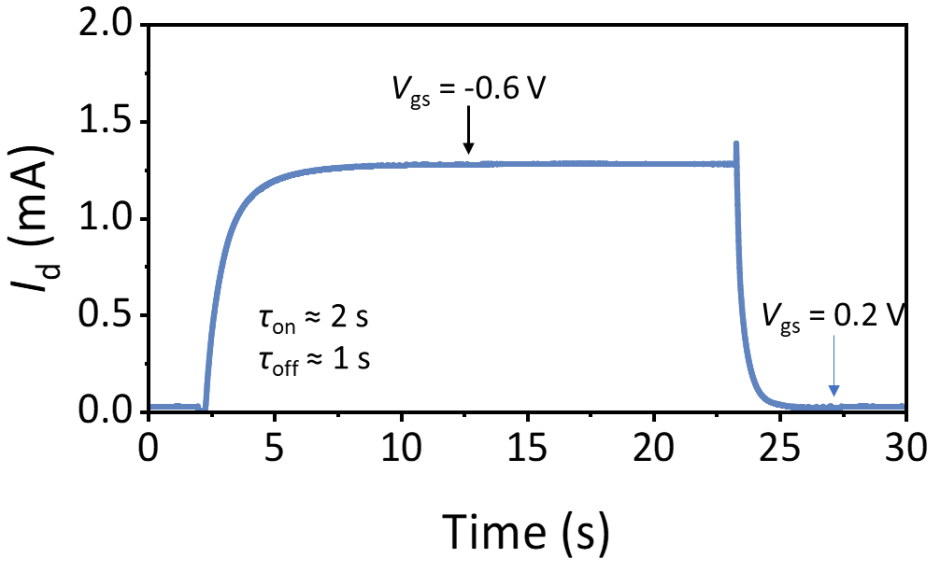


Fig. S59. Operation speed test of hydrogel-based OECT. The turn-on and turn-off times were determined as the time needed to reach 90% and 10% of the on-current, respectively. V_{ds} = -0.8 V. The response speed is mainly limited by the large area (12 mm^2) of transferred hydro-SC film.

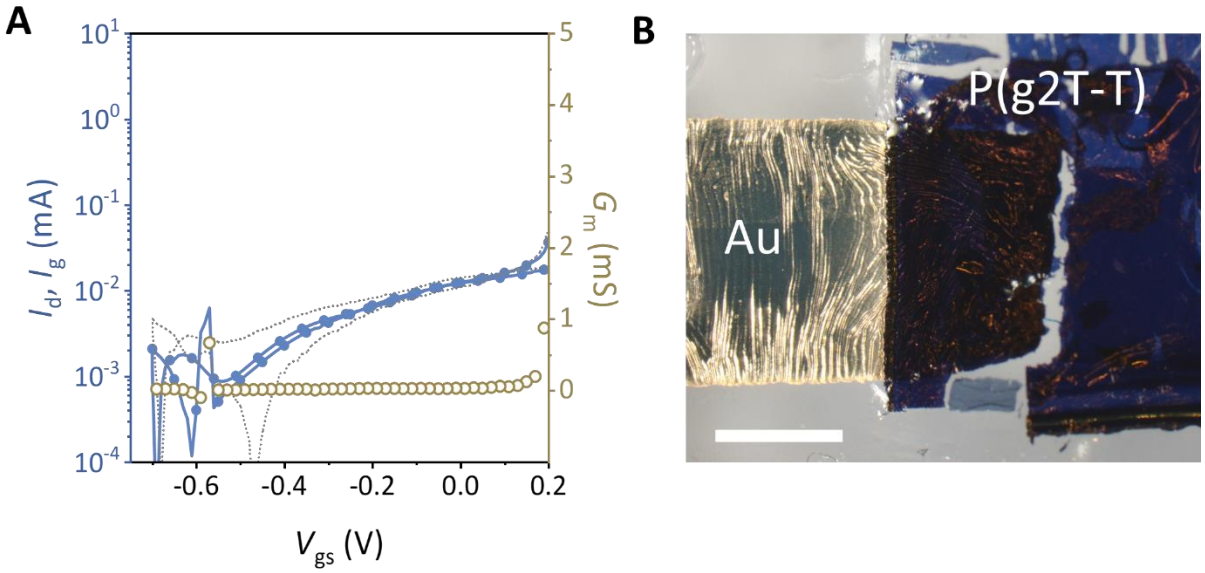


Fig. S60. P(g2T-T)-based hydro-OECT. (**A**) Transfer curve of p(g2T-T)-based hydro-OECT. V_{ds} = -0.8 V. The blue and grey curves refer to I_d and I_g, respectively. (**B**) Microscopic image showing the cracking of p(g2T-T) film as the result of hydrogel substrate swelling. Scale bar, 1 mm.

Movie S1. Floating-on-water test for measuring the mechanical properties of a swelled hydro-SC thin film.

Movie S2. Heating of a hydro-SC film on wet skin through the photothermal effect.

References and Notes

1. H. Yuk, B. Lu, X. Zhao, Hydrogel bioelectronics. *Chem. Soc. Rev.* **48**, 1642–1667 (2019). doi:10.1039/C8CS00595H Medline

2. A. C. Daly, L. Riley, T. Segura, J. A. Burdick, Hydrogel microparticles for biomedical applications. *Nat. Rev. Mater.* **5**, 20–43 (2020). doi:10.1038/s41578-019-0148-6 Medline

3. J. Lou, D. J. Mooney, Chemical strategies to engineer hydrogels for cell culture. *Nat. Rev. Chem.* **6**, 726–744 (2022). doi:10.1038/s41570-022-00420-7 Medline

4. K. Y. Lee, D. J. Mooney, Hydrogels for tissue engineering. *Chem. Rev.* **101**, 1869–1879 (2001). doi:10.1021/cr000108x Medline

5. J. Shi, Y. Lin, P. Li, P. Mickel, C. Sun, K. Parekh, J. Ma, S. Kim, B. Ashwood, L. Meng, Y. Luo, S. Chen, H.-M. Tsai, C. M. Cham, J. Zhang, Z. Cheng, J. A. Abu-Halimah, J. Chen, P. Griffin, E. B. Chang, P. Král, J. Yue, B. Tian, Monolithic-to-focal evolving biointerfaces in tissue regeneration and bioelectronics. *Nat. Chem. Eng.* **1**, 73–86 (2024). doi:10.1038/s44286-023-00008-y

6. Y. Liang, J. He, B. Guo, Functional hydrogels as wound dressing to enhance wound healing. *ACS Nano* **15**, 12687–12722 (2021). doi:10.1021/acsnano.1c04206 Medline

7. J. Deng, H. Yuk, J. Wu, C. E. Varela, X. Chen, E. T. Roche, C. F. Guo, X. Zhao, Electrical bioadhesive interface for bioelectronics. *Nat. Mater.* **20**, 229–236 (2021). doi:10.1038/s41563-020-00814-2 Medline

8. J. Ma, J. Zhong, F. Sun, B. Liu, Z. Peng, J. Lian, X. Wu, L. Li, M. Hao, T. Zhang, Hydrogel sensors for biomedical electronics. *Chem. Eng. J.* **481**, 148317 (2024). doi:10.1016/j.cej.2023.148317

9. P. Li, J. Zhang, H. Hayashi, J. Yue, W. Li, C. Yang, C. Sun, J. Shi, J. Huberman-Shlaes, N. Hibino, B. Tian, Monolithic silicon for high spatiotemporal translational photostimulation. *Nature* **626**, 990–998 (2024). doi:10.1038/s41586-024-07016-9 Medline

10. J. Liu, N. Liu, Y. Xu, M. Wu, H. Zhang, Y. Wang, Y. Yan, A. Hill, R. Song, Z. Xu, M. Park, Y. Wu, J. L. Ciatti, J. Gu, H. Luan, Y. Zhang, T. Yang, H.-Y. Ahn, S. Li, W. Z. Ray, C. K. Franz, M. R. MacEwan, Y. Huang, C. W. Hammill, H. Wang, J. A. Rogers, Bioresorbable shape-adaptive structures for ultrasonic monitoring of deep-tissue homeostasis. *Science* **383**, 1096–1103 (2024). doi:10.1126/science.adk9880 Medline

11. X. Strakosas, H. Biesmans, T. Abrahamsson, K. Hellman, M. S. Ejneby, M. J. Donahue, P. Ekström, F. Ek, M. Savvakis, M. Hjort, D. Bliman, M. Linares, C. Lindholm, E. Stavrinidou, J. Y. Gerasimov, D. T. Simon, R. Olsson, M. Berggren, Metabolite-induced in vivo fabrication of substrate-free organic bioelectronics. *Science* **379**, 795–802 (2023). doi:10.1126/science.adc9998 Medline

12. H. Yuk, J. Wu, X. Zhao, Hydrogel interfaces for merging humans and machines. *Nat. Rev. Mater.* **7**, 935–952 (2022). doi:10.1038/s41578-022-00483-4

13. S. Naahidi, M. Jafari, M. Logan, Y. Wang, Y. Yuan, H. Bae, B. Dixon, P. Chen, Biocompatibility of hydrogel-based scaffolds for tissue engineering applications. *Biotechnol. Adv.* **35**, 530–544 (2017). doi:10.1016/j.biotechadv.2017.05.006 Medline

14. E. Axpe, D. Chan, G. S. Offeddu, Y. Chang, D. Merida, H. L. Hernandez, E. A. Appel, A multiscale model for solute diffusion in hydrogels. *Macromolecules* **52**, 6889–6897 (2019). doi:10.1021/acs.macromol.9b00753 Medline

15. D. Liu, C. Huyan, Z. Wang, Z. Guo, X. Zhang, H. Torun, D. Mulvihill, B. B. Xu, F. Chen, Conductive polymer based hydrogels and their application in wearable sensors: A review. *Mater. Horiz.* **10**, 2800–2823 (2023). doi:10.1039/D3MH00056G Medline

16. B. Lu, H. Yuk, S. Lin, N. Jian, K. Qu, J. Xu, X. Zhao, Pure PEDOT:PSS hydrogels. *Nat. Commun.* **10**, 1043 (2019). doi:10.1038/s41467-019-09003-5 Medline

17. T. Zhou, H. Yuk, F. Hu, J. Wu, F. Tian, H. Roh, Z. Shen, G. Gu, J. Xu, B. Lu, X. Zhao, 3D printable high-performance conducting polymer hydrogel for all-hydrogel bioelectronic interfaces. *Nat. Mater.* **22**, 895–902 (2023). doi:10.1038/s41563-023-01569-2 Medline

18. T. Dvir, B. P. Timko, M. D. Brigham, S. R. Naik, S. S. Karajanagi, O. Levy, H. Jin, K. K. Parker, R. Langer, D. S. Kohane, Nanowired three-dimensional cardiac patches. *Nat. Nanotechnol.* **6**, 720–725 (2011). doi:10.1038/nnano.2011.160 Medline

19. Y. Lu, G. Yang, S. Wang, Y. Zhang, Y. Jian, L. He, T. Yu, H. Luo, D. Kong, Y. Xianyu, B. Liang, T. Liu, X. Ouyang, J. Yu, X. Hu, H. Yang, Z. Gu, W. Huang, K. Xu, Stretchable graphene–hydrogel interfaces for wearable and implantable bioelectronics. *Nat. Electron.* **7**, 51–65 (2024). doi:10.1038/s41928-023-01091-y

20. Y. Zheng, S. Zhang, J. B.-H. Tok, Z. Bao, Molecular design of stretchable polymer semiconductors: Current progress and future directions. *J. Am. Chem. Soc.* **144**, 4699–4715 (2022). doi:10.1021/jacs.2c00072 Medline

21. L. Ding, Z.-D. Yu, X.-Y. Wang, Z.-F. Yao, Y. Lu, C.-Y. Yang, J.-Y. Wang, J. Pei, Polymer semiconductors: Synthesis, processing, and applications. *Chem. Rev.* **123**, 7421–7497 (2023). doi:10.1021/acs.chemrev.2c00696 Medline

22. K. M. Sim, S. Yoon, S.-K. Kim, H. Ko, S. Z. Hassan, D. S. Chung, Surfactant-induced solubility control to realize water-processed high-precision patterning of polymeric semiconductors for full color organic image sensor. *ACS Nano* **14**, 415–421 (2020). doi:10.1021/acsnano.9b06076 Medline

23. J. Cho, S. Yoon, K. Min Sim, Y. Jin Jeong, C. Eon Park, S.-K. Kwon, Y.-H. Kim, D. S. Chung, Universal selection rule for surfactants used in miniemulsion processes for eco-friendly and high performance polymer semiconductors. *Energy Environ. Sci.* **10**, 2324–2333 (2017). doi:10.1039/C7EE01943B

24. B. Kang, Z. Wu, M. J. Kim, H. Y. Woo, J. H. Cho, Aqueous-alcohol-processable high-mobility semiconducting copolymers with engineered oligo(ethylene glycol) side chains. *Chem. Mater.* **32**, 1111–1119 (2020). doi:10.1021/acs.chemmater.9b04004

25. P. Li, W. Sun, J. Li, J.-P. Chen, X. Wang, Z. Mei, G. Jin, Y. Lei, R. Xin, M. Yang, J. Xu, X. Pan, C. Song, X.-Y. Deng, X. Lei, K. Liu, X. Wang, Y. Zheng, J. Zhu, S. Lv, Z. Zhang, X. Dai, T. Lei, N-type semiconducting hydrogel. *Science* **384**, 557–563 (2024). doi:10.1126/science.adj4397 Medline

26. Z. Xu, K. S. Park, J. J. Kwok, O. Lin, B. B. Patel, P. Kafle, D. W. Davies, Q. Chen, Y. Diao, Not all aggregates are made the same: Distinct structures of solution aggregates drastically modulate assembly pathways, morphology, and electronic properties of conjugated polymers. *Adv. Mater.* **34**, e2203055 (2022). doi:10.1002/adma.202203055 Medline

27. C. G. Bischak, L. Q. Flagg, D. S. Ginger, Ion exchange gels allow organic electrochemical transistor operation with hydrophobic polymers in aqueous solution. *Adv. Mater.* **32**, e2002610 (2020). doi:10.1002/adma.202002610 Medline

28. L. A. Galuska, E. S. Muckley, Z. Cao, D. F. Ehlenberg, Z. Qian, S. Zhang, S. Rondeau-Gagné, M. D. Phan, J. F. Ankner, I. N. Ivanov, X. Gu, SMART transfer method to directly compare the mechanical response of water-supported and free-standing ultrathin polymeric films. *Nat. Commun.* **12**, 2347 (2021). doi:10.1038/s41467-021-22473-w Medline

29. Z. Yan, D. Xu, Z. Lin, P. Wang, B. Cao, H. Ren, F. Song, C. Wan, L. Wang, J. Zhou, X. Zhao, J. Chen, Y. Huang, X. Duan, Highly stretchable van der Waals thin films for adaptable and breathable electronic membranes. *Science* **375**, 852–859 (2022). doi:10.1126/science.abl8941 Medline

30. S. Capuani, G. Malgir, C. Y. X. Chua, A. Grattoni, Advanced strategies to thwart foreign body response to implantable devices. *Bioeng. Transl. Med.* **7**, e10300 (2022). doi:10.1002/btm2.10300 Medline

31. N. Li, Y. Li, Z. Cheng, Y. Liu, Y. Dai, S. Kang, S. Li, N. Shan, S. Wai, A. Ziaja, Y. Wang, J. Strzalka, W. Liu, C. Zhang, X. Gu, J. A. Hubbell, B. Tian, S. Wang, Bioadhesive polymer semiconductors and transistors for intimate biointerfaces. *Science* **381**, 686–693 (2023). doi:10.1126/science.adg8758 Medline

32. R. Sridharan, B. Cavanagh, A. R. Cameron, D. J. Kelly, F. J. O'Brien, Material stiffness influences the polarization state, function and migration mode of macrophages. *Acta Biomater.* **89**, 47–59 (2019). doi:10.1016/j.actbio.2019.02.048 Medline

33. C.-L. Chang, W.-C. Lin, L.-Y. Ting, C.-H. Shih, S.-Y. Chen, T.-F. Huang, H. Tateno, J. Jayakumar, W.-Y. Jao, C.-W. Tai, C.-Y. Chu, C.-W. Chen, C.-H. Yu, Y.-J. Lu, C.-C. Hu, A. M. Elewa, T. Mochizuki, H.-H. Chou, Main-chain engineering of polymer photocatalysts with hydrophilic non-conjugated segments for visible-light-driven hydrogen evolution. *Nat. Commun.* **13**, 5460 (2022). doi:10.1038/s41467-022-33211-1 Medline

34. N. Wang, A. Yang, Y. Fu, Y. Li, F. Yan, Functionalized organic thin film transistors for biosensing. *Acc. Chem. Res.* **52**, 277–287 (2019). doi:10.1021/acs.accounts.8b00448 Medline

35. X. Ji, X. Lin, J. Rivnay, Organic electrochemical transistors as on-site signal amplifiers for electrochemical aptamer-based sensing. *Nat. Commun.* **14**, 1665 (2023). doi:10.1038/s41467-023-37402-2 Medline

36. G. E. Fenoy, R. Hasler, C. Lorenz, J. Movilli, W. A. Marmisollé, O. Azzaroni, J. Huskens, P. Bäuerle, W. Knoll, Interface engineering of "clickable" organic electrochemical transistors toward biosensing devices. *ACS Appl. Mater. Interfaces* **15**, 10885–10896 (2023). doi:10.1021/acsami.2c21493 Medline

37. D. Ohayon, G. Nikiforidis, A. Savva, A. Giugni, S. Wustoni, T. Palanisamy, X. Chen, I. P. Maria, E. Di Fabrizio, P. M. F. J. Costa, I. McCulloch, S. Inal, Biofuel powered glucose detection in bodily fluids with an n-type conjugated polymer. *Nat. Mater.* **19**, 456–463 (2020). doi:10.1038/s41563-019-0556-4 Medline

38. I. Migneault, C. Dartiguenave, M. J. Bertrand, K. C. Waldron, Glutaraldehyde: Behavior in aqueous solution, reaction with proteins, and application to enzyme crosslinking. *Biotechniques* **37**, 790–802 (2004). doi:10.2144/04375RV01 Medline

39. M. Pasta, F. La Mantia, Y. Cui, Mechanism of glucose electrochemical oxidation on gold surface. *Electrochim. Acta* **55**, 5561–5568 (2010). doi:10.1016/j.electacta.2010.04.069

40. Y. Jiang, S. Ji, J. Sun, J. Huang, Y. Li, G. Zou, T. Salim, C. Wang, W. Li, H. Jin, J. Xu, S. Wang, T. Lei, X. Yan, W. Y. X. Peh, S.-C. Yen, Z. Liu, M. Yu, H. Zhao, Z. Lu, G. Li, H. Gao, Z. Liu, Z. Bao, X. Chen, A universal interface for plug-and-play assembly of stretchable devices. *Nature* **614**, 456–462 (2023). doi:10.1038/s41586-022-05579-z Medline

41. C. B. Nielsen, A. Giovannitti, D.-T. Sbircea, E. Bandiello, M. R. Niazi, D. A. Hanifi, M. Sessolo, A. Amassian, G. G. Malliaras, J. Rivnay, I. McCulloch, Molecular design of semiconducting polymers for high-performance organic electrochemical transistors. *J. Am. Chem. Soc.* **138**, 10252–10259 (2016). doi:10.1021/jacs.6b05280 Medline

42. S. Griggs, A. Marks, D. Meli, G. Rebetez, O. Bardagot, B. D. Paulsen, H. Chen, K. Weaver, M. I. Nugraha, E. A. Schafer, J. Tropp, C. M. Aitchison, T. D. Anthopoulos, N. Banerji, J. Rivnay, I. McCulloch, The effect of residual palladium on the performance of organic electrochemical transistors. *Nat. Commun.* **13**, 7964 (2022). doi:10.1038/s41467-022-35573-y Medline

43. D. Bernards, G. Malliaras, Steady-state and transient behavior of organic electrochemical transistors. *Adv. Funct. Mater.* **17**, 3538–3544 (2007). doi:10.1002/adfm.200601239

44. D. Moia, A. Giovannitti, A. A. Szumska, I. P. Maria, E. Rezasoltani, M. Sachs, M. Schnurr, P. R. F. Barnes, I. McCulloch, J. Nelson, Design and evaluation of conjugated polymers with polar side chains as electrode materials for electrochemical energy storage in aqueous electrolytes. *Energy Environ. Sci.* **12**, 1349–1357 (2019). doi:10.1039/C8EE03518K

45. A. Giovannitti, D.-T. Sbircea, S. Inal, C. B. Nielsen, E. Bandiello, D. A. Hanifi, M. Sessolo, G. G. Malliaras, I. McCulloch, J. Rivnay, Controlling the mode of operation of organic transistors through side-chain engineering. *Proc. Natl. Acad. Sci. U.S.A.* **113**, 12017–12022 (2016). doi:10.1073/pnas.1608780113 Medline

46. A. Giovannitti, I. P. Maria, D. Hanifi, M. J. Donahue, D. Bryant, K. J. Barth, B. E. Makdah, A. Savva, D. Moia, M. Zetek, P. R. F. Barnes, O. G. Reid, S. Inal, G. Rumbles, G. G. Malliaras, J. Nelson, J. Rivnay, I. McCulloch, The role of the side chain on the performance of N-type conjugated polymers in aqueous electrolytes. *Chem. Mater.* **30**, 2945–2953 (2018). doi:10.1021/acs.chemmater.8b00321 Medline

47. J. Mun, Y. Ochiai, W. Wang, Y. Zheng, Y.-Q. Zheng, H.-C. Wu, N. Matsuhisa, T. Higashihara, J. B.-H. Tok, Y. Yun, Z. Bao, A design strategy for high mobility stretchable polymer semiconductors. *Nat. Commun.* **12**, 3572 (2021). doi:10.1038/s41467-021-23798-2 Medline

48. R. Peng, B. Pang, D. Hu, M. Chen, G. Zhang, X. Wang, H. Lu, K. Cho, L. Qiu, An ABA triblock copolymer strategy for intrinsically stretchable semiconductors. *J. Mater. Chem. C* **3**, 3599–3606 (2015). doi:10.1039/C4TC02476A

49. B. O'Connor, E. P. Chan, C. Chan, B. R. Conrad, L. J. Richter, R. J. Kline, M. Heeney, I. McCulloch, C. L. Soles, D. M. DeLongchamp, Correlations between mechanical and electrical properties of polythiophenes. *ACS Nano* **4**, 7538–7544 (2010). doi:10.1021/nn1018768 Medline

50. Y.-C. Lin, Y.-W. Huang, C.-C. Hung, Y.-C. Chiang, C.-K. Chen, L.-C. Hsu, C.-C. Chueh, W.-C. Chen, Backbone engineering of diketopyrrolopyrrole-based conjugated polymers through random terpolymerization for improved mobility-stretchability property. *ACS Appl. Mater. Interfaces* **12**, 50648–50659 (2020). doi:10.1021/acsami.0c14592 Medline

51. J. Y. Oh, S. Rondeau-Gagné, Y.-C. Chiu, A. Chortos, F. Lissel, G. N. Wang, B. C. Schroeder, T. Kurosawa, J. Lopez, T. Katsumata, J. Xu, C. Zhu, X. Gu, W.-G. Bae, Y. Kim, L. Jin, J. W. Chung, J. B.-H. Tok, Z. Bao, Intrinsically stretchable and healable semiconducting polymer for organic transistors. *Nature* **539**, 411–415 (2016). doi:10.1038/nature20102 Medline

52. J. Mun, J. Kang, Y. Zheng, S. Luo, H.-C. Wu, N. Matsuhisa, J. Xu, G. N. Wang, Y. Yun, G. Xue, J. B.-H. Tok, Z. Bao, Conjugated carbon cyclic nanorings as additives for intrinsically stretchable semiconducting polymers. *Adv. Mater.* **31**, e1903912 (2019). doi:10.1002/adma.201903912 Medline

53. S. Savagatrup, X. Zhao, E. Chan, J. Mei, D. J. Lipomi, Effect of broken conjugation on the stretchability of semiconducting polymers. *Macromol. Rapid Commun.* **37**, 1623–1628 (2016). doi:10.1002/marc.201600377 Medline

54. J. Mun, G.-J. N. Wang, J. Y. Oh, T. Katsumata, F. L. Lee, J. Kang, H.-C. Wu, F. Lissel, S. Rondeau-Gagné, J. B.-H. Tok, Z. Bao, Effect of nonconjugated spacers on mechanical properties of semiconducting polymers for stretchable transistors. *Adv. Funct. Mater.* **28**, 1804222 (2018). doi:10.1002/adfm.201804222

55. Y.-W. Huang, Y.-C. Lin, H.-C. Yen, C.-K. Chen, W.-Y. Lee, W.-C. Chen, C.-C. Chueh, High mobility preservation of near amorphous conjugated polymers in the stretched states enabled by biaxially-extended conjugated side-chain design. *Chem. Mater.* **32**, 7370–7382 (2020). doi:10.1021/acs.chemmater.0c02258

56. Y. Dai, S. Dai, N. Li, Y. Li, M. Moser, J. Strzalka, A. Prominski, Y. Liu, Q. Zhang, S. Li, H. Hu, W. Liu, S. Chatterji, P. Cheng, B. Tian, I. McCulloch, J. Xu, S. Wang, Stretchable redox-active semiconducting polymers for high-performance organic electrochemical transistors. *Adv. Mater.* **34**, e2201178 (2022). doi:10.1002/adma.202201178 Medline

57. E. L. Melenbrink, K. M. Hilby, K. Choudhary, S. Samal, N. Kazerouni, J. L. McConn, D. J. Lipomi, B. C. Thompson, Influence of acceptor side-chain length and conjugation-break spacer content on the mechanical and electronic properties of semi-random polymers. *ACS Appl. Polym. Mater.* **1**, 1107–1117 (2019). doi:10.1021/acsapm.9b00115

58. X. Yu, L. Chen, C. Li, C. Gao, X. Xue, X. Zhang, G. Zhang, D. Zhang, Intrinsically stretchable polymer semiconductors with good ductility and high charge mobility through reducing the central symmetry of the conjugated backbone units. *Adv. Mater.* **35**, 2209896 (2023). doi:10.1002/adma.202209896 Medline

59. H.-C. Wu, C.-C. Hung, C.-W. Hong, H.-S. Sun, J.-T. Wang, G. Yamashita, T. Higashihara, W.-C. Chen, Isoindigo-based semiconducting polymers using carbosilane side chains for high performance stretchable field-effect transistors. *Macromolecules* **49**, 8540–8548 (2016). doi:10.1021/acs.macromol.6b02145

60. M. Y. Lee, S. Dharmapurikar, S. J. Lee, Y. Cho, C. Yang, J. H. Oh, Regular H-bonding-containing polymers with stretchability up to 100% external strain for self-healable plastic transistors. *Chem. Mater.* **32**, 1914–1924 (2020). doi:10.1021/acs.chemmater.9b04574

61. H.-F. Wen, H.-C. Wu, J. Aimi, C.-C. Hung, Y.-C. Chiang, C.-C. Kuo, W.-C. Chen, Soft poly(butyl acrylate) side chains toward intrinsically stretchable polymeric semiconductors for field-effect transistor applications. *Macromolecules* **50**, 4982–4992 (2017). doi:10.1021/acs.macromol.7b00860

62. C. Müller, S. Goffri, D. W. Breiby, J. W. Andreasen, H. D. Chanzy, R. A. J. Janssen, M. M. Nielsen, C. P. Radano, H. Sirringhaus, P. Smith, N. Stingelin-Stutzmann, Tough, semiconducting polyethylene-poly(3-hexylthiophene) diblock copolymers. *Adv. Funct. Mater.* **17**, 2674–2679 (2007). doi:10.1002/adfm.200601248

63. Y.-C. Chiang, H.-C. Wu, H.-F. Wen, C.-C. Hung, C.-W. Hong, C.-C. Kuo, T. Higashihara, W.-C. Chen, Tailoring carbosilane side chains toward intrinsically stretchable semiconducting polymers. *Macromolecules* **52**, 4396–4404 (2019). doi:10.1021/acs.macromol.9b00589

64. J.-T. Wang, S. Takshima, H.-C. Wu, C.-C. Shih, T. Isono, T. Kakuchi, T. Satoh, W.-C. Chen, Stretchable conjugated rod–coil poly(3-hexylthiophene)-*block*-poly(butyl acrylate) thin films for field effect transistor applications. *Macromolecules* **50**, 1442–1452 (2017). doi:10.1021/acs.macromol.6b02722

65. Y. Li, W. K. Tatum, J. W. Onorato, Y. Zhang, C. K. Luscombe, Low elastic modulus and high charge mobility of low-crystallinity indacenodithiophene-based semiconducting polymers for potential applications in stretchable electronics. *Macromolecules* **51**, 6352–6358 (2018). doi:10.1021/acs.macromol.8b00898

66. G. N. Wang, L. Shaw, J. Xu, T. Kurosawa, B. C. Schroeder, J. Y. Oh, S. J. Benight, Z. Bao, Inducing elasticity through oligo-siloxane crosslinks for intrinsically stretchable semiconducting polymers. *Adv. Funct. Mater.* **26**, 7254–7262 (2016). doi:10.1002/adfm.201602603

67. S. Zhang, Y.-H. Cheng, L. Galuska, A. Roy, M. Lorenz, B. Chen, S. Luo, Y.-T. Li, C.-C. Hung, Z. Qian, P. B. J. St. Onge, G. T. Mason, L. Cowen, D. Zhou, S. I. Nazarenko, R. F. Storey, B. C. Schroeder, S. Rondeau-Gagné, Y.-C. Chiu, X. Gu, Tacky elastomers to enable tear-resistant and autonomous self-healing semiconductor composites. *Adv. Funct. Mater.* **30**, 2000663 (2020). doi:10.1002/adfm.202000663

68. A. Peña-Alcántara, S. Nikzad, L. Michalek, N. Prine, Y. Wang, H. Gong, E. Ponte, S. Schneider, Y. Wu, S. E. Root, M. He, J. B.-H. Tok, X. Gu, Z. Bao, Effect of molecular

weight on the morphology of a polymer semiconductor–thermoplastic elastomer blend. *Adv. Electron. Mater.* **9**, 2201055 (2023). doi:10.1002/aelm.202201055

69. J. Xu, S. Wang, G. N. Wang, C. Zhu, S. Luo, L. Jin, X. Gu, S. Chen, V. R. Feig, J. W. F. To, S. Rondeau-Gagné, J. Park, B. C. Schroeder, C. Lu, J. Y. Oh, Y. Wang, Y.-H. Kim, H. Yan, R. Sinclair, D. Zhou, G. Xue, B. Murmann, C. Linder, W. Cai, J. B.-H. Tok, J. W. Chung, Z. Bao, Highly stretchable polymer semiconductor films through the nanoconfinement effect. *Science* **355**, 59–64 (2017). doi:10.1126/science.aah4496 Medline

70. M. Shin, J. Y. Oh, K.-E. Byun, Y.-J. Lee, B. Kim, H.-K. Baik, J.-J. Park, U. Jeong, Polythiophene nanofibril bundles surface-embedded in elastomer: A route to a highly stretchable active channel layer. *Adv. Mater.* **27**, 1255–1261 (2015). doi:10.1002/adma.201404602 Medline

71. D. Devadiga, M. Selvakumar, P. Shetty, M. S. Santosh, The integration of flexible dye-sensitized solar cells and storage devices towards wearable self-charging power systems: A review. *Renew. Sustain. Energy Rev.* **159**, 112252 (2022). doi:10.1016/j.rser.2022.112252

72. K. Musaie, S. Abbaszadeh, V. Nosrati-Siahmazgi, M. Qahremani, S. Wang, M. R. Eskandari, S. V. Niknezhad, F. Haghi, Y. Li, B. Xiao, M.-A. Shahbazi, Metal-coordination synthesis of a natural injectable photoactive hydrogel with antibacterial and blood-aggregating functions for cancer thermotherapy and mild-heating wound repair. *Biomater. Sci.* **11**, 2486–2503 (2023). doi:10.1039/D2BM01965E Medline

73. Y. Wang, H.-M. Meng, G. Song, Z. Li, X.-B. Zhang, Conjugated-polymer-based nanomaterials for photothermal therapy. *ACS Appl. Polym. Mater.* **2**, 4258–4272 (2020). doi:10.1021/acsapm.0c00680